دوسية المهندسة نبال الزعبي

لمادة أنظمة وإشارات



Team ZerOne

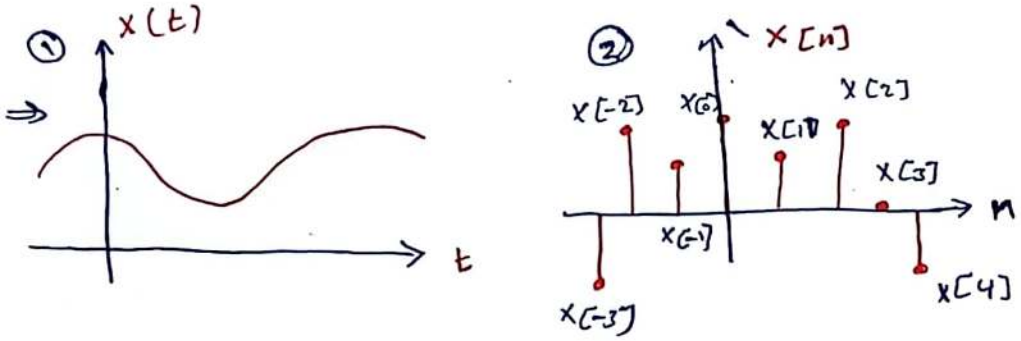
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* Signals & systems.

→ [1] Continuous-time signals. $X(t)$ → t: time.

[2] Discrete-time signals. $X[n]$ → n: Discrete time

n : is integer value



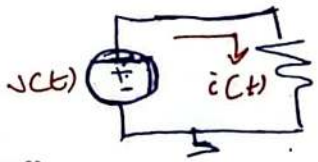
* Signal Energy & Power.

$[x(t)]$

① ⇒ A signal with finite energy which means that the amplitude → 0 at time → ∞ is said to be "Energy signal". ⇒ must have zero power.

② ⇒ A signal $x(t)$ with finite and non zero power → is said to be power signal $\boxed{0 < P < \infty}$ then. of necessity Energy = ∞

③ signals which nither power nor energy are finite.



⇒ $P(t) = v(t)\, i(t)$. ⇒ The instantaneous power.

$= \frac{v^2(t)}{R}$

$R = 1\ \Omega$.

$P(t) = v^2(t)$ ⇒ normalized est. power.

$v(t)$ is a signal.

$\boxed{P(t) = x^2(t).}$

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⇒ The total energy of a continuous-time signal $x(t)$ is defined as:-

$$E = \int_{-\infty}^{\infty} x^2(t)\, dt.$$

⇒ if $x(t)$ complex valu

$$E = \int_{-\infty}^{\infty} |x(t)|^2\, dt.$$

⇒ $0 < E < \infty$ → Energy signal.

⇒ Like since the power is the time average of energy the power is defined as:-

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{T/2} x^2(t)\, dt.$$

OR $\lim_{T\to\infty} \frac{1}{T} \int_{0}^{T} x^2(t)\, dt$

$$P = \lim_{T\to\infty} \frac{E}{2T}$$

$0 < P < \infty$ → power signal.

⇒ For discrete-time signal. $x[n]$, $n_1 \le n \le n_2$.

The energy $E = \sum_{n=n_1}^{n_2} x[n]^2$ ⇒ real signal or $\sum_{n=-\infty}^{\infty} x[n]$

$$E = \sum_{n=n_1}^{n_2} |x[n]|^2 \Rightarrow \text{complex signal.}$$

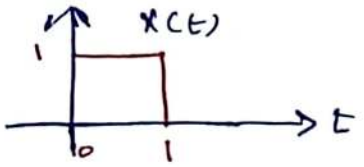
→ The total power



$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} x[n]^2$$

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Ex(1) $x(t) = 1, \quad 0 \le t \le 1$



Sol: $E = \int_{-\infty}^{\infty} x(t)^2 \, dt = \int_{0}^{1} 1 \, dt$

$\boxed{E = 1} \Rightarrow$ finite energy.

$$P = \lim_{T \to \infty} \frac{E}{2T}$$

$$P = \lim_{T \to \infty} \frac{1}{2T} = 0$$

$\neq$ Energy Signal.

Ex(2): $x(t) = 2t$.

$$E = \int_{-\infty}^{\infty} (2t)^2 \, dt$$

$$= 4 \int_{-\infty}^{\infty} t^2 \, dt.$$

$$E = \frac{4}{3} t^3 \Big]_{-\infty}^{\infty} = \boxed{\infty}$$

$$P = \lim_{T \to \infty} \frac{E}{2T} = \infty$$

$\boxed{P = \infty}$

$\#$ Nithar power nor Energy

Ex(3):- $x[n] = 2$.

$$E = \sum_{-\infty}^{\infty} (2)^2 = \sum_{-\infty}^{\infty} 4 = \infty$$

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} (2)^2 = \lim_{N \to \infty} \frac{\sum_{-N}^{N} 4}{2N+1}$$

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$\Rightarrow P = \lim_{N \to \infty} 4$

$\boxed{P = 4}$ # Power signal.

⇒ Periodic & Aperiodic (non periodic) signal.

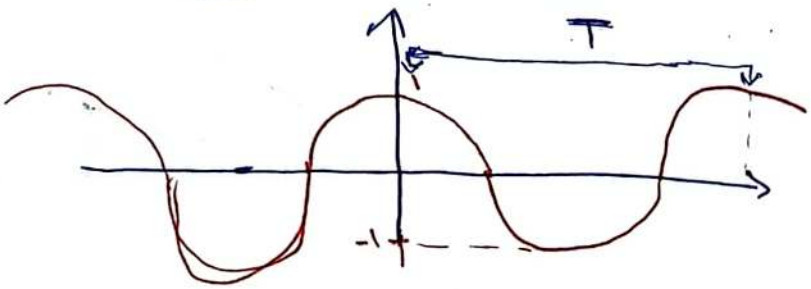
A signal $x(t)$ is said to be periodic signal with period T if it is exhibits periodicity.

$(T \neq 0)$
positive &
non zero value

⇒ $x(t) = x(t \pm T)$ - for all values of t.

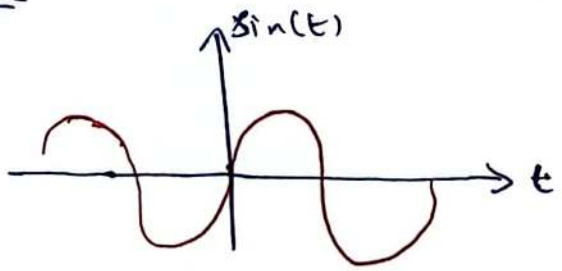
↳ T: is a period of the signal

⇒ ex:- $\cos(t)$



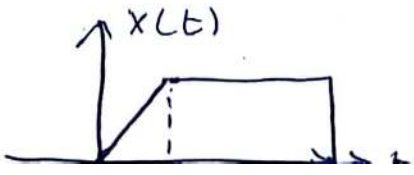
$\cos(t \pm T) = \cos(t)\cos(T) \pm \sin(t)\sin(T)$

ex: $x(t) = \sin(t)$



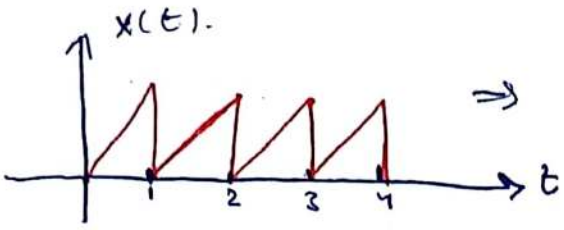
$\sin(t \pm T) = \sin t \cos T \pm \cos t \sin T$

⇒ signal dose not satisfy the $x(t) = x(t \pm T)$ is nonperiodic



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ex:



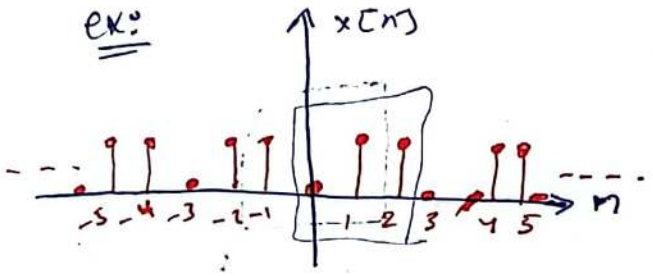
$\Rightarrow T_0 = 1$

The fundamental period of $x(t)$ T_0 is the smallest positive value of T

$\Rightarrow$ for discrete-time signal $x[n]$ is periodic with period N, where N is a positive integer if

$$x[n] = x[n+N] \quad \text{for all values of } n$$

ex:



$N_0 = 3$

* Even & odd signals.

→ Even signal if.

$x(t) = x(-t)$. → Continuous-time signal.

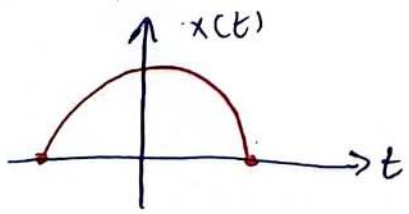
or $x[n] = x[-n]$ → Discrete-time signal.

→ Odd signal if.

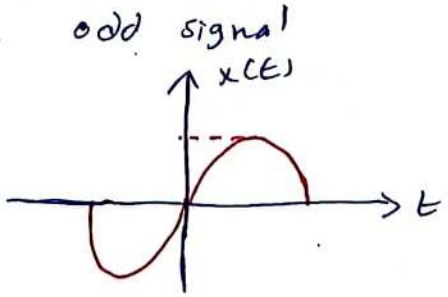
$x(-t) = -x(t)$. → Continuous-time signal.

$x[-n] = -x[n]$. → Discrete-time signal.

ex:- Even signal.



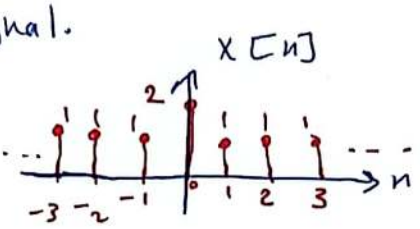
odd signal



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Ex: Discrete-time signal.

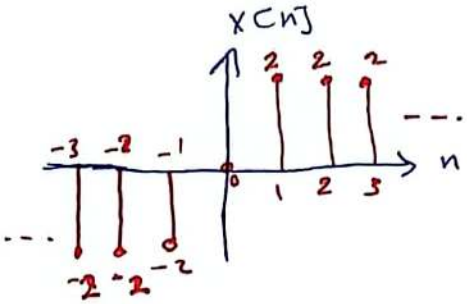
$$\Rightarrow x[n] = \begin{cases} 1, & n > 0 \\ 2, & n = 0 \\ 1, & n < 0 \end{cases}$$



Even signal

$x[n] = x[-n] \Rightarrow x_e[n]$.

$$\Rightarrow x[n] = \begin{cases} 2, & n > 0 \\ 0, & n = 0 \\ -2, & n < 0 \end{cases}$$



odd signal

$x[-n] = -x[n] \Rightarrow x_o[n]$.

$\Rightarrow x[n] = x_o[n] + x_e[n]$.

* Any signal can be broken into a sum of two part (even & odd) ;-

$$x_e[n] = \frac{1}{2}\left[x[n] + x[-n]\right]$$
$$x_o[n] = \frac{1}{2}\left[x[n] - x[-n]\right]$$

} $\Rightarrow$ for discrete-time

$\Rightarrow$ for continuous-time signal.

$$x(t) = x_e(t) + x_o(t).$$

$$x_e(t) = \frac{1}{2}\left[x(t) + x(-t)\right]$$

$$x_o(t) = \frac{1}{2}\left[x(t) - x(-t)\right].$$

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* ① Multiplication of any [even signal] with [odd signal]

⇒ [odd signal]. *

Proof:- $y(t) = x_e(t) * x_o(t)$ ⇒ $x_e(t)$ ⇒ even signal. $x_o(t)$ ⇒ odd signal

let $y(-t) = x_e(-t) * x_o(-t)$

$y(-t) = x_e(t)\,(-x_o(t))$

$y(-t) = -x_e(t) * x_o(t)$

$\boxed{y(-t) = -y(t)}$

* $y(t)$ is odd signal.

② Multiplication any of [even signal] with [even signal]

⇒ Even signal. *

③ Multiplication any of [odd signal] with [odd signal]

⇒ Even signal. *

④ Transformation of the independent variable.

[1] Time shift.

- For continuous-time signal

$x(t) \Rightarrow x(t - t_0)$

- For discrete-time signal.

$x[n] = x[n - n_0]$

⇒ if (t_0 or n_0) is positive [$(t_0 > 0 , n_0 > 0)$]

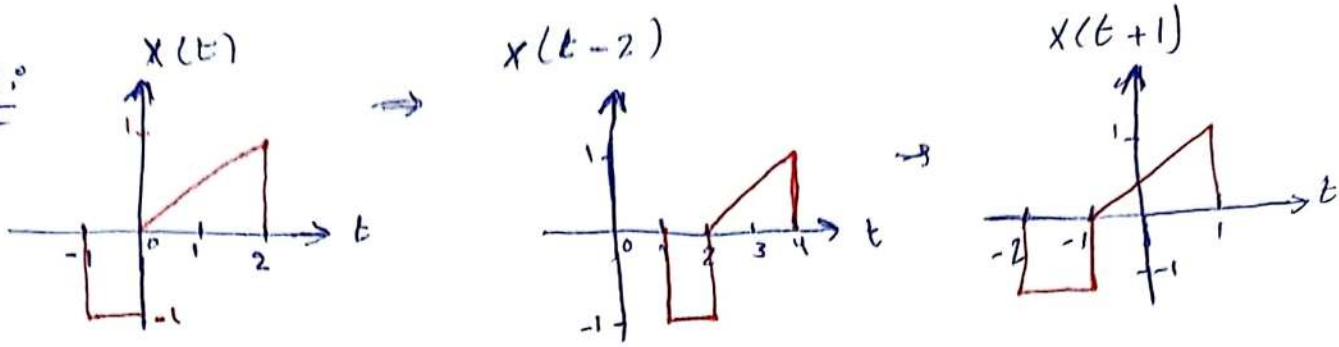
⇒ the signal [is delayed]

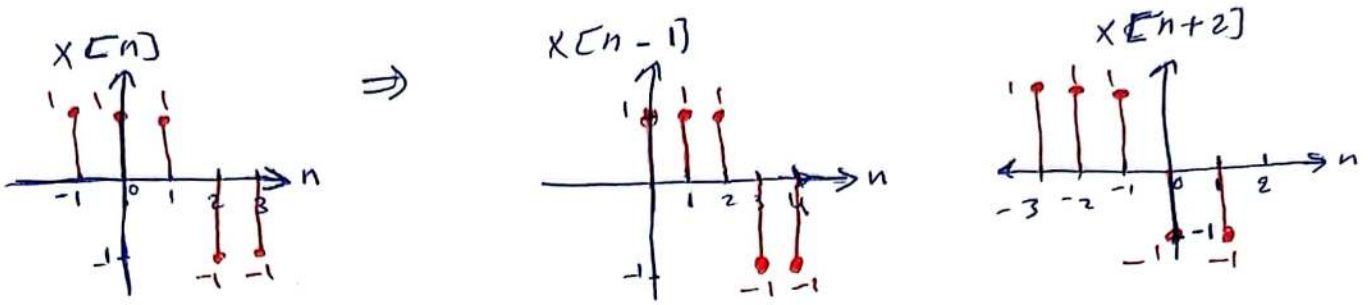
⇒ if (t_0 or n_0) is negative [$(t_0 < 0 , n_0 < 0)$]

⇒ the signal [is advanced]

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Ex:



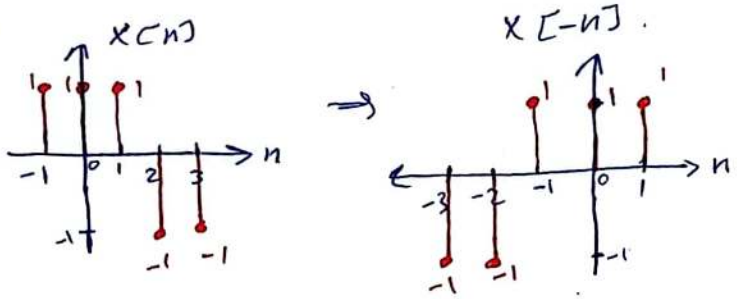


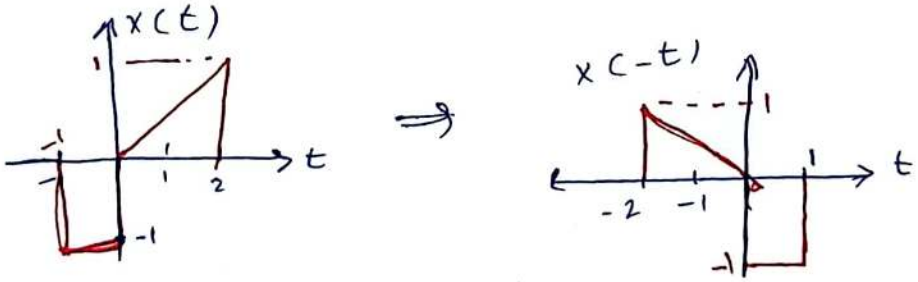
2 Time reversal..

$x(t) \Rightarrow x(-t)$

$x[n] \Rightarrow x[-n]$

Ex:





3 Time scaling

$x(t) \Rightarrow x(at)$

or $x[n] \Rightarrow x[an]$

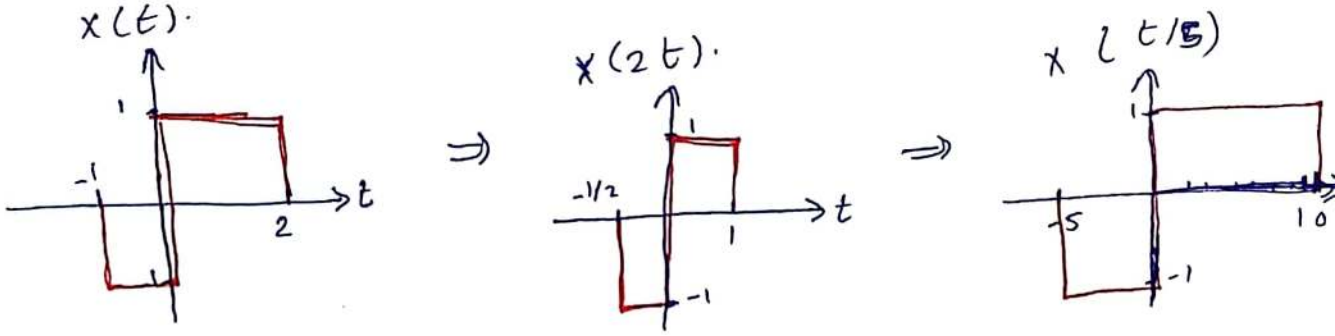
* n: is only integer value

① if $|a| > 1$ ⇒ the time scaled signal is compressed in time

② if $|a| < 1$ ⇒ the time scaled signal is expanded in time

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Ex:

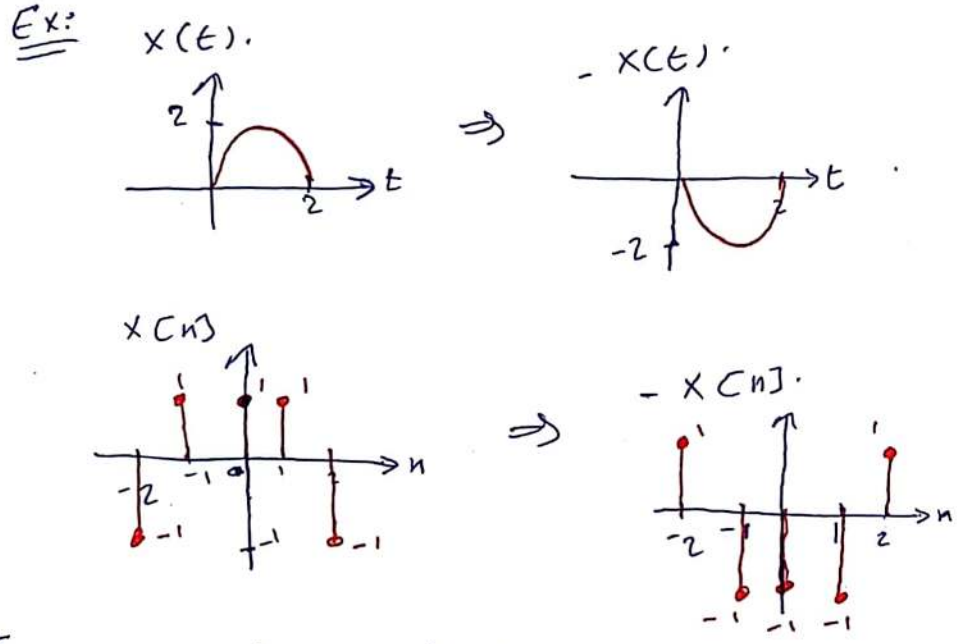


4 Amplitude reversal.

$$x(t) \Rightarrow -x(t).$$

$$x[n] \Rightarrow -x[n].$$

Ex:



5 Amplitude scaling.

$$x(t) \Rightarrow A\,x(t).$$

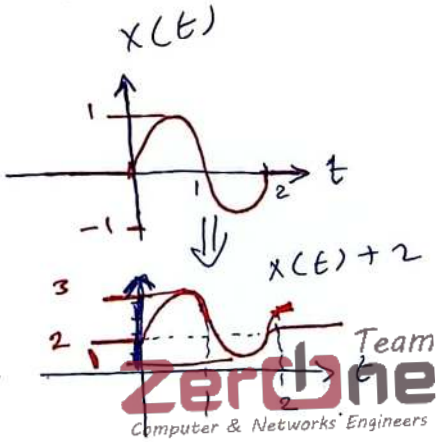
$$x[n] \Rightarrow A\,x[n].$$

6 Amplitude shifting.

$$x(t) \Rightarrow x(t) + B.$$

B ⇒ positive (up shift) ↑

B ⇒ negative (down shift) ↓

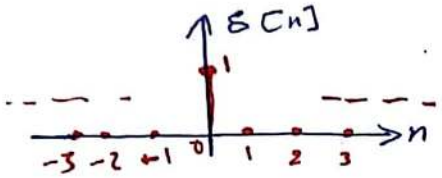


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* The unit impulse and unit step functions

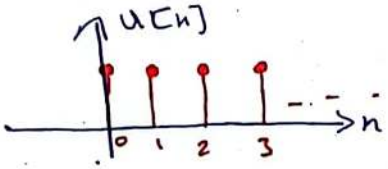
→ Discrete-time unit impuls. OR (unit sample).

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



→ Discrete-time unit step.

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



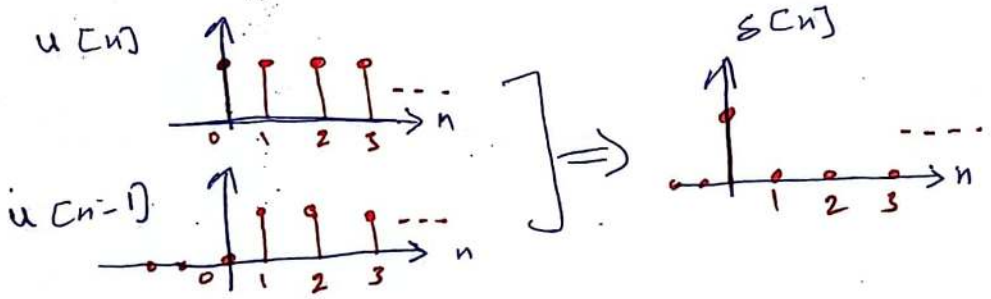
→ shifted time.

$$u[n-n_0] = \begin{cases} 1, & n > n_0 \\ 0, & n < n_0 \end{cases}$$

$$\Rightarrow \quad u(t-t_0) = \left(u(t-t_0)\right)^k$$

for k: any positive integer.

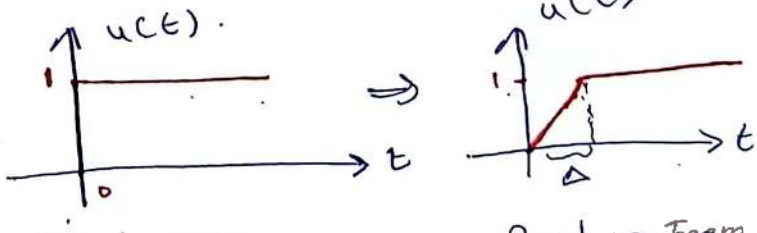
(*) $\delta[n] = u[n] - u[n-1]$



* $u[n] = \sum_{k=0}^{\infty} \delta[n-k]$.

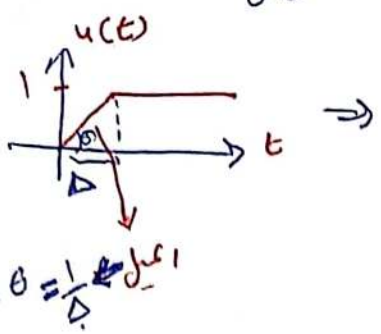
⇒ Continuous-time functions.

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

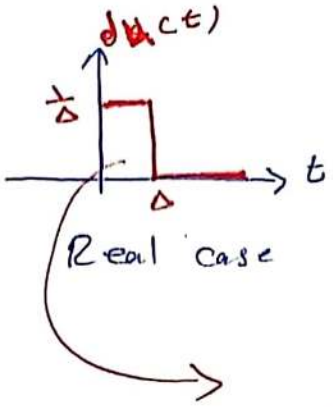


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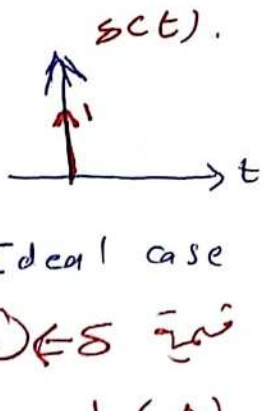
$\Rightarrow \quad \delta(t) = \frac{d\,u(t)}{dt}$ $\Rightarrow$ First derivative.



$\Rightarrow$



$\lim_{\Delta \to 0}$



① ⇐ δ نبحث

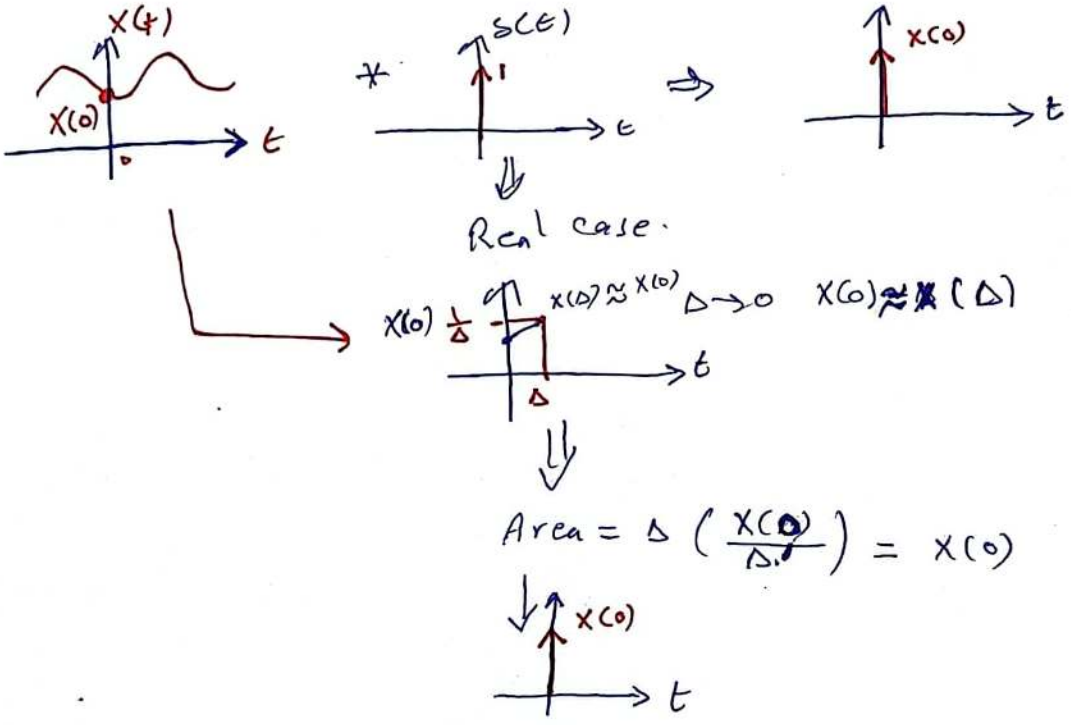
$$area = \lim_{\Delta \to 0} \frac{1}{\Delta}(\Delta) = 1$$

$$\Rightarrow u(t) = \int_0^{\infty} \delta(t-\tau)\, d\tau$$

✱ Properties of $\delta(t)$

① Sifting.

$$x(t)\,\delta(t) = x(0)\,\delta(t)$$



② $x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$

$t = t_0$

③ $x(t-t_0)\,\delta(t) = x(-t_0)\,\delta(t)$

$t = 0$

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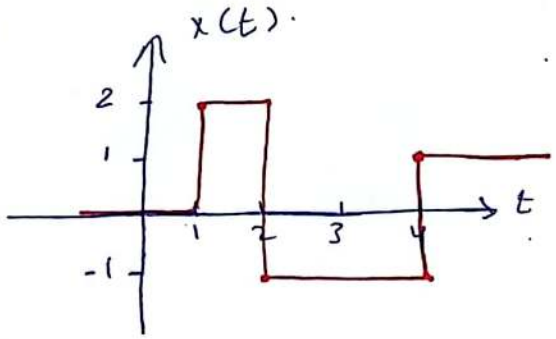
(4) $x(t-t_1)\,\delta(t-t_2) = x(t_2-t_1)\,\delta(t-t_2)$

$t = t_2$

(5) $\delta(t) = \delta(-t) \Rightarrow$ Even function.

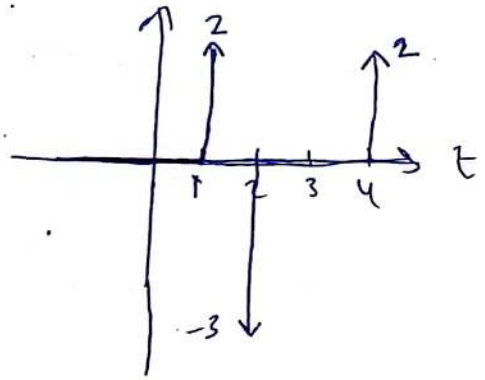
Just for $\delta(t) \neq 0$

(6) Ex:



$x(t) = 2u(t-1) - 3u(t-2) + 2u(t-4)$

$\frac{dx(t)}{dt}$



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* 6 $\delta(at+b) = \delta(a(t+\frac{b}{a}))$

$= \frac{1}{|a|}\delta(t+\frac{b}{a})$

Area.

* $u(at+b) = u(t+\frac{b}{a})$

Proof:-

$$\int_{-\infty}^{\infty} \delta(at+b)\,dt = \frac{1}{|a|}$$

let. $at+b = u$

$a\,dt = du$

at $t = \pm\infty \Rightarrow u = \begin{cases} \pm\infty, & a>0 \\ \mp\infty, & a<0 \end{cases}$

$\Rightarrow$ Two case :-

① $a>0$

$$\int_{-\infty}^{\infty} \delta(u)\frac{du}{a} = \frac{1}{a}\underbrace{\int_{-\infty}^{\infty}\delta(u)\,du}_{1} = \frac{1}{|a|}$$

$\Rightarrow a = \begin{cases} |a|, & a>0 \\ -|a|, & a<0 \end{cases}$

② $a<0$

$$\int_{\infty}^{-\infty} \delta(u)\frac{du}{a} = \frac{-1}{a}\underbrace{\int_{-\infty}^{\infty}\delta(u)\,du}_{1}$$

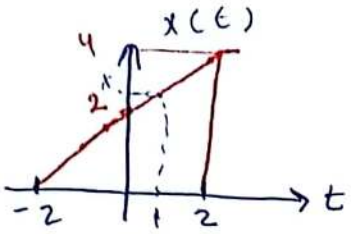
$$= \frac{-1}{a}$$

$$= \frac{-1}{-|a|} = \frac{1}{|a|}$$ *

$$\Rightarrow \int_{-\infty}^{\infty} \delta(at+b)\,dt = \frac{1}{|a|}$$ *

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* Ex:



① $\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = x(0)\int_{-\infty}^{\infty}\delta(t)\,dt$ ($t=0$)

$= x(0)$

$= 2.$

② $\int_{-\infty}^{\infty} x(t)\,\delta(t-1)\,dt = \int_{-\infty}^{\infty} x(1)\,\delta(t-1)\,dt$ ($t=1$)

$= x(1)$

$\Rightarrow \frac{x(t)-0}{t+2} = \frac{x(t)}{t+2} = 3$

slop: $\frac{x(t)}{t+2} = \frac{4-0}{2-(-2)} = 1$

$x(t) = t+2$

$x(1) = 3$

③ $\int_{-\infty}^{\infty} x(t-1)\,\delta(t-1)\,dt$ ($t=1$)

$= x(0) = 2.$

④ $\int_{-\infty}^{\infty} x(t)\,\delta(2t-1)\,dt.$

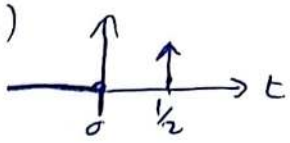
$= \int_{-\infty}^{\infty} x(t)\,\delta\left(2\left(t-\frac{1}{2}\right)\right)dt$

$= \int_{-\infty}^{\infty} x(t)\,\frac{1}{|2|}\,\delta\left(t-\frac{1}{2}\right)dt$ ($t=\frac{1}{2}$)

$= \frac{x(1/2)}{|2|} = \frac{2.5}{2} = \frac{5}{4}$

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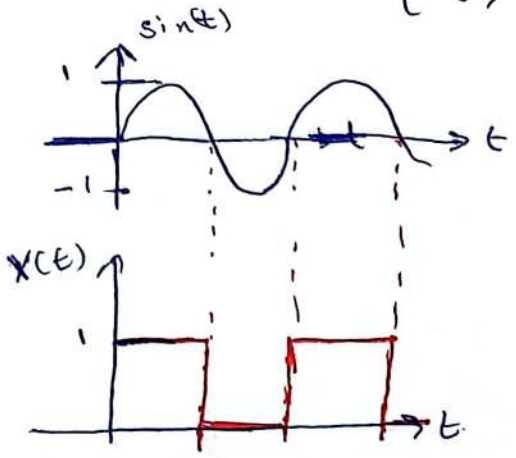
(5) $\int_{-\infty}^{0} x(t)\,\delta(2t-1)\,dt = 0$ → $\delta(2t-1)$



$2t-1 = 0$ not in the interval $(-\infty, 0)$

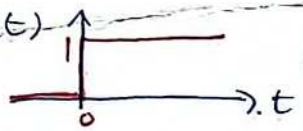
Ex: plot $x(t) = u(\sin t)$

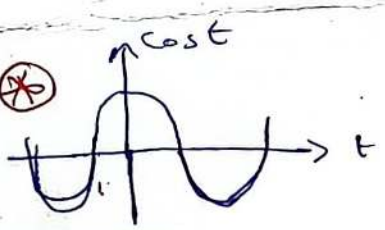
$$\Rightarrow u(\sin t) = \begin{cases} 1, & \sin t > 0 \\ 0, & \sin t < 0 \end{cases}$$

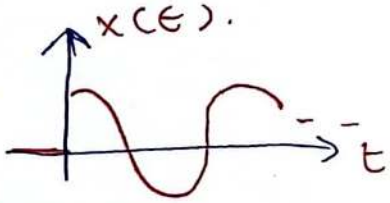


Ex: plot $x(t) = u(t)\cos(t)$.

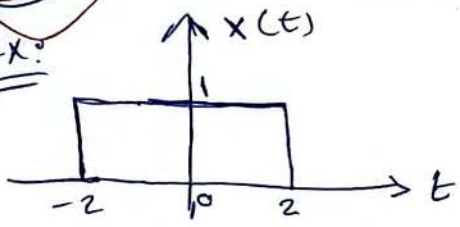
Sol: $u(t)$





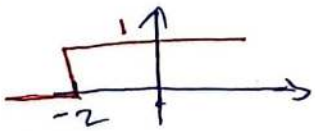


Ex:

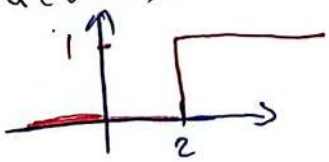


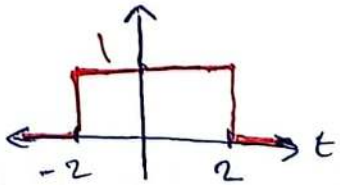
→ Express x(t) as a function of u(t).

Sol: $u(t+2)$



$u(t-2)$



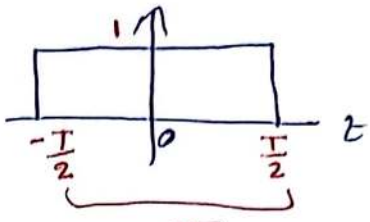


$$x(t) = u(t+2) - u(t-2)$$

Rect ⇒ $\text{rect}\left(\frac{t}{4}\right)$

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$$\text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & -\frac{T}{2} < t < \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$



T width.

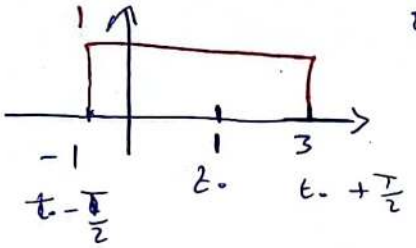
$$\Rightarrow \text{rect}\left(\frac{t}{T}\right) = \begin{cases} u\left(t+\frac{T}{2}\right) - u\left(t-\frac{T}{2}\right) \\ u\left(\frac{T}{2}-t\right) - u\left(-\frac{T}{2}-t\right) \\ u\left(\frac{T}{2}+t\right) \oplus u\left(T/2-t\right) \end{cases}$$

$\Rightarrow$ Even function.

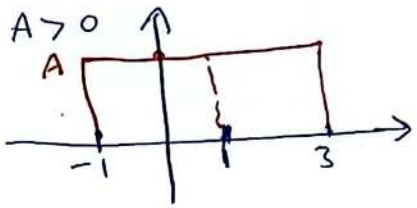
$$\rightarrow \text{rect}\left(\frac{t-t_0}{T}\right) = \begin{cases} 1, & t_0 - \frac{T}{2} < t < t_0 + \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

Ex: plot $\text{rect}\left(\frac{t-1}{4}\right)$ $\Rightarrow$ ※ scaling

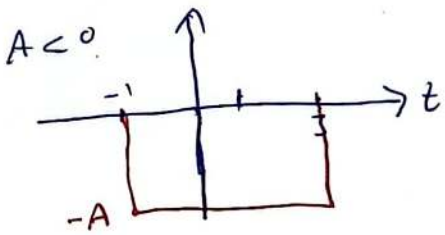
$T = 4$, $t_0 = 1$



$\Rightarrow$ Ⓐ $\text{rect}\left(\frac{t-1}{4}\right)$ const.

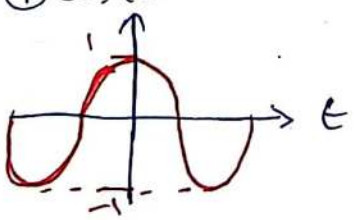


Ex: plot $x(t) = \cos(t)\,\text{rect}\left(\frac{t-\pi}{2\pi}\right)$



Sol:

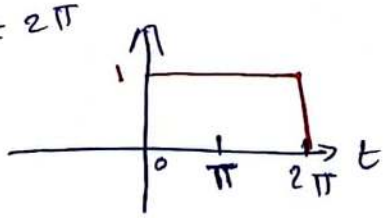
① $\cos(t)$



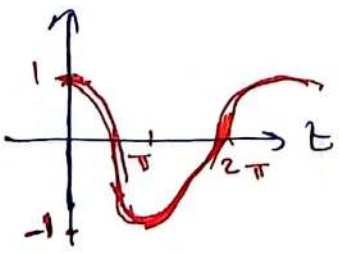
② $\text{rect}\left(\frac{t-\pi}{2\pi}\right)$

$t_0 = \pi$

$T = 2\pi$

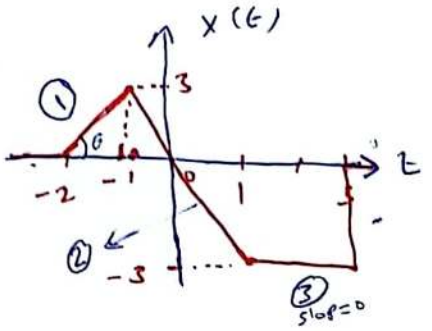


① * ②



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Ex: Express $x(t)$ as a function of $u(t)$



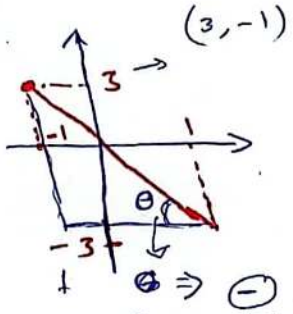
Sol: ① $\text{slop}_1 = \tan\theta$.

$= \frac{3}{1}$

$= 3$

$\Rightarrow \frac{x_1(t) - 0}{t+2} = 3$

$\Rightarrow \boxed{x_1(t) = 3(t+2)}$



② $\text{sbp}_2 = \tan\theta$.

$= (-)\left(\frac{6}{2}\right)$

$= -3$

$\frac{x_2(t) - 3}{t+1} = -3$

$\Rightarrow x_2(t) = -3(t+1) + 3$

$\boxed{x_2(t) = -3t}$

③ $x_3(t) = -3 \Rightarrow$ constant.

$x_1(t)$ ① ②

$x(t) = \underbrace{3(t+2)[u(t+2) - u(t+1)]}_{①} + \underbrace{-3t[u(t+1) - u(t-1)]}_{②}$

$+ \underbrace{-3[u(t-1) - u(t-3)]}_{③}$

$\Rightarrow x(t) = (3t)u(t+2) - (3t)u(t+1) + 6u(t+2) - 6u(t+1)$

$+ (3t)u(t+1) + (3t)u(t-1) - 3u(t-1) + 3u(t-3)$

$\Rightarrow x(t) = u(t+2)[3t+6] + u(t+1)[-6](t+1)$

$+ u(t-1)[3t-3] + 3u(t-3)$

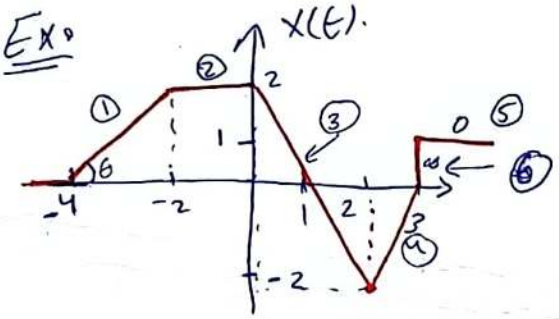
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$$\Rightarrow x(t) = 3(t+2)\,u(t+2) + -6(t+1)\,u(t+1) + 3(t-1)\,u(t-1) + 3\,u(t-3)$$

OR

$$x(t) = \overbrace{3}^{(slop_{cu} - slop_{pr})}(t+2)\,u(t+2) + \underbrace{(-3-3)}_{-6}(t+1)\,u(t+1) + 3(t-1)\,u(t-1) + 3\,u(t-3)$$

Ex:



Sol:

① $slop_1 = 1$

② $slop_2 = 0$

③ $slop_3 = -2$

④ $slop_4 = 2$

⑤ $slop_5 = 0$

⑥ vertical line.

$$\Rightarrow x(t) = (t+4)\,u(t+4) + (t+2)\,u(t+2) - 2(t)\,u(t) + 4(t-2)\,u(t-2) + u(t-3)\,[2(t-3)\,u(t-3)]$$

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* Periodic & non periodic signals.

⇒ Periodic if it repeates itself after a regular interval of time (T)

⇒ if $x(t \pm nT) = x(t)$.

n: integer
T: period

⇒ T_o: smallest value of time
↓
fundamental period. ⇒ $f_o = \frac{1}{T_o}$ Hz. ⇒ fundamental frequency.

Angular freq. ω_o

$\omega_o = 2\pi f_o$

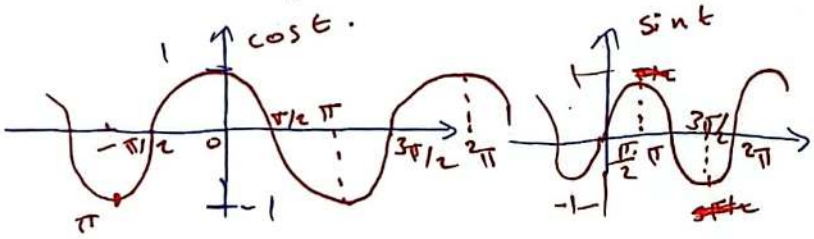
$\omega_o = \frac{2\pi}{T_o}$ rad/sec

OR for discrete-time signal.

$x[n] = x[n \pm mN]$.

m: integer.
N: period.
N_o: smallest value of time.

* cost & sint ⇒ are two periodic signals.



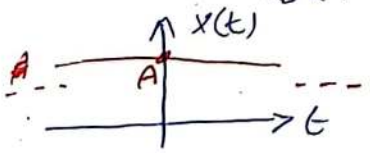
* $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

* $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

* if $x(t)$ is constant.

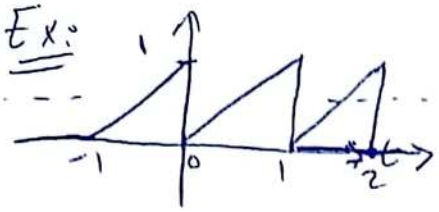
$x(t) = A$ → periodic signal.

but the T_o ⇒ is undefined.
↓
$f_o = 0$



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Ex:



Periodic signal.

$T_0 = 1$

$f_0 = 1$

$\omega_0 = \frac{2\pi}{T_0}$

$\omega_0 = 2\pi$ rad/sec

Ex: $x(t) = e^{\sin t}$

$x(t) \stackrel{?}{=} x(t+T)$.

$e^{\sin t} \stackrel{?}{=} e^{\sin(t+T)}$

At $T = 2\pi$.

$e^{\sin t} \stackrel{?}{=} e^{\sin(t+2\pi)}$

$= e^{\sin t \cos 2\pi + \cos t \sin 2\pi}$

$\cos 2\pi = 1$, $\sin 2\pi = 0$

$e^{\sin t} = e^{\sin t}$

$e^0 = 1$

$\therefore x(t)$ is periodic signal

Ex: $x(t) = t\, e^{\sin t}$.

$x(t+T) = (t+T)\, e^{\sin(t+T)}$

$= (t+2\pi)\, e^{\sin t}$

At $T = 2\pi$

$e^{\sin(t+2\pi)} = e^{\sin t}$

$\Rightarrow x(t) \neq x(t+T) \Rightarrow$ Non periodic signal.

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* The sum of Continuous-time periodic signals is periodic if and only if the ratios of the periodic of the individual signals are ratios of integer.

if sum of N periodic signals is periodic, the fundamental period can be found as follow:—

① Convert each period ratio $\frac{T_1}{T_i}$, $2 \le i \le N$ to a ratio of integers.

$T_1 \rightarrow$ the period of the first signal.

$T_i \Rightarrow$ The period of the other $(N-1)$ signals.

$\Rightarrow$ if one of these ratios is not rational

$\Rightarrow$ The sum is not periodic

$\frac{T_1}{T_i} = \frac{a}{b}$ for all $2 \le i \le N-1$ } periodic

~ are integers

$\frac{T_1}{T_{2,or\,3}} \neq \frac{a}{b} \Rightarrow$ not periodic.

② Eliminate common factors from the numerator and denominator of each ratio of integers.

③ The fun. period T_0 of the sum of the signals is $T_0 = k T_1$, where k is the least common factors from the numerator and denominator of each ratio of integers.

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Ex: 1 $x(t) = \cos\left(\frac{\pi}{3}t\right) + \sin\left(\frac{\pi}{4}t\right)$.

- Determine if the signal is periodic, if it is periodic determine the T_o :-

Sol: $x_1(t) = \cos\left(\frac{2\pi}{3}t\right) = \cos(\omega t)$

$x_2(t) = \sin\left(\frac{\pi}{4}t\right) = \sin(\omega t)$.

$\Rightarrow \omega_1 = \frac{2\pi}{T_1} \Rightarrow T_1 = \frac{2\pi}{\omega_1} \Rightarrow \frac{2\pi}{3} = \omega_1$

$T_1 = 3$ sec.

$\Rightarrow \omega_2 = \frac{2\pi}{T_2} \rightarrow T_2 = \frac{2\pi}{\pi}4$

$\omega_2 = \frac{\pi}{4}$

$T_2 = 8$ sec.

$\Rightarrow \frac{T_1}{T_2} = \frac{3}{8} \Rightarrow$ integer $\Rightarrow$ periodic signal $x(t)$

rational

$T_o = 8T_1$ OR $3T_2$

$T_o = 24$ sec

2 $x(t) = \sin(\sqrt{2}t) + \cos(t)$.

$x_1(t) = \sin(\sqrt{2}t) = \sin(\omega t)$

$x_2(t) = \cos(t) = \cos(\omega t)$

$T_1 = \frac{2\pi}{\omega_1} \rightarrow \omega_1 = \sqrt{2}$, $T_2 = \frac{2\pi}{\omega_2}$

$T_1 = \frac{2\pi}{\sqrt{2}}$

$\omega_2 = 1$

$T_2 = 2\pi$

$\Rightarrow \frac{T_1}{T_2} = \frac{2\pi}{\sqrt{2}\,2\pi} = \frac{1}{\sqrt{2}} \Rightarrow$ irrational

$\Rightarrow x(t)$ is non periodic signal

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3 $x(t) = \sin^2(t)$.

$$\sin^2(t) = \frac{1}{2}(1 - \cos 2t)$$

$$\therefore x(t) = \frac{1}{2} - \frac{1}{2}\cos(2t)$$

$$\Rightarrow T_0 = \frac{2\pi}{\omega}, \quad \omega = 2$$

$$\boxed{T_0 = \pi} \Rightarrow f_0 = \frac{1}{T_0} = \frac{1}{\pi} \text{ Hz}.$$

4 $x[n] = \cos\left[\frac{1}{4}n\right]$.

$$= \cos[\omega n].$$

$$N_0 = \frac{2\pi}{\omega} m, \quad \omega = \frac{1}{4}, \quad m: \text{integer value}.$$

$$\frac{N_0}{m} = 8\pi \Rightarrow \text{irrational}$$

$$\boxed{\frac{N_0}{m} = 8\pi} \Rightarrow x(t) \text{ is nonperiodic}$$

5 $x(t) = e^{\sin t}$

$$x(t+T) \stackrel{?}{=} x(t)$$

$$e^{\sin(t+T)} = e^{\sin t \cos T + \sin T \cos t}$$

at 2π: $\cos T = 1$, $\sin T = 0$

at $T = 2\pi$

$$\Rightarrow e^{\sin(t+T)} = e^{\sin(t)}$$

$$x(t+T) = x(t).$$

$\Rightarrow x(t)$ is periodic

$\Rightarrow$ if $x(t) = t\, e^{\sin t}$.

$$x(t+T) = (t+T)\, e^{\sin(t+T)}$$

$$x(t+T) = (t+T)\, e^{\sin t}$$

$$x(t+T) \neq x(t).$$

$\Rightarrow x(t)$ is non periodic.

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6 if $x(t) = \cos(3.5t + 30^\circ) + \sin(2t) + 2\cos(\frac{7}{6}t)$.

is $x(t)$ periodic signal or not?

Sol:

⇒ ① $x_1(t) = \cos(3.5t) = \cos(\omega t)$

$T_1 = \frac{2\pi}{\omega}$, $\omega = 3.5$

$$T_1 = \frac{2\pi}{3.5} = \frac{4}{7}\pi$$

② $x_2(t) = \sin(2t) = \sin(\omega t)$.

$T_2 = \frac{2\pi}{\omega}$, $\omega = 2$

$$T_2 = \pi$$

③ $x_3(t) = 2\cos(\frac{7t}{6}) = 2\cos(\omega t)$

$T_3 = \frac{2\pi}{\omega}$, $\omega = \frac{7}{6}$

$$T_3 = \frac{12}{7}\pi$$

⇒ $\frac{T_1}{T_2} = \frac{4}{7}\pi / \pi = \frac{4}{7}$ ⇒ rational.

$\frac{T_1}{T_3} = \frac{4\pi}{7} / \frac{12\pi}{7} = \frac{1}{3}$ ⇒ rational.

⇒ $x(t)$ is periodic.

⇒ $T_0 = k T_1$ ⇒ least Common factor] $3*7 = 21$

$T_0 = 21 \frac{4\pi}{7}$

$$T_0 = 12\pi$$

⇒ if we add

$x_4(t) = \sin(\frac{5\pi}{6}t)$

⇒ $T_4 = \frac{2\pi}{\omega}$, $\omega = \frac{5\pi}{6}$

$T_4 = \frac{2\pi}{\frac{5\pi}{6}} = \frac{12}{5}$

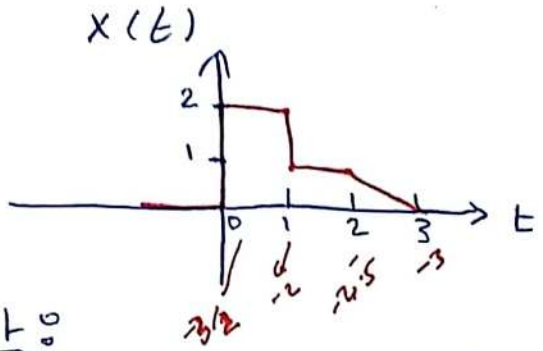
$\frac{T_1}{T_4} = \frac{4\pi}{7} \cdot \frac{5}{12} = \frac{5}{21}$ irrational.

⇒ $x(t)$ is non periodic

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⇒ Examples :-

⇒ ① Time shift.
② Time scale.
③ Time reverse
④ Amplitud scal or reverse
⑤ Amplitude shift.



* plot :

⊗ 2 x(-2t-3).

Sol: let $\tau = -2t-3$.

⇒ 2 x(τ).

→ $y(t) = 2\,x(\tau)$.

① At $\tau = 0$, ② $\tau = 1$, ③ $\tau = 2$

$-2t-3 = 0$ | $1 = -2t-3$ | $2 = -2t-3$

$-2t = 3$ | $-2t = 4$ | $-2t = 5$

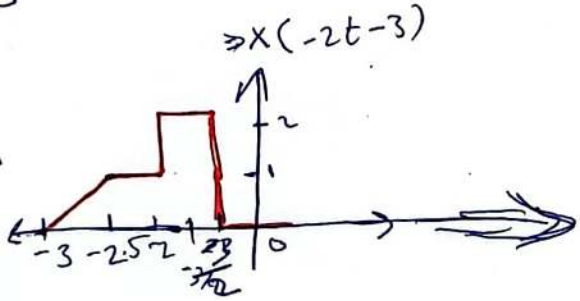
$t = -\frac{3}{2}$ | $t = -2$ | $t = -\frac{5}{2}$

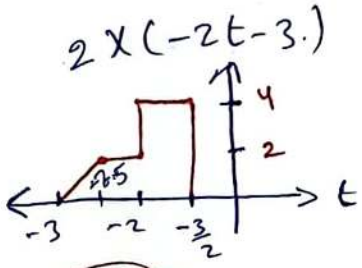
④ $\tau = 3$

$3 = -2t-3$

$-2t = 6$

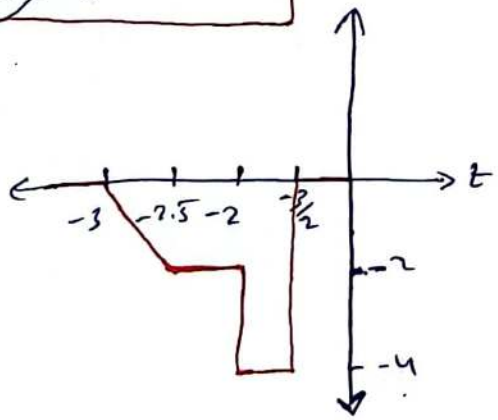
$t = -3$





#1.

If $-2x(-2t-3)$



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Continuous-time Systems.

⇒ A system transfors the input signals into output signals.

⇒ $x(t)$ or $x[n]$ → System → $y(t)$ or $y[n]$. ⇒ Block diagram.

System: $h(t)$ impuls response or Unit response.

⇒ $y(t) = T[x(t)]$ ⇒ $T[\cdot]$ ⇒ Transformation.

$x(t)$ ⇒ cause or input or excitation

$y(t)$ ⇒ effect or output or response

⇒ ① Continuous-time system. ⇒ All signals are continuous time (AM & FM radio)

② Discrete-time system ⇒ All signals are discrete-time. (PC)

③ Hybrid system ⇒ Both continuous and discrete-time signals appear in system.

↓

A/D, D/A] Converters.

⇒ The relations between the input and output signals are expressed as equations (system model).

⇒ Continuous-time ~~signals~~ systems.

⇒ The system-transformation notation $T[\cdot]$ ⇒

① Block ● → $x(t)$ → $T[\cdot]$ → $y(t)$.

② circle represents a summing junction. $x_1(t)$ → ⊕ ← $x_2(t)$ → $y(t) = x_1 + x_2$

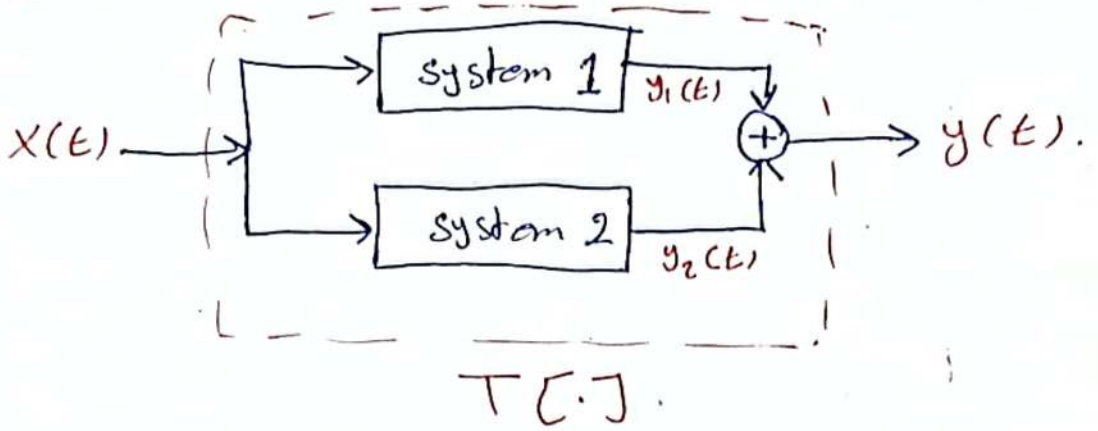
③ circle represents a product junction. $x_1(t)$ → ⊗ ← $x_2(t)$ → $y(t) = x_1(t) * x_2(t)$

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Interconnecting systems.

— Two basic connections.

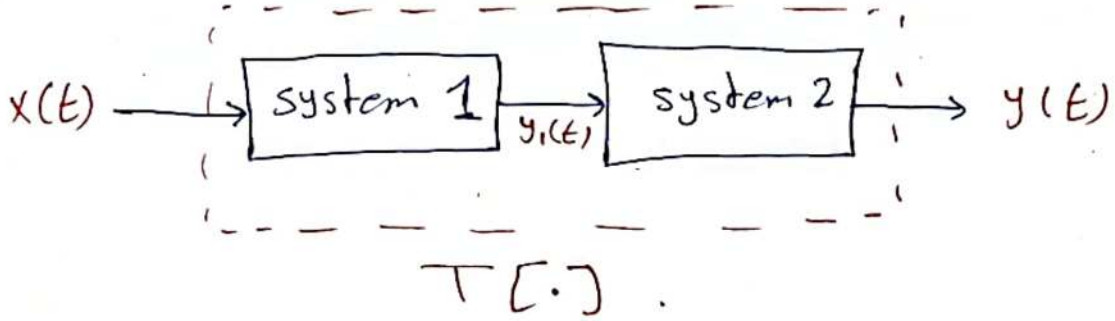
1. Parallel connection.



$$y(t) = T[x(t)] = y_1(t) + y_2(t).$$

$$= T_1[x(t)] + T_2[x(t)].$$

2. series (cascade) connection.

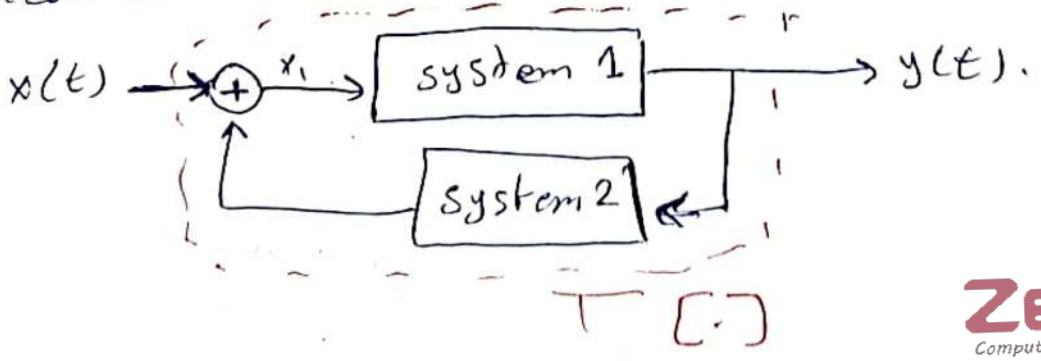


$$\Rightarrow y(t) = T_2[y_1(t)]$$

$$= T_2[T_1[x(t)]]$$

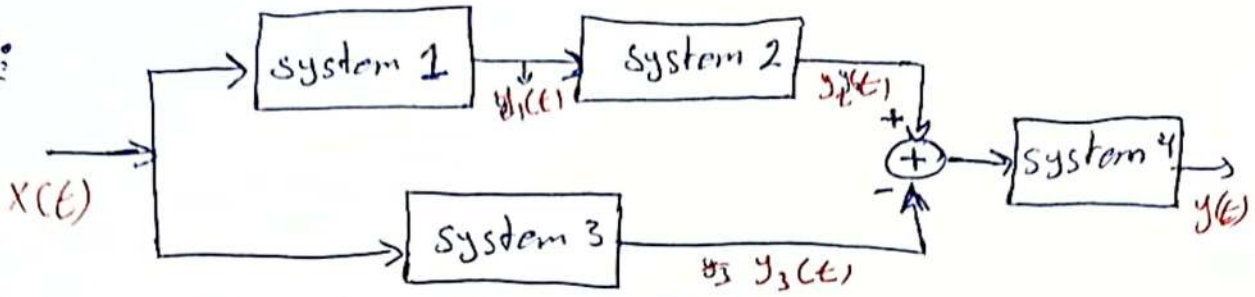
$$y(t) = T[x(t)]$$

3. Feedback Connection



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⇒ Ex:



Sol:

$$y(t) = T_4\left[y_2(t) - y_3(t)\right] \quad , \quad y_3(t) = T_3\left[x(t)\right]$$

$$y_2(t) = T_2\left[y_1(t)\right] = T_2\left[T_1\left[x(t)\right]\right]$$

$$y(t) = T_4\left[T_2\left[T_1\left[x(t)\right]\right] - T_3\left[x(t)\right]\right]$$

⇒ Properties of continuous-time systems.

① Memory ⇒ At t_0 : current value. ⇒ with memory / memoryless

if $y(t_0)$ depends only on $x(t_0)$

⇒ then the system is memoryless.

otherwise the system is memory. (with)

Ex: ① $y(t) = 2x(t) + 10$

⇒ At t_0

$y(t_0) = 2x(t_0) + 10$ ⇒ memoryless system.

② $y(t) = -x(t) + 2x(t-2)$

At t_0

$y(t_0) = -x(t_0) + 2x(t_0 - 2)$ past value.

⇒ with memory system

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(3) $y(t) = x^2(t)$.

$y(t_0) = x^2(t_0) \Rightarrow$ memoryless.

(4) $y(t) = x(t+2)$

$y(t_0) = x(t_0+2) \Rightarrow$ with memory.

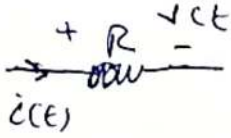
(5) $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau \Rightarrow$ with memory.

(6) $y(t) = (2x(t) - x^2(t))^2 \Rightarrow$ memoryless.

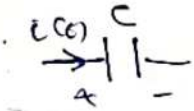
$y(t_0) = (2x(t_0) - x^2(t_0))^2$

$\Rightarrow$ Resistor is a memoryless system:

$v(t) = i(t) R$



$\Rightarrow$ Capacitor is with memory system.



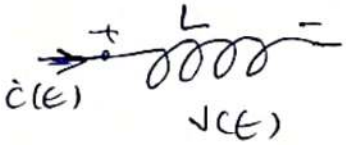
$\rightarrow i_c = C \frac{dv}{dt}$.

$v(t) = \frac{1}{C} \int_{-\infty}^{t} i(\tau)\, d\tau$.

$v(t_0) = \frac{1}{C} \int_{-\infty}^{t_0} i(\tau)\, dt \rightarrow i(t \le t_0)$

$\Rightarrow$ Inductor is with memory system

$\rightarrow v = L \frac{di}{dt}$.



$i(t) = \frac{1}{L} \int_{-\infty}^{t} v(\tau)\, d\tau$

$i(t_0) = \frac{1}{L} \int_{-\infty}^{t_0} v(\tau)\, d\tau \rightarrow v(t \le t_0)$

with memory.

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2 Invertibility. → invertible. / → not invertible.

if distinct inputs give distinct outputs
⇒ the system is invertible.

otherwise the system is not invertible.

⇒ if $y(t) = x^2(t)$

$y(2) = 4$
$y(-2) = 4$] → not invertible.

distinct input (2 & -2)
⇒ the same output. (4).

⇒ if $y(t) = 5\,x(t)$ ⇒ Amplifier.

$y(2) = 10$
$y(-2) = -10$] ⇒ distinct input ⇒ distinct output.
$y(1) = 5$

invertible system.

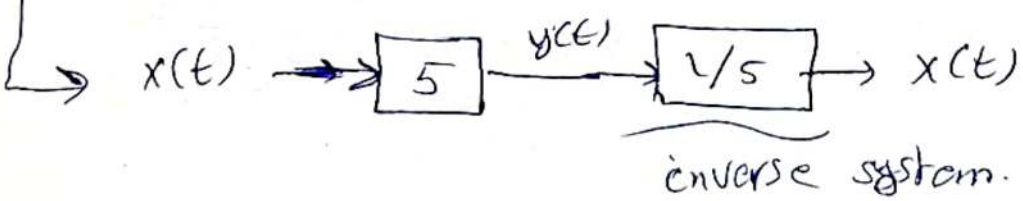
⇒ $x(t) = \frac{y(t)}{5}$ ⇒ inverse system. is invertible system.

Ex:

① $y(t) = 0$ → non invertible.

② $y(t) = x^3(t)$ → invertible.

③ $y(t) = x(t)$ → invertible.



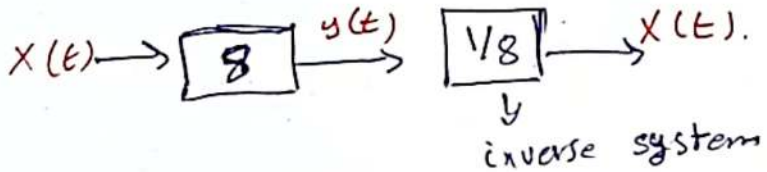
inverse system.

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⇒ The inverse of the system ($T_i[\cdot]$) is a system that when connected with the system $(T[\cdot])$ yields the identity system

→ $y(t) = x(t)$.

The inverse system is invertible system



3] Causality

- Causal
- non Causal.

* If $y(t_0)$ depends on $x(t \le t_0)$

⇒ the system is causal.

otherwise the system is non causal.

Ex: ① $y(t) = x(t)$ → causal.

② $y(t) = x(t-1)$ → causal.

③ $y(t) = x(t+2)$ → non causal.

④ $y(t) = x(t) + x(t+2)$ ⇒ non causal.

⇒ All memoryless systems are causal.

Ex: $y(t) = x(t)\cos(t+1)$ ⇒ causal system and memoryless.

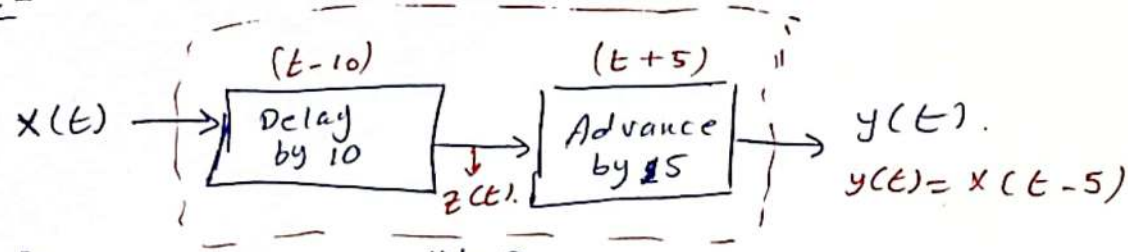
$t_0 \rightarrow y(t_0) = x(t_0)\cos(t_0+1)$

$x(t_0)$ ↙ input.

$\cos(t_0+1)$ not input. is a function.

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Ex:-



* Check the system causality?

⇒ Sol:-

(t+5) ⇒ non causal component

y(t) = x(t-5) ⇒ Causal system.

⇒ if y(t) = (t+2) x(t-3)

⇒ The system is causal.

⇒ y(t) = x(-t).

⇒ non causal system

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(4) stability → stable / unstable.

⇒ If a bounded input (BI) gives bounded output (BO)

⇒ the system is stable.

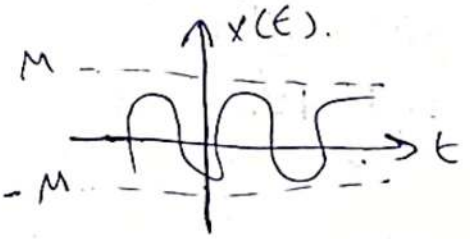
otherwise the system is unstable.

⇒ BI ⟶ BO ⇒ stable system.

BI ↛ BO ⇒ unstable system.

⇒ BI: bounded input $-M \le x(t) \le M$

OR $|x(t)| \le M$



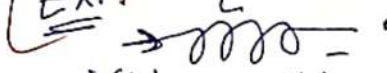
⇒ BO : bounded output.

$|y(t)| \le R$.

if $|y(t)| \le R$ for $|x(t)| \le M$ for all t.

⇒ stable system.

Ex:1



⇒ check stability if $v(t) = u(t)$.

$$i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\, d\tau.$$

Sol:

① if $v(t) = u(t)$ → input.

$0 \le u(t) \le 1$ ⇒ **BI**



② if $v(t) = e^{-at}u(t)$. $a > 0$

$i(t) = \frac{1}{aL}(1 - e^{-at})\, u(t)$

⇒ BI → marginal BO

⇒ the system is marginally stable

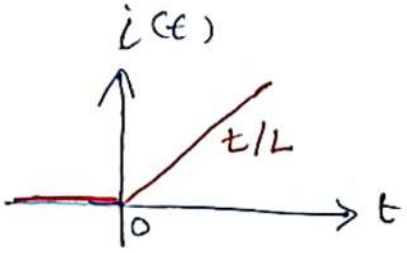
(3)

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$$\Rightarrow i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,d\tau$$

$$= \frac{1}{L}\int_{0}^{t} u(t)\,d\tau \quad (1)$$

$$= \frac{1}{L}\int_{0}^{t} d\tau$$



$$i(t) = \frac{t}{L}\,u(t), \quad t \geq 0$$

output

$\Rightarrow$ output not bounded.

$\Rightarrow$ BI $\not\Rightarrow$ BO $\Rightarrow$ the system is unstable.

Ex: $y(t) = e^{x(t)}$. $\rightarrow$ check the system stability.

Sol:

Assume.

$-A \leq x(t) \leq A \rightarrow |x(t)| \leq A \Rightarrow$ BI

Then.

$e^{-A} \leq y(t) \leq e^{A} \Rightarrow$ BO

$\Rightarrow$ The system is stable

Ex: $y(t) = t\,x(t)$.

Sol: if $x(t)$ is constant $= 1$ or $x(t) = u(t)$ (BI)

$|x(t)| \leq A$

$y(t) = tA \Rightarrow$ unstable.

$t \to \infty$

$y(t) \to \infty$

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5 **Time inveriance.**

- Time inveriant. (TI)
- Time variant. (Not TI)

⇒ If the time shift in the input signal results only in the same time shift in the output signal

⇒ the system is time inveriant (TI)

otherwise the system is time variant. or (Not TI)

⇒ if $y(t) = T[x(t)]$

$\Rightarrow y(t-t_0) = T[x(t-t_0)]$ ⇒ if $y(t)\big|_{x(t-t_0)} = y(t-t_0)$

(TI) system.

⇒ Time inveriant system also called fixed system.

Ex: if $y(t) = e^{x(t)}$. ⇒ the system? check time inveriance.?

Sol: $y(t)\big|_{x(t-t_0)} \stackrel{?}{=} y(t-t_0)$

$y(t) = e^{x(t-t_0)} \stackrel{?}{=} e^{x(t-t_0)}$

$\therefore \Rightarrow y(t)\big|_{x(t-t_0)} = y(t-t_0) \Rightarrow$ time inveriant system.

(TI)

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Ex: if $y(t) = e^{-t} x(t)$.

Sol: $y(t)\big|_{x(t-t_0)} \stackrel{?}{=} y(t-t_0)$

$= e^{-t} x(t-t_0) \stackrel{?}{=} e^{-(t-t_0)} x(t-t_0)$

$\neq$

$\Rightarrow$ [Time varying system (Not TI)]

Ex: $y(t) = x(t-2)$. Check for time-inveriance.

Sol:

$y(t)\big|_{t-t_0} = x(t-t_0-2)$.

$y(t)\big|_{x(t-t_0)} = x(t-t_0-2)$

$\Rightarrow y(t)\big|_{t-t_0} = y(t)\big|_{x(t-t_0)} \Rightarrow$ [time-inveriant system. (TI)]

Ex: $y(t) = x(-t)$

$\Rightarrow y(t)\big|_{(t-t_0)} = x(-(t-t_0)) = x(-t+t_0)$

$y(t)\big|_{x(t-t_0)} = x(-t-t_0)$ $\Big\} \neq$

[$\Rightarrow$ time-varying system]

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6| Linearity.

Linear — non linear

⇒ A system is linear if superposition principle applies.

⇒ if $x_1(t) \longrightarrow y_1(t)$.

& $x_2(t) \longrightarrow y_2(t)$

⇒ if $a_1 x_1(t) + a_2 x_2(t) \Longrightarrow a_1 y_1(t) + a_2 y_2(t)$

a_1 & a_2: constants.

⇒ the system is linear.

Ex:① $y(t) = k\,x(t)$. , k: constant. (simple amplifer system).

Sol:

$x_1(t) \longrightarrow y_1(t) = k\,x_1(t)$.

$x_2(t) \longrightarrow y_2(t) = k\,x_2(t)$

$\Rightarrow x_1(t) + x_2(t) \longrightarrow k[x_1(t) + x_2(t)]$

$= k x_1(t) + k x_2(t)$

$= y_1 + y_2$

$a_1 x_1(t) + a_2 x_2(t) \longrightarrow k\,[a_1 x_1(t) + a_2 x_2(t)]$

$\Rightarrow k a_1 x_1(t) + k a_2 x_2(t)$

$= a_1 y_1(t) + a_2 y_2(t)$.

⇒ [The system is linear.]

Ex: $y(t) = x^2(t)$.

$x_1(t) \longrightarrow y_1(t) = x_1^2(t)$

$x_2(t) \longrightarrow y_2(t) = x_2^2(t)$

⇒ $[x_1(t) + x_2(t)] \Rightarrow y(t) = [x_1(t) + x_2(t)]^2$

$= x_1^2(t) + 2x_1(t)x_2(t) + x_2^2(t)$

$= y_1(t) + y_2(t) + 2x_1(t)x_2(t)$

⇒ Non linear system.

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Ex: $y(t) = x(t) + 5$

$x_1(t) \Rightarrow y_1(t) = x_1(t) + 5$.

$x_2(t) \rightarrow y_2(t) = x_2(t) + 5$

$a_1 x_1(t) + a_2 x_2(t) \Rightarrow [a_1 x_1(t) + a_2 x_2(t)] + 5$

$\Rightarrow \neq a_1 y_1(t) + a_2 y_2(t)$.

$\Rightarrow$ [The system is non linear.].

Ex: Determine the properties of a particular system.

$y(t) = \sin(2t)\, x(t)$

$\Rightarrow$ Sol:

① $y(t)|_{t_0} = \sin(2t_0)\, x(t_0)$

$\Rightarrow$ [memoryless system.] $\Rightarrow y(t_0)$ depends only on $x(t_0)$

② ⇓ [Causal system.] $\Rightarrow y(t_0)$ dependens on $x(t \le t_0)$

③ Non invertible system.

at $t=0 \rightarrow y(t)=0$
$t=\pi \rightarrow y(\pi)=0$
} distinct inputs $\not\Rightarrow$ distinct output.

④ Assume $|x(t)| \le M \rightarrow BI$

$y(t) = x(t) \sin(2t)$

$|y(t)| = |x(t)|\, |\sin(2t)|$

$= \le M \qquad 0 \le \sin(2t) \le 1$

$\therefore |y(t)| \le R \rightarrow BO$

[BI $\rightarrow$ BO $\Rightarrow$ stable system.]

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(5) $y(t)\Big|_{(t-t_0)} = x(t-t_0)\sin(2(t-t_0))$ ✗

$y(t)\Big|_{x(t-t_0)} = x(t-t_0)\sin(2t)$

⇒ [Time-varying system.]

(6)

$x_1(t) \rightarrow x_1(t)\sin(2t)$

$x_2(t) \rightarrow x_2(t)\sin(2t)$

$a_1 x_1(t) + a_2 x_2(t) \Rightarrow (a_1 x_1(t) + a_2 x_2(t))\sin(2t)$

$\Rightarrow a_1 x_1(t)\sin(2t) + a_2 x_2(t)\sin(2t)$

$\Rightarrow a_1 y_1(t) + a_2 y_2(t)$

⇒ [Linear system.]

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✱✱ Continuous-time linear time invariant systems. (LTI)

⇒ LTI
- Linear ⇒ superposition applies.
- Time-invariant ⇒ $x(t-t_0) \Rightarrow y(t-t_0)$

⇒ $x(t)$ (input) → [LTI, $h(t)$] → $y(t)$ (output).

⇒ if $x(t) = \delta(t)$ ⇒ impulse function.

$y(t) = h(t)$ ⇒ impulse response. for LTI system

if $\delta(t-t_0) \Rightarrow h(t-t_0)$ ⇒ Time-invariant.

⇒ $y(t) = x(t) * h(t)$. → convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

OR $$\int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau.$$

⇒ $y(t) = x(t) * h(t)$

$= h(t) * x(t)$

The convolution is commutative.

⇒ if $x(t) = \delta(t)$.

$$y(t) = \int_{-\infty}^{\infty} \delta(\tau)\, h(t-\tau)\, d\tau.$$

$\tau = 0$

$$y(t) = h(t) \int_{-\infty}^{\infty} \delta(\tau)\, d\tau$$

$y(t) = h(t)$ $\quad = 1$ output.

impulse response

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$\Rightarrow \quad \delta(t) * h(t) = h(t).$

$\delta(t-t_0) * h(t) = h(t-t_0)$

$\delta(t) * h(t-t_0) = h(t-t_0).$

$\delta(t-t_1) * h(t-t_2) = h(t-t_2-t_1)$

→ properties of convolution.

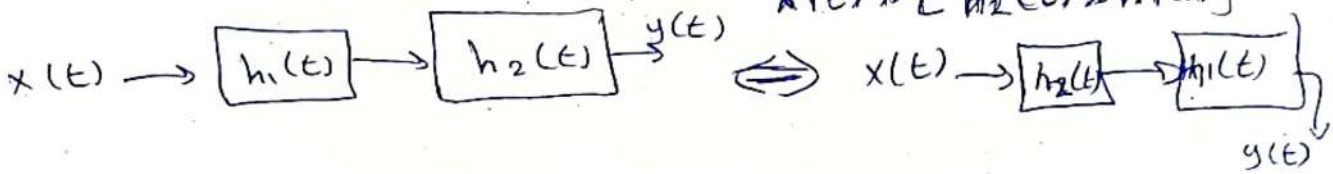
① commutative

$$x(t) * h(t) = h(t) * x(t).$$



② Associative.

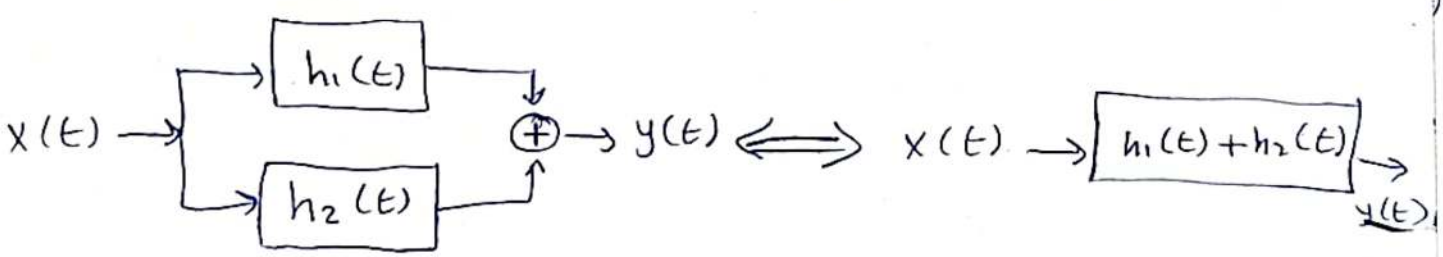
$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$
$$= x(t) * [h_2(t) * h_1(t)]$$





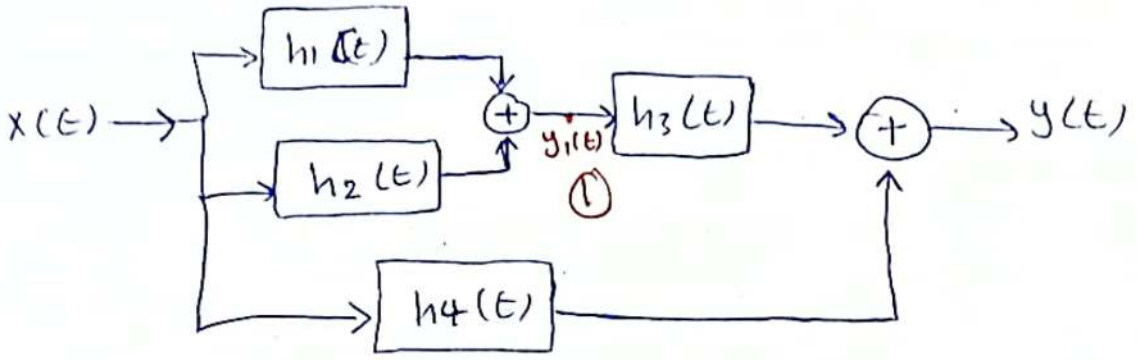
③ Distributive.

$$x(t) * h_1(t) + x(t) * h_2(t) = x(t) * [h_1(t) + h_2(t)]$$



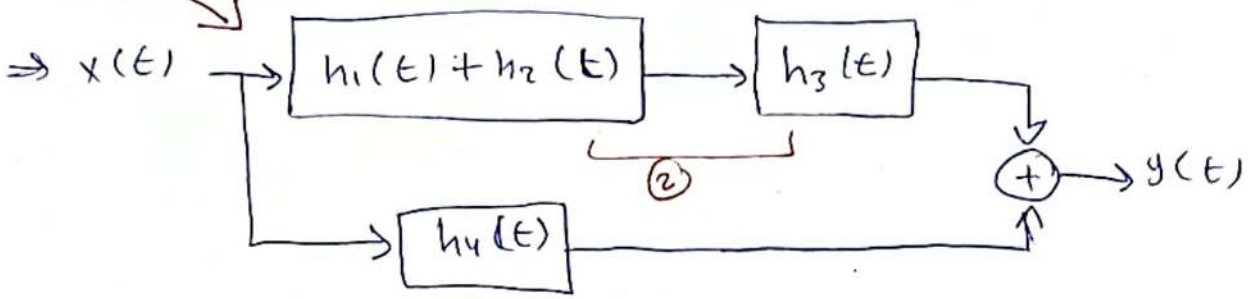
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Ex:

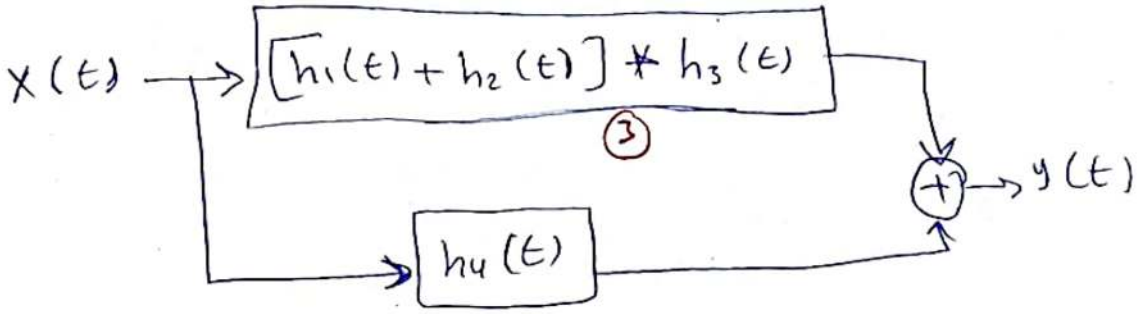


$\Rightarrow y_1(t) = x(t) * h_1(t) + x(t) * h_2(t)$

$= [x(t) * [h_1(t) + h_2(t)]]$

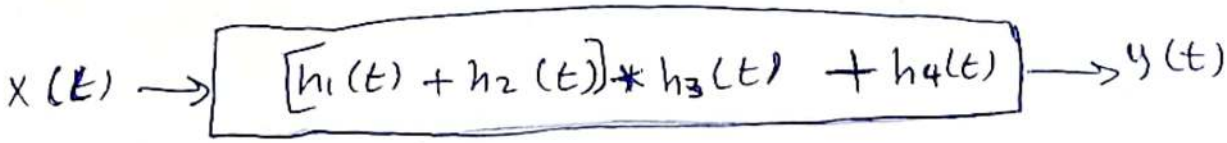


$[h_1(t) + h_2(t)] * h_3(t)$



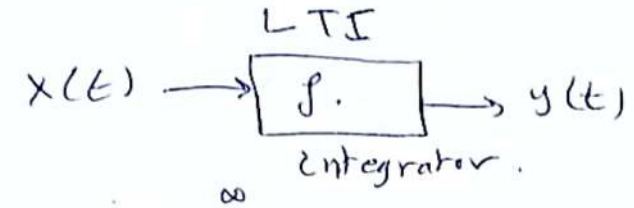
(3) $\Rightarrow x(t) * [[h_1(t) + h_2(t)] * h_3(t)] + x(t) * h_4(t) = y(t)$

$\Rightarrow y(t) = x(t) * [[h_1(t) + h_2(t)] * h_3(t) + h_4(t)]$



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Ex:



$$\Rightarrow y(t) = \int_{-\infty}^{\infty} x(t)\, dt.$$

⇒ (a) Find the impulse response $h(t)$.

(b) Find $y(t)$ when $x(t) = u(t)$.

(c) Find $y(t)$ when $x(t) = e^{-at} u(t)$, $a > 0$

① $h(t) = y(t)\Big|_{x(t)=\delta(t)}$

$$\Rightarrow y(t) = \int_{-\infty}^{\infty} \delta(\tau)\, d\tau.$$

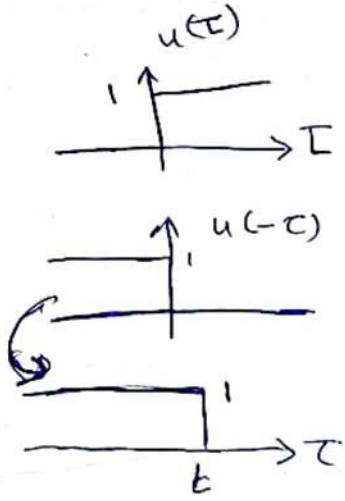
$$y(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

$$y(t) = u(t) = h(t).$$

② $y(t) = x(t) * h(t).$

$$= u(t) * u(t).$$

$$= \int_{-\infty}^{\infty} u(\tau)\, u(t-\tau)\, d\tau.$$



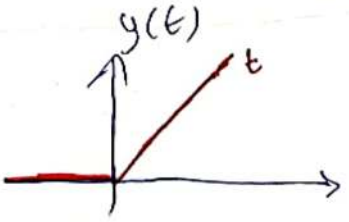
① $y(t)$, $t < 0$

$$= \int_{-\infty}^{0} 0\, d\tau = 0$$

② $y(t)$, $t > 0$

$$y(t) = \int_{0}^{t} 1\, d\tau = t.$$

$$y(t) = t\, u(t) \Rightarrow$$



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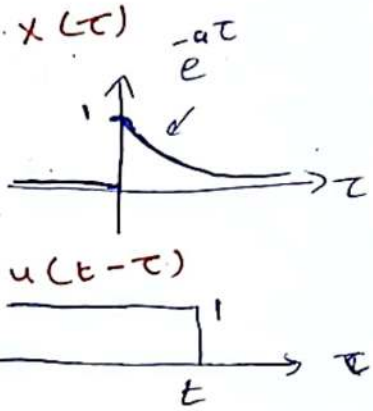
(e) $y(t) = x(t) * h(t)$.

$= e^{-at} u(t) * u(t)$.

OR

$= u(t) * e^{-at} u(t)$.

$$y(t) = \int_{-\infty}^{\infty} e^{-a\tau} u(\tau)\, u(t-\tau)\, d\tau .$$



① For $t < 0$

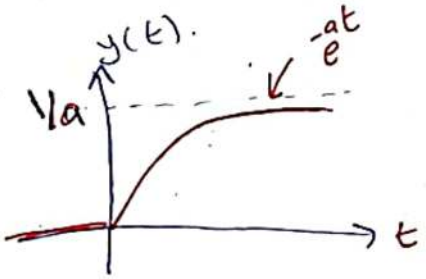
$y(t) = 0$

② For $t \geq 0$

$$y(t) = \int_0^t e^{-a\tau} \cdot 1 \, d\tau$$

$$= \frac{e^{-a\tau}}{-a} \Big|_0^t$$

$$y(t) = \frac{1}{a}\left(1 - e^{-at}\right) u(t), \quad t \geq 0.$$



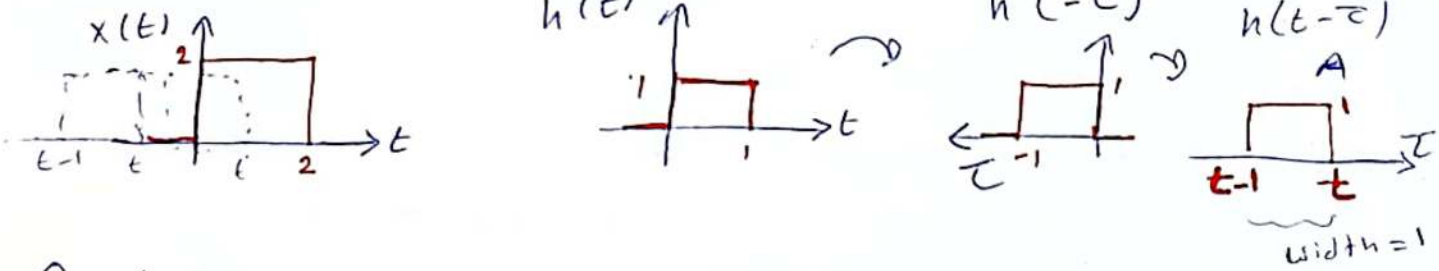
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Ex: if $x(t) = \begin{cases} 2, & 0 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$

and $h(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$

Find $y(t)$.

Sol: $y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$, $x(t) * h(t)$.



$\Rightarrow$ ① $t < 0$

$y(t) = 0$

② $0 < t < 1$

$$y(t) = \int_0^t 2(1)\, d\tau$$

$$y(t) = 2t$$

③ $1 < t \le 2$

$$y(t) = \int_{t-1}^{t} 2(1)\, d\tau$$

$$= 2(t - t + 1)$$

$$= 2$$

④ $2 < t \le 3$

$$y(t) = \int_{t-1}^{2} 2\, d\tau$$

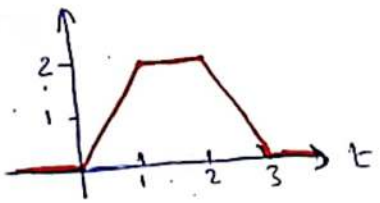
$$= 2(2 - t + 1)$$

$$= 2(3 - t)$$

⑤ $t > 3$

$y(t) = 0$

$\Rightarrow$ $y(t)$

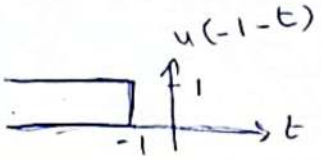


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Ex: $x(t) = e^{t}\, u(-1-t)$

$h(t) = e^{-t}\, u(t-1)$

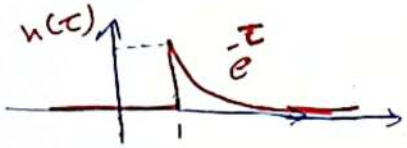
$\Rightarrow$ Find $y(t)$.



$\Rightarrow y(t) = x(t) * h(t).$

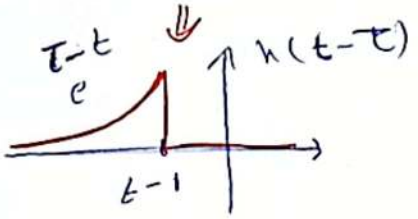


$$\Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau.$$



① $t-1 < -1$

$t < 0$



$$y(t) = \int_{-\infty}^{t-1} e^{\tau}\, e^{\tau-t}\, d\tau.$$

$$= e^{-t} \int_{-\infty}^{t-1} e^{2\tau}\, d\tau$$

$$= e^{-t} \left[\frac{e^{2\tau}}{2} \right]_{-\infty}^{t-1}$$

$$= e^{-t} \left(\frac{e^{2t-2}}{2} - 0 \right) = \left(\frac{e^{-2}}{2} \right) e^{t}$$

② $t-1 > -1$

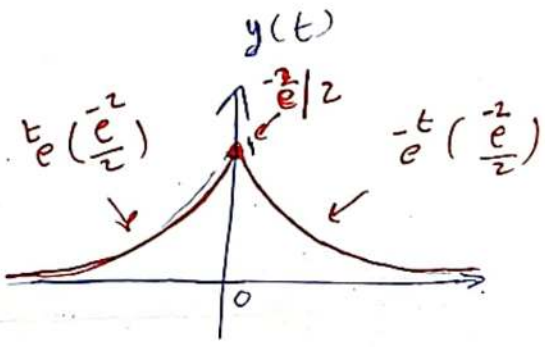
$t > 0$

$$y(t) = \int_{-\infty}^{t-1} e^{\tau}\, e^{\tau-t}\, d\tau.$$

$$= e^{-t} \int_{-\infty}^{-1} e^{2\tau}\, d\tau$$

$$= e^{-t} \left[\frac{e^{2\tau}}{2} \right]_{-\infty}^{-1}$$

$$y(t) = \frac{e^{-2}}{2} \left(e^{-t} \right)$$



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* Properties of continuous-time LTI system.

⇒ $y(t) = x(t) * h(t)$.

$y(t)\big|_{x(t)=\delta(t)} = h(t)$ ⇒ impulse response.

1. Memory
   - → Memory.
   - → Memoryless.

⇒ An LTI system is memoryless if and only if

$$h(t) = k\,\delta(t)$$

⇒ $y(t) = k\,x(t)$ ⇒ Ideal Amplifier. for memoryless LTI system.

if $k = 1$

$y(t) = x(t)$ ⇒ identity system.

for $\delta(t)$: $\int_{-\infty}^{\infty} x(t)\,\delta(t - t_0)\,dt = x(t_0)$

(##) For LTI → $y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$

if $h(t) = k\,\delta(t)$

⇒ $y(t) = \int_{-\infty}^{\infty} x(\tau)\,k\,\delta(t-\tau)\,d\tau$

at t_1 → $y(t_1) = \int_{-\infty}^{\infty} x(\tau)\,k\,\delta(t_1-\tau)\,d\tau$

$$\boxed{y(t_1) = k\,x(t_1)}$$ ##

⇒ Memoryless LTI system

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2 Invertibility → invertible / non invertible.

⇒ An LTI system (with impulse response $h(t)$) is invertible if its input can be determined from its output.

$$x(t) * h(t) * h_i(t) = x(t)$$

~~Inverse~~.

⇒ $h_i(t)$: impulse response of the inverse system.

→ $x(t) * \delta(t) = x(t)$.

then $h(t) * h_i(t) = \delta(t)$ ※.

3 Causality → causal / non causal.

⇒ An LTI system is causal if the current value of the output depends on only the current & past values of the input.

→ $h(t)$ → at $x(t) = \delta(t)$ → it zero on all time except at $t=0$

⇒ so $h(t)$ of a causal system must be zero for $t<0$.

⇒ For causal LTI system.

$$y(t) = \int_0^{\infty} x(t-\tau)\, h(\tau)\, d\tau = \int_{-\infty}^{t} x(\tau)\, h(t-\tau)\, d\tau$$

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(4) Stability → Stable. / → Unstable.

For LTI system.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau .$$

if $|x(t)| \leq M$ → BI

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau \right|$$

$$|y(t)| \leq \int_{-\infty}^{\infty} |x(t-\tau)|\,|h(\tau)\, d\tau|$$

$$|y(t)| \leq \int_{-\infty}^{\infty} M\, |h(\tau)|\, d\tau$$

⇒ $y(t)$ → bounded if

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$$ ⇒ must be absolutely integrable. #

⇒

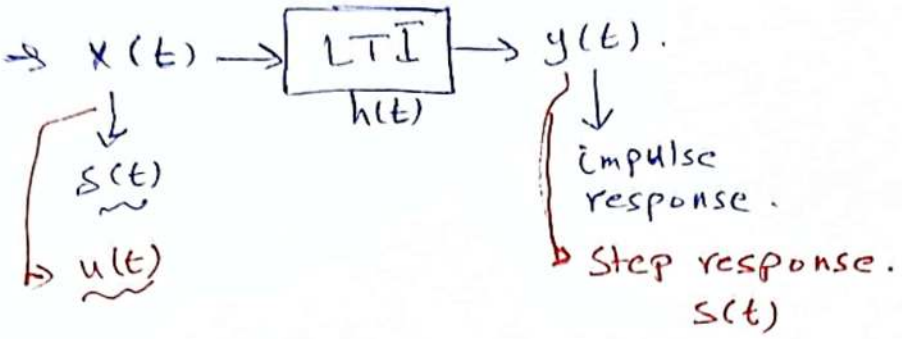
⇒ for causal LTI system.

$$\int_{0}^{\infty} |h(t)|\, d\tau < \infty$$

⇒ BIBO → stable.

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Unit step response.



Step response.

$$\underbrace{y(t)}_{s(t)} = u(t) * h(t)$$

$$\Rightarrow \underbrace{s(t)}_{-\infty \to t} = \int_{-\infty}^{\infty} h(\tau)\, \underbrace{u(t-\tau)}_{-\infty \to t}\, d\tau$$

$$\boxed{s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau} \Rightarrow \boxed{\text{step response.}}$$

$$\textcircled{*} \quad \frac{d\,s(t)}{dt} = h(t).$$

$$\Rightarrow \quad \underbrace{u(t)}_{\text{For LTI}} = \int_{-\infty}^{t} s(\tau)\, d\tau$$

$$\boxed{s(t) = \int_{-\infty}^{\infty} h(\tau)\, d\tau}$$

$$\Rightarrow \text{LTI} \quad s(t) = \frac{d\,u(t)}{dt}$$

$$\boxed{h(t) = \frac{d\,s(t)}{dt}}$$

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* Ex: $h(t) = e^{-at} u(t)$, $a > 0$

(*) Determine $s(t)$

Sol:

$$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

$$= \int_{-\infty}^{t} e^{-a\tau} u(\tau)\, d\tau$$

$$= \int_{0}^{t} e^{-a\tau}\, d\tau$$

$$= \frac{e^{-a\tau}}{-a} \Big]_0^t = \frac{e^{-at}}{-a} - \frac{-1}{a}$$

$$s(t) = \frac{1}{a}\left(1 - e^{-at}\right) u(t)$$

Ex: Determine the stability of the causal LTI system.

$h(t) = e^{-3t} u(t)$.

Sol:

$$\Rightarrow \int_{-\infty}^{\infty} |h(t)|\, dt = \int_{0}^{\infty} e^{-3t}\, dt$$

$$= \frac{e^{-3t}}{-3} \Big]_0^\infty$$

$$\left[= \frac{1}{3} < \infty\right]$$

$\Rightarrow$ this system is

Stable ##

for the sam system if $h(t) = e^{5t} u(t)$.

$$\Rightarrow \int_{-\infty}^{\infty} |h(t)|\, dt = \int_{0}^{\infty} e^{5t}\, dt$$

$$\left[= \frac{e^{5t}}{5} \Big]_0^\infty = \infty\right] \text{ unbounded.}$$

$\Rightarrow$ **Unstable System** ##

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Ex: For LTI system. if $h(t)=u(t)$.

the input-output relationship.

$y(t)=\int_{-\infty}^{t} x(\tau)\,d\tau$. Determine the stability.

Sol:

$h(t)=0$ for $t<0$

$\Rightarrow$ causal system.

$$\Rightarrow \int_{0}^{\infty} |h(t)|\,dt = \int_{0}^{\infty} u(t)\,dt$$

$$= t\Big]_{0}^{\infty}$$

$= \infty \Rightarrow$ unbounded

$\Rightarrow$ Unstable system

#

Ex: For $h(t)=t\,u(t)$ $\Rightarrow$ stable? causal?

Sol:

① $h(t)=0$ for $t<0$

$\Rightarrow$ causal.

② $$\int_{-\infty}^{\infty} |h(t)|\,dt = \int_{0}^{\infty} t\,u(\tau)\,d\tau$$

$$= \frac{t^2}{2}\Big]_{0}^{\infty}$$

$= \infty$ unbounded.

$\Rightarrow$ Unstable system

#

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Ch 4: Fourier series

⇒ Partitioning technique ⟶ partitioning of complex problems into simpler ones.

⇒ Taylor's series.

$$f(t) = f(0) + \dot{f}(0)\frac{t}{1!} + \ddot{f}(0)\frac{t^2}{2!} + \cdots$$

$$= \sum_{n=0}^{\infty} f^{(n)}(0)\frac{t^n}{n!}$$

⇒ where $f^{(n)}$: n-th derivative.

$$f^{(n)}(0) = \frac{d^n f(t)}{dt^n}\Big|_{t=0}$$

⇒ Fourier series:-

Fourier series expansion is used for periodic signals to expand them in terms of their harmonics which are sinusoidal and orthogonal to one another.

⇒ ① Periodic signal.

$$x(t) = x(t \pm nT) \quad n \hookrightarrow \text{integer}$$

$\cos(\omega t)$, $\omega = \frac{2\pi}{T}$ → fundamental period. T_0

$$\omega_0 = \frac{2\pi}{T_0} = 2\pi F_0 \quad \omega_0: \text{rad/s}, \quad F_0: \text{Hz}.$$

⇒ ② Harmonic & orthogonal / sinusoidal

$$\int_0^{2\pi} \cos(\omega t)\cos(2\omega t)\,dt = 0 \Rightarrow \text{orthogonal.}$$

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original signal or 1st harmonic

$\Rightarrow x(t) = \cos(\omega t) + 2\cos(2\omega t) + .7\cos(3\omega t) + \dots$

$\omega = \frac{2\pi}{T_0}$, $\omega_0 = 2\pi f_0$

2nd harmonic

third harmonic

$\Rightarrow$ $x(t)$ is periodic $\Rightarrow$ we can expanded into a fourier series.

3 forms:

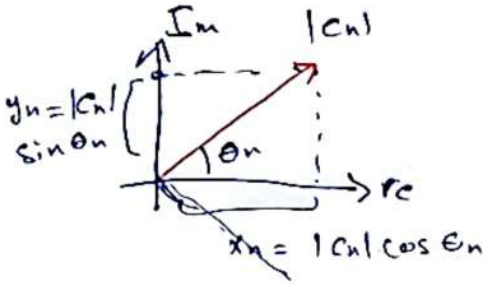
1. The complex exponential form
2. The combined Trigonometric form
3. The Trigonometric form

$\Rightarrow$ (1) The complex exponential form.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$\Rightarrow$ C_n : fourier series complex coefficients.

$$C_n = X_n + jY_n = |C_n| e^{j\theta_n}$$



$$|C_n| = \sqrt{(X_n)^2 + (Y_n)^2}$$

$$\theta_n = \tan^{-1}\left(\frac{Y_n}{X_n}\right)$$

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$\Rightarrow |C_{-n}| = \sqrt{(x_{-n})^2 + (y_{-n})^2}$, $C_n = x_{-n} + j\, y_{-n}$

$|C_n| = |C_{-n}|$

$\theta_n = -\theta_{-n}$

$C_n = C_{-n}^*$ → conjugate.

$C_n = 5 + j3 \rightarrow C_{-n} = 5 - j3$

$\Rightarrow e^{jw_0 t}$ ⇒ complex exponential.

→ w_0 is fundamental frequency.

$w_0 = \frac{2\pi}{T_0}$ (rad/s)

nw_0 ⇒ n-th harmonic.

is the n-th integer multiple of w_0

$\Rightarrow x(t) = \cdots + C_{-2} e^{-2jw_0 t} + C_{-1} e^{-jw_0 t} + C_0 + C_1 e^{jw_0 t} + C_2 e^{j2w_0 t} + \cdots$

C_0 : Dc-value or average value (real).

$\Rightarrow C_{-1} e^{-jw_0 t} + C_1 e^{jw_0 t} = 2|C_1| \cos(w_0 t + \theta_1)$, $\cos(t) = \frac{e^{jt} + e^{-jt}}{2}$

$C_1 = |C_1| e^{j\theta_1}$

$C_{-1} = |C_{-1}| e^{j\theta_{-1}}$

$C_{-1} = |C_1| e^{-j\theta_1}$

↓ first harmonic with frequency ⇒ w_0 rad/s

$\Rightarrow C_{-2} e^{-j2w_0 t} + C_2 e^{j2w_0 t} = 2|C_2| \cos(2w_0 t + \theta_2)$

second harmonic

⇒ $2w_0$

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$\Rightarrow$ Sum all harmonic sinusoidal $\Rightarrow x(t)$.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$\Rightarrow C_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jn\omega_0 t} \, dt.$$

(2) The combined Trigonometric form.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$= \sum_{-\infty}^{-1} C_n e^{jn\omega_0 t} + C_0 + \sum_{1}^{\infty} C_n e^{jn\omega_0 t}$$

$$= \sum_{-1}^{-\infty} C_n e^{jn\omega_0 t} + C_0 + \sum_{1}^{\infty} C_n e^{jn\omega_0 t}$$

$$= \sum_{1}^{\infty} C_{-n} e^{j-n\omega_0 t} + C_0 + \sum_{1}^{\infty} C_n e^{jn\omega_0 t}$$

$$= C_0 + \sum_{1}^{\infty} \left(C_{-n} e^{-jn\omega_0 t} + C_n e^{jn\omega_0 t} \right)$$

$$\Rightarrow C_{-n} = |C_{-n}| e^{j\theta_{-n}}$$

$$C_n = |C_n| e^{j\theta_n}$$

$$C_n = C_{-n}^{*}$$

$$|C_n| = |C_{-n}|$$

$$\theta_n = -\theta_{-n}$$

$$\Rightarrow x(t) = C_0 + \sum_{n=1}^{\infty} 2|C_n| \cos(n\omega_0 t + \theta_n)$$

(1)

$$C_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jn\omega_0 t} \, dt$$

$$= x_n + j y_n$$

$$|C_n| = \sqrt{x_n^2 + y_n^2}$$

$$\theta_n = \tan^{-1}\left(\frac{y_n}{x_n}\right)$$

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$$\Rightarrow C_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt. \Rightarrow \text{real value.}$$

(3) The Trigonometric form.

$$x(t) = C_0 + \sum_{n=1}^{\infty} 2|C_n|\cos(n\omega_0 t + \theta_n)$$

$$\Rightarrow \boxed{\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B}$$

$$\Rightarrow x(t) = C_0 + \sum_{n=1}^{\infty} \underbrace{2|C_n|}_{A_n}\cos(n\omega_0 t)\cos\theta_n - \underbrace{2|C_n|}_{B}\sin(n\omega_0 t)\sin\theta_n$$

let $A_n = 2|C_n|\cos\theta_n$

$B_n = -2|C_n|\sin\theta_n$

$$\Rightarrow \boxed{x(t) = C_0 + \sum_{n=1}^{\infty} A_n\cos(n\omega_0 t) + B_n\sin(n\omega_0 t)} \quad (3)$$

$$\Rightarrow C_n = \frac{1}{T_0}\int_0^{T_0} x(t)\,e^{-jn\omega_0 t}\,dt. \Rightarrow C_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt.$$

$$2C_n = 2|C_n|\,e^{j\theta_n}$$

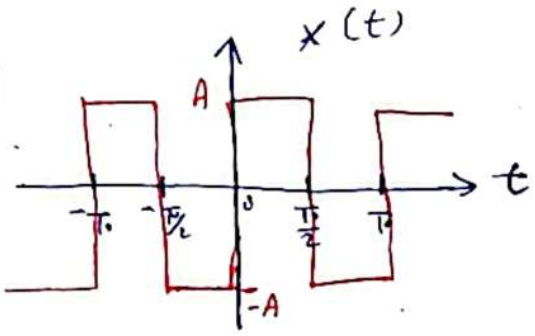
$$= 2|C_n|(\cos\theta_n + j\sin\theta_n)$$

$$2C_n = \underbrace{2|C_n|\cos\theta_n}_{A_n} + j\underbrace{2|C_n|\sin\theta_n}_{-B_n}$$

$$\Rightarrow \boxed{2C_n = A_n - jB_n}$$

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Ex:



- Expand $x(t)$ into a fourier series in its 3 forms.

Sol: ① The complex exponential form.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$\rightarrow C_n = \frac{1}{T_0} \int_0^{T_0} x(t)\, e^{-jn\omega_0 t}\, dt.$$

$$= \frac{1}{T_0}\left[\int_0^{T_0/2} A e^{-jn\omega_0 t}\, dt + \int_{T_0/2}^{T_0} -A e^{-jn\omega_0 t}\, dt\right]$$

$$= \frac{A}{T_0}\left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0}\Big]_0^{T_0/2} - \frac{e^{-jn\omega_0 t}}{-jn\omega_0}\Big]_{T_0/2}^{T_0}\right]$$

$$= \frac{A}{jn2\pi}\left[1 - e^{-jn\pi} + e^{-j2\pi n} - e^{-jn\pi}\right]$$

(①)

$$= \frac{A}{jn2\pi}\left[2 - 2e^{-jn\pi}\right]$$

$$C_n = \begin{cases} 0, & n \text{ even} \\ \frac{2A}{jn\pi}, & n \text{ odd} \end{cases}$$

$$\begin{cases} \omega_0 T_0 = 2\pi \\ \omega_0 \frac{T_0}{2} = \pi \\ e^{-j2n\pi} = \cos(-2n\pi) - j\sin(2n\pi) \end{cases}$$

$\sin(2n\pi) = 0$

$$e^{-jn2\pi} = 1$$

$$e^{-jn\pi} = \cos(-n\pi) - j\sin(n\pi)$$

$$= \cos(n\pi) - j\sin(n\pi)$$

$\sin(n\pi) = 0$

$$e^{-jn\pi} = \begin{cases} 1, & n \text{ even} \\ -1, & n \text{ odd} \end{cases}$$

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$$\Rightarrow \quad x(t) = \sum_{\substack{n=-\infty \\ n:\text{odd}}}^{\infty} \frac{2A}{jn\pi} e^{jn\omega_0 t}$$

$$x(t) = \sum_{\substack{n=-\infty \\ n:\text{odd}}}^{\infty} \frac{2A}{n\pi} e^{j\left(n\omega_0 t - \frac{\pi}{2}\right)}$$

$$\frac{1}{j} = -j$$

$$-j = e^{-j\pi/2}$$

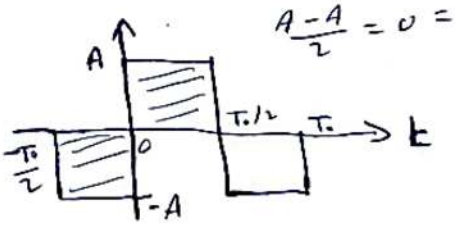
$$j = e^{j\pi/2}$$

(2) The combind trigonometric form.

$$x(t) = C_0 + \sum_{n=1}^{\infty} 2|C_n| \cos(n\omega_0 t + \theta_n).$$

$$\left[C_n = \begin{cases} \frac{2A}{jn\pi}, & n \text{ odd.} \\ 0, & n \text{ even.} \end{cases} \right]$$

$$\left[C_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)\, dt = 0 \right]$$



$$\frac{A-A}{2} = 0 = C_0$$

$$|C_n| = \begin{cases} \left|\frac{2A}{n\pi}\right|, & n \text{ odd.} \\ 0, & n \text{ even.} \end{cases}$$

$$|C_n| = \sqrt{\left(\frac{2A}{n\pi}\right)^2}$$

$$|C_n| = \left|\frac{2A}{n\pi}\right|$$

$$\theta_n = \tan^{-1}\left(\frac{-2A}{\frac{n\pi}{0}}\right) = \tan^{-1}(-\infty) = -\frac{\pi}{2}.$$

$$\theta_n = \begin{cases} -\frac{\pi}{2}, & n \text{ odd.} \\ \text{Undefined}, & n \text{ even.} \end{cases}$$

$$\tan\left(-\frac{\pi}{2}\right) = \frac{\operatorname{sen}\left(-\frac{\pi}{2}\right)}{\cos\left(-\frac{\pi}{2}\right)}$$

$$\tan\left(-\frac{\pi}{2}\right) = \frac{-1}{0} = -\infty$$

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$$\Rightarrow x(t) = \sum_{\substack{n=1 \\ n:odd}}^{\infty} \frac{4A}{n\pi} \cos\left(n\omega_0 t - \frac{\pi}{2}\right).$$

(3) The Trigonometric form:

$$x(t) = C_0 + \sum_{n=1}^{\infty} \left[A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t) \right]$$

$\Rightarrow$ C_0 = Zero.

$2C_n = A_n - jB_n$.

$$\begin{cases} \frac{4A}{jn\pi}, & n \text{ odd} \\ 0, & n \text{ even.} \end{cases} = A_n - jB_n.$$

real part = 0 $\Rightarrow A_n = 0$ for all n

$$\text{img. part} = \begin{cases} -j\frac{4A}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even.} \end{cases} \Rightarrow B_n = \begin{cases} \frac{4A}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even.} \end{cases}$$

$\Rightarrow$ A_n = Zero

$x(t) \rightarrow$ odd Signal $\Rightarrow$ $A_n \cos(n\omega_0 t)$ = Zero

$B_n \sin(n\omega_0 t)$ ✓

$$\Rightarrow x(t) = \sum_{\substack{n=1 \\ n=odd}}^{\infty} \frac{4A}{n\pi} \sin(n\omega_0 t)$$

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Ex: If $x(t) = \cos^2(2t) + 3\sin(8t)$

Expand $x(t)$ into a Fourier series in 3 forms

Sol:

① The complex exponential form.

$$\Rightarrow x(t) = \left(\frac{e^{j2t} + e^{-j2t}}{2}\right)^2 + 3\left(\frac{e^{+j8t} - e^{-j8t}}{2j}\right)$$

$$X(t) = \frac{1}{4}\left(e^{j4t} + e^{-j4t} + 2e^{jut}e^{-jut}\right) + \frac{3}{2j}e^{j8t} - \frac{3}{2j}e^{-j8t}$$

$$X(t) = \frac{1}{4}e^{j4t} + \frac{1}{4}e^{-j4t} + \frac{1}{2} + \frac{3}{2j}e^{j8t} - \frac{3}{2j}e^{-8t}$$

$\Rightarrow w_o = 4$ rad/s

$$c_o = \frac{1}{2}$$

$$c_n = \begin{cases} \frac{1}{4}, & n = \pm 1 \\ \frac{1}{2}, & n = 0 \\ \frac{3}{2j}, & n = 2 \\ -\frac{3}{2j}, & n = -2 \\ 0, & \text{otherwise.} \end{cases}$$

② The Combind Trigonometric form.

$$x(t) = \cos^2(2t) + 3\sin(8t)$$

$$= \frac{1 + \cos(4t)}{2} + 3\cos\left(8t - \frac{\pi}{2}\right)$$

$$= \frac{1}{2} + \frac{1}{2}\cos(4t) + 3\cos\left(8t - \frac{\pi}{2}\right)$$

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$\Rightarrow \quad C_0 = \frac{1}{2}$

$C_1 = |C_1| e^{j\theta_1} = \frac{1}{4} e^{j0} = \frac{1}{4}$

$2|C_1| = \frac{1}{2}$

$|C_1| = \frac{1}{4}$

$C_2 = |C_2| e^{j\theta_2} = \frac{3}{2} e^{-j\pi/2} = -j\frac{3}{2}$

$2|C_2| = 3$

$|C_2| = \frac{3}{2}$

$e^{-j\pi/2} = \cos(-\frac{\pi}{2}) + j\sin(\frac{-\pi}{2})$

$= 0 - j\sin(\pi/2)$

$e^{-j\pi/2} = -j$

(3) The Trigonomatric form.

$x(t) = \cos^2(2t) + 3\sin(8t)$

$= \frac{1}{2} + \frac{\cos(4t)}{2} + 3\sin(8t)$

$\frac{1}{2}$: C_0 ; $\frac{\cos(4t)}{2}$: A_1, ω_0 ; 3 : B_2 ; $\sin(8t)$: $2\omega_0$

$C_0 = \frac{1}{2}$

$A_n = \begin{cases} \frac{1}{2}, & n = 1 \\ 0, & \text{otherwise.} \end{cases}$

$B_n = \begin{cases} 3, & n = 2 \\ 0, & \text{otherwise.} \end{cases}$

$2C_n = A_n - jB_n \Rightarrow C_n = \frac{A_n}{2} - j\frac{B_n}{2}$

$\Rightarrow C_1 = \frac{1}{4}$

$C_2 = -j\frac{3}{2}$

$C_{-1} = C_1^*$

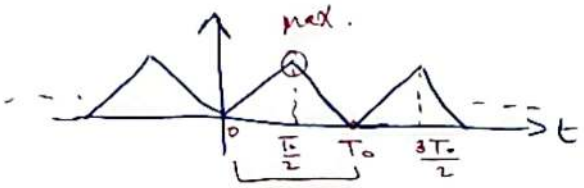
$\Rightarrow C_n = \begin{cases} \frac{1}{2}, & n = 0 \\ \frac{1}{4}, & n = \pm 1 \\ -\frac{3}{2}j, & n = 2 \\ \frac{3}{2}j, & n = -2 \\ 0, & \text{otherwise.} \end{cases}$

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✱ conditions for existance of Fourier series.

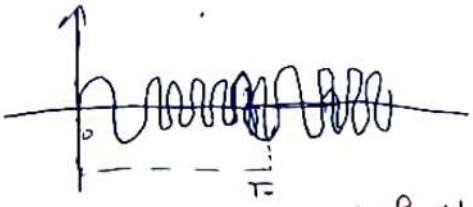
⇒ For periodic signal $x(t)$ we can use the Fourier series expansion ✱✱ :-

① The signal should have finite numbers of maxima and minima over the range of time period.



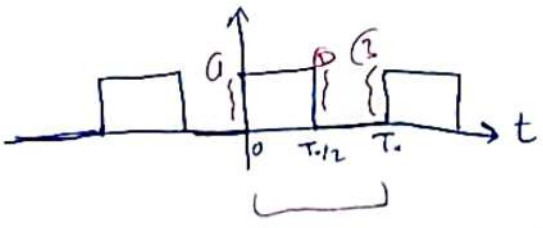
$T_0 \rightarrow$ 1 maxima.

$\frac{T_0}{2} \rightarrow \frac{3T_0}{2} \rightarrow$ 1 minima. ⇒ we can use Fourier series ⇒ Fourier series exist.

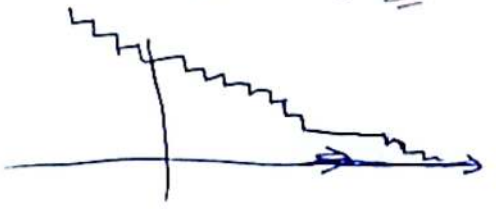


$0 \rightarrow T_0 \rightarrow$ infinite number of maxima & minima ⇒ we can't use Fourier series.

② The signal should have a finite number of discontinuities over the range of time period.



$0 - T_0 \Rightarrow$ 3 discontinuities ⇒ we can use the Fourier series.

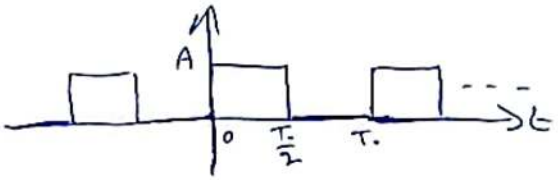


⇒ infinit number of discontinuities ⇒ we can't use the Fourier series.

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(3) The signal should be absolutely integrable over the range of time period

$$\Rightarrow \int_{T_0} x(t)\,dt < \infty$$

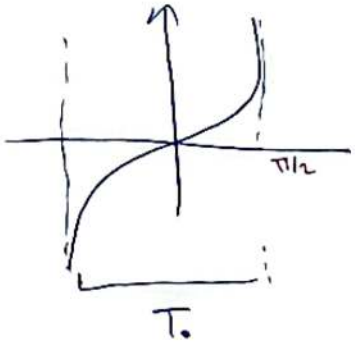


$$\int_0^{T_0/2} x(t)\,dt = A\left(\frac{T_0}{2}\right) \Rightarrow \text{absolutely integrable.}$$

$$\leq \infty$$

$\Rightarrow$ we can use the Fourier series.

$x(t) = \tan t$



$\tan\left(\frac{\pi}{2}\right) = \infty$ $\rightarrow$ we can't use the Fourier series or Fourier series doesn't exists

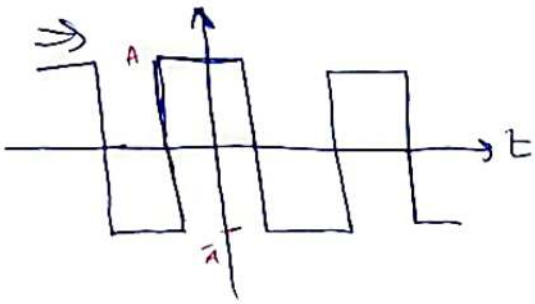
$\Rightarrow$ Remarks:

① If $x(t)$ is symmetrical about X-axis $\Rightarrow a_0 =$ zero.

② If $x(t)$ is even signal $\Rightarrow B_n = 0 \Rightarrow$ no sine term.

③ If $x(t)$ is odd signal $\Rightarrow A_n = 0 \Rightarrow$ no cosine terms

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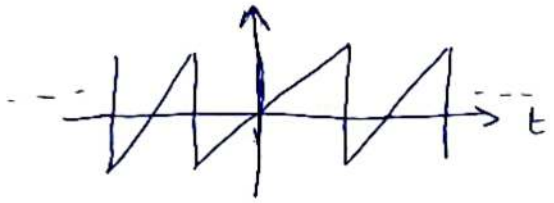


④ Symmetrical about X-axis $\Rightarrow$ $a_0 = \text{zero}$.

also " " y-axis $\Rightarrow$ Even signal. $\Rightarrow$ $B_n = \text{zero}$.

$x(t) = x(-t)$.

$a_n \neq 0$

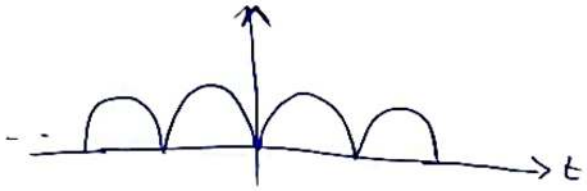


* Symmetrical about X-axis $\Rightarrow$ $a_0 = \text{zero}$.

not symmetrical about y-axis $\Rightarrow$ ODD signal.

$\Rightarrow$ $A_n = \text{zero}$.

$\therefore B_n \neq 0$



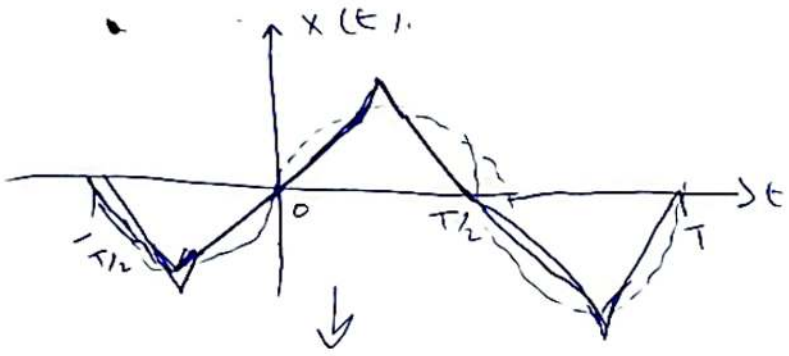
$\Rightarrow$ Not symmetric about X-axis $\Rightarrow$ $a_0 \neq 0$

Symmetric about y-axis $\Rightarrow$ Even signal $\Rightarrow$ $B_n = 0$

$A_n \neq 0$

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Approximating periodic functions.



$B_1 \sin \omega_0 t \Rightarrow$ Approximate function.

$\Rightarrow e(t) = x(t) - B_1 \sin \omega_0 t$ $\Rightarrow$ error function of approximation.

$\Rightarrow J[e(t)] \Rightarrow$ the function of error to be minimized.

$\Rightarrow$ Very common function used in minimization is the mean-square error.

$$J[e(t)] = \frac{1}{T_0} \int_0^{T_0} e^2(t)\, dt$$

$$= \frac{1}{T_0} \int_0^{T_0} \left(x(t) - B_1 \sin(\omega_0 t)\right)^2 dt$$

$J[e(t)] \Rightarrow$ is called the cost function.

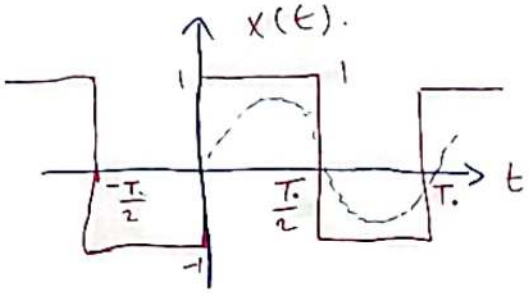
$\Rightarrow B_1 \Rightarrow$ no other value will result in a smaller error function.

$\Rightarrow$ For minimization.

$$\frac{dJ[e(t)]}{dB_1} = 0 \Rightarrow B_1 = \frac{2}{T_0} \int_0^{T_0} x(t) \sin(\omega_0 t)\, dt$$

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Ex: Find the best approximation, in a mean-square sense, of the square wave, by a sine wave.



$\Rightarrow$ Approximate function $B_1 \sin(\omega_0 t)$

$$\Rightarrow B_1 = \frac{2}{T_0}\int_0^{T_0} x(t)\sin(\omega_0 t)\,dt$$

$$= \frac{2}{T_0}\int_0^{T_0/2} \sin(\omega_0 t)\,dt + \int_{T_0/2}^{T_0} -\sin(\omega_0 t)\,dt$$

$$= \frac{2}{T_0}\left[-\frac{\cos(\omega_0 t)}{\omega_0}\Big]_0^{T_0/2} + \frac{\cos(\omega_0 t)}{\omega_0}\Big]_{T_0/2}^{T_0} \right]$$

$$= \frac{2}{\omega_0 T_0}\left[-\cos\left(\omega_0 \tfrac{T_0}{2}\right) - (-1) + \cos(\omega_0 T_0) - \cos\left(\omega_0 \tfrac{T_0}{2}\right)\right]$$

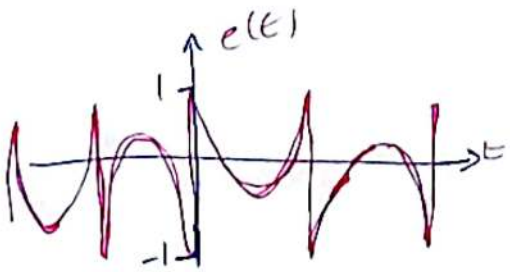
($\omega_0 \frac{T_0}{2} = \pi$, $\omega_0 T_0 = 2\pi$, $-\cos(\pi) = -(-1)$)

$$= \frac{2}{2\pi}\left[1 + 1 + 1 + 1\right]$$

$$\omega_0 T_0 = 2\pi \quad \boxed{B_1 = \frac{4}{\pi}} \quad \#$$

$$\omega_0 \frac{T_0}{2} = \pi$$

$$\Rightarrow e(t) = x(t) - \frac{4}{\pi}\sin(\omega_0 t)$$



$\Rightarrow$ Error

* other error functions can give a resonable results in minimization procedure $\Rightarrow$ the average value of the magnitude of e(t)

$$J_m[e(t)] = \frac{1}{T_0}\int_0^{T_0} |e(t)|\,dt$$

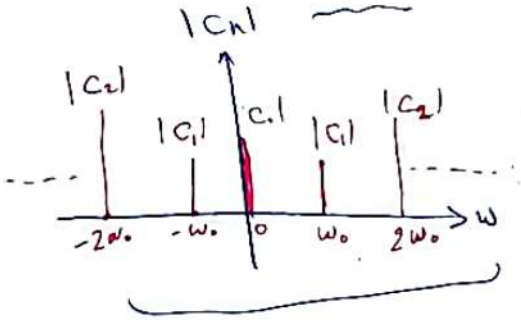
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* Frequency spectra.

⇒ C_n : Fourier series complex coefficients.

$$C_n = \underbrace{x_n + j y_n}_{\text{rectangular form}} = \underbrace{|C_n| e^{j\theta_n}}_{\text{polar form}}$$

⇒ Represent C_n ⟹ $|C_n|$, θ_n

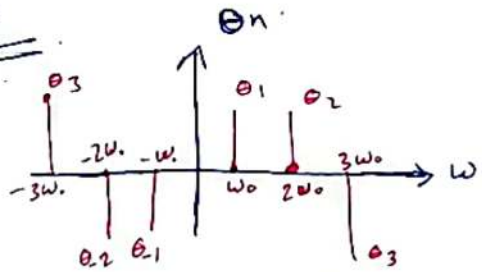


Magnitude spectrum

or (Discrete spectrum)

or (Line spectrum)

(even representation)



phase spectrum

(discrete spectrum
Line spectrum)

odd

$\theta_{-n} = -\theta_n$

$|C_n| = |C_{-n}|$, $\theta_n = -\theta_{-n}$

$C_n = C_{-n}^*$

Ex: $x(t) = 10 + 3\cos(w_0 t) + 5\cos(2w_0 t + 30^\circ) + 4\sin(3w_0 t)$

Sol: *Complex exponential form.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jnw_0 t}$$

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$$\Rightarrow x(t) = 10 + \frac{3}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) + \frac{5}{2}\left(e^{j2\omega_0 t + \pi/6} + e^{-j2\omega_0 t + \pi/6}\right) + \frac{4}{2j}\left(e^{j3\omega_0 t} - e^{-j3\omega_0 t}\right)$$

$$\Rightarrow x(t) = 10 + \frac{3}{2}e^{j\omega_0 t} + \frac{3}{2}e^{-j\omega_0 t} + \frac{5}{2}e^{j2\omega_0 t}\,e^{j\pi/6} + \frac{5}{2}e^{-j\pi/6}\,e^{-j2\omega_0 t} + \frac{2}{j}e^{j3\omega_0 t} - \frac{2}{j}e^{-j3\omega_0 t}$$

ω_0 ✱

$\Rightarrow C_0 = 10$

$C_1 = \frac{3}{2}$

$C_{-1} = \frac{3}{2}$

$C_2 = \frac{5}{2}e^{j\pi/6}$

$C_{-2} = \frac{5}{2}e^{-j\pi/6}$

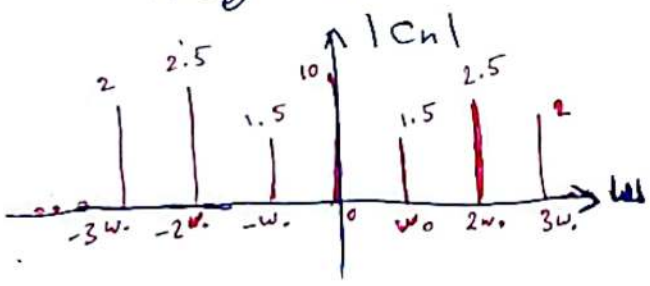
$C_3 = \frac{2}{j} = -j2 = 2e^{-j\pi/2}$ ($\tan^{-1}\left(\frac{-2}{0}\right) = \tan^{-1}(-\infty) = -\pi/2$)

$C_{-3} = \frac{-2}{j} = j2 = 2e^{j\pi/2}$ ($\tan^{-1}\left(\frac{2}{0}\right) = \tan^{-1}(\infty) = \pi/2$)

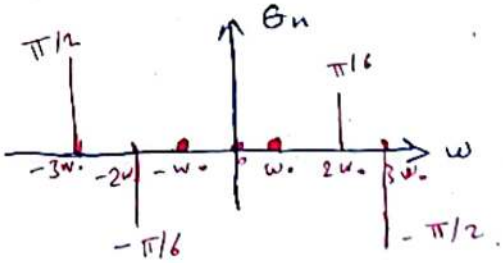
$C_n = 0$, otherwise

$$\Rightarrow x(t) = \sum_{n=-3}^{3} C_n e^{jn\omega_0 t}$$

Magnitude spectrum.



phase spectrum.

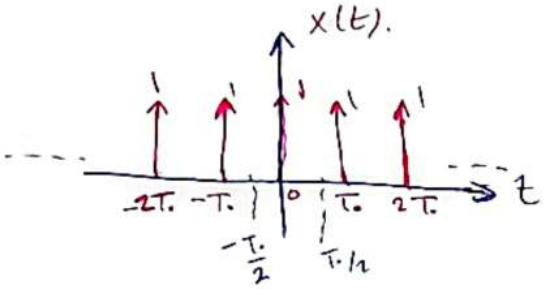


$$C_n = \begin{cases} 10, & n = 0 \\ \frac{3}{2}, & n = 1, -1 \\ \frac{5}{2}, & n = 2, -2 \\ -2j, & n = 3 \\ 2j, & n = -3 \\ 0, & \text{otherwise} \end{cases}$$

$$\theta_n = \begin{cases} 0, & n = 0 \\ 0, & n = 1, -1 \\ 0, & n = 2, -2 \\ -\pi/2, & n = 3 \\ \pi/2, & n = -3 \\ 0, & \text{otherwise} \end{cases}$$

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⇒ Fourier series for impulse train.



⇒ Expand x(t) using Fourier series.

① complex exponential form.

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}.$$

$$C_n = \frac{1}{T_0} \int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt.$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0} \underbrace{e^{-jn\omega_0(0)}}_{1} \underbrace{\int_{-T_0/2}^{T_0/2} \delta(t)\, dt}_{1}$$

$$C_n = \frac{1}{T_0}$$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T_0} e^{jn\omega_0 t}$$

② The combined trigonometric form.

$$x(t) = C_0 + \sum_{n=1}^{\infty} 2|C_n| \cos(n\omega_0 t + \theta_n)$$

$$= \frac{1}{T_0} + \sum_{n=1}^{\infty} \frac{2}{T_0} \cos(n\omega_0 t)$$

③ The Trigonometric form.

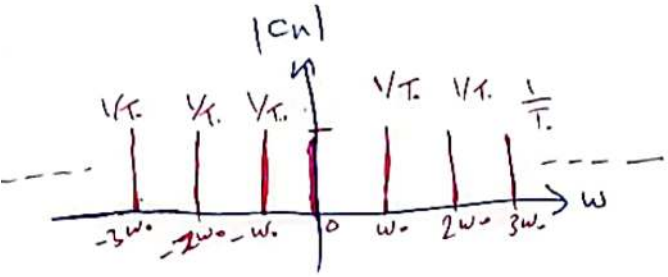
B_n = Zero → $x(t)$ → even

$$\Rightarrow x(t) = C_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) \rightarrow A_n = 2|C_n| = 2\,\frac{1}{T_0}$$

$$\Rightarrow x(t) = \frac{1}{T_0} + \sum_{n=1}^{\infty} \frac{2}{T_0} \cos(n\omega_0 t) \neq \text{the same as form ②}$$

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⇒ $C_n \rightarrow$ is real ⇒ no phase spectrum.

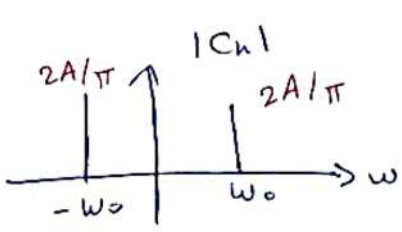


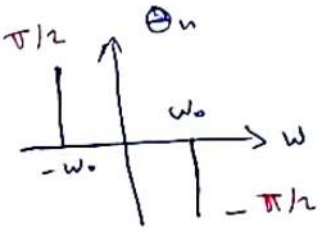
Magnitude spectrum.

⇒ Back to Table 4.3 page 168 Ch4 [recored back]

Fourier series for Common signals.

Ex: if





* Find x(t) ?!

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jnw_o t} \qquad n=1 \qquad w_o$$

$$= C_1 e^{jw_o t} + C_{-1} e^{-jw_o t}$$

$$\begin{cases} |C_1| = \frac{2A}{\pi} e^{-j\pi/2} \\ |C_{-1}| = \frac{2A}{\pi} e^{j\pi/2} \end{cases} \Rightarrow x(t) = \frac{2A}{\pi} e^{jw_o t} \cdot e^{-j\pi/2} + \frac{2A}{\pi} e^{-jw_o t} \cdot e^{j\pi/2}$$

$$= \frac{2A}{\pi} \left[e^{j(w_o t - \frac{\pi}{2})} + e^{-j(w_o t - \frac{\pi}{2})} \right]$$

$$= \frac{2A}{\pi} \left(2\cos(w_o t - \frac{\pi}{2}) \right)$$

$$x(t) = \frac{4A}{\pi} \cos(w_o t - \frac{\pi}{2}) \qquad \cos(\theta - \frac{\pi}{2}) = \sin\theta$$

$$\boxed{x(t) = \frac{4A}{\pi} \sin(w_o t)}$$

//

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Ch 5: The Fourier Transform (FT).

⇒ $x(t) \xrightarrow{\mathcal{F}} X(\omega)$

time domain

Frequency domain.

$x(t) \xleftarrow{\mathcal{F}^{-1}} X(\omega)$

⇒ $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$

⇒ Mathmatical tool used for frequency analysis of signals.

$X(\omega)$: is the Fourier transform of $x(t)$.

$x(t)$: is the inverse Fourier transform of $X(\omega)$

⇒ $x(t) \Leftrightarrow X(\omega)$ are called a Fourier transform pair.

⇒ Definition.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt. \Rightarrow \mathcal{F}\{x(t)\}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega. \Rightarrow \mathcal{F}^{-1}[X(\omega)]$$

$\Rightarrow 2\pi f = \omega = \frac{2\pi}{t}.$

⇒ 2 Advantages of F.T:-

⇒ ① For LTI system.

$x(t) \rightarrow [\text{LTI}] \rightarrow y(t)$.

$\Rightarrow y(t) = x(t) * h(t).$

$x(t) \xrightarrow{\text{F.T}} X(\omega)$

$y(t) \xrightarrow{\text{F.T}} Y(\omega)$

$y(t) = x(t) * h(t)$

$Y(\omega) = X(\omega) \cdot H(\omega) \Rightarrow \mathcal{F}^{-1}\{Y(\omega)\} = y(t)$

$H(\omega) = \frac{Y(\omega)}{X(\omega)} \Rightarrow$ Transfer function.

$h(t) = \mathcal{F}^{-1}\{H(\omega)\}$

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⇒ For Laplase transform.

$X(s) \Rightarrow s = \sigma + j\omega$

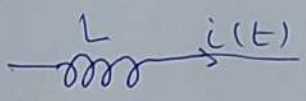
if σ = zero.

$s = j\omega \Rightarrow X(s) = X(j\omega) = X(\omega)$.

$\Rightarrow X(\omega) = X(s)\big|_{s=j\omega}$

⇒ ✳ The Fourier transform is a special case of the laplase transform.

(2) circuit theory.

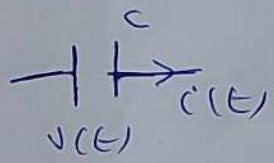


$$i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\, d\tau$$

F.T ↓

$$I(\omega) = \frac{1}{L}\,\frac{V(\omega)}{j\omega}$$

$$\boxed{\frac{V(\omega)}{I(\omega)} = j\omega L} \Rightarrow \text{indactive impedance.}$$



$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\, d\tau$$

↓

F.T.

$$V(\omega) = \frac{1}{C}\,\frac{I(\omega)}{j\omega}$$

$$\boxed{\frac{V(\omega)}{I(\omega)} = \frac{1}{j\omega C}} \Rightarrow \text{capacitive impedance.}$$

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⇒ Conditions for existence of Fourier transform

① Signals should have finite number of maxima & minima over any finite interval

② Signal should have finite number of discontinuities over any finite interval

③ Signal should be absolutely integrable.

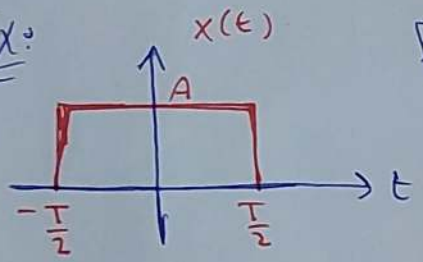
$$\int_{-\infty}^{\infty} |x(t)|\,dt < \infty$$

→ For Energy signals. ✓

→ For Power signals ✗ (= have F.T.)

⇒ These conditions are sufficient but not necessary.

Ex:



Find $X(\omega)$. ⇒ $x(t) = A\,\text{rect}\left(\frac{t}{T}\right)$

⇒ Sol:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-T/2}^{T/2} A\, e^{-j\omega t}\, dt$$

$$= A\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T/2}^{T/2} = A\left(\frac{e^{j\omega(\frac{T}{2})} - e^{-j\omega(\frac{T}{2})}}{j\omega}\right)$$

$$= 2A\frac{T}{2}\left(\frac{e^{j\omega\frac{T}{2}} - e^{-j\omega\frac{T}{2}}}{2j\omega T/2}\right)$$

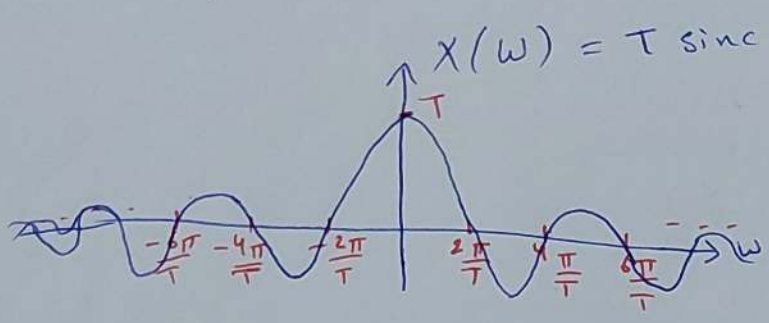
$$= AT\,\frac{\sin(\omega\frac{T}{2})}{\omega(T/2)}$$

$$\boxed{X(\omega) = AT\,\text{sinc}\left(T\frac{\omega}{2}\right)}$$ #-

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$\Rightarrow A\,\text{rect}\left(\frac{t}{\tau}\right) \overset{\mathcal{F}}{\longleftrightarrow} A\tau\,\text{sinc}\left(\tau\frac{\omega}{2}\right)$.

$\text{rect}\left(\frac{t}{\tau}\right) \overset{\mathcal{F}}{\longleftrightarrow} \tau\,\text{sinc}\left(\tau\frac{\omega}{2}\right)$.



$\Rightarrow \text{sinc}\left(\tau\frac{\omega}{2}\right) = 0$

$\tau\frac{\omega}{2} = \pm n\pi$

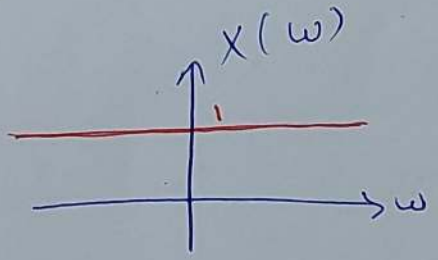
$\omega = \pm\frac{2n\pi}{\tau}$

Ex: Find $X(\omega)$ for $x(t) = \delta(t)$

Sol: $X(\omega) = \int_{-\infty}^{\infty} \delta(t)\, e^{-j\omega t}\, dt$

$= \int_{-\infty}^{\infty} \delta(t)\, e^{0}\, dt$

$X(\omega) = 1$



⇒ white spectrum.

$\Rightarrow \delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1$

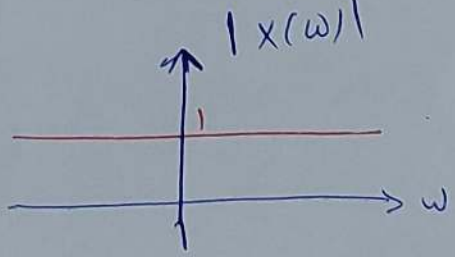
Ex: if $x(t) = \delta(t - t_0)$

Sol: $X(\omega) = \int_{-\infty}^{\infty} \delta(t - t_0)\, e^{-j\omega t}\, dt$

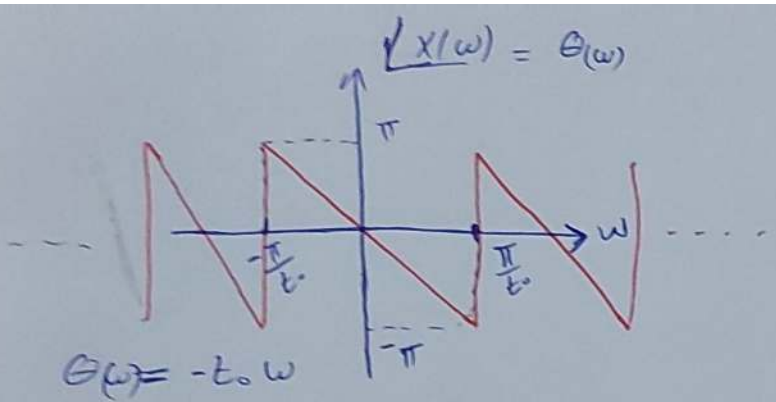
$= \int_{-\infty}^{\infty} \delta(t - t_0)\, e^{-j\omega t_0}\, dt$

$X(\omega) = e^{-j\omega t_0}$

$\Rightarrow \delta(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_0}$



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$\theta(w) = -t_0 w$

$X(w) = e^{-jwt_0}$

$= \cos(wt_0) - j\sin(wt_0)$

$|X(w)| = 1$

$\theta(w) = -t_0 w$

$\theta + 2\pi = \theta \Rightarrow \boxed{\theta \pm 2n\pi = \theta}$.

Periodic with period $= 2\pi$

$\Rightarrow \theta(w) \Rightarrow$ Periodic $\rightarrow w = 2\pi$

$\Rightarrow -t_0 w \pm 2n\pi = 0$

$n = 1$

$-t_0 w + 2\pi = 0$

$w = \frac{2\pi}{t_0}$

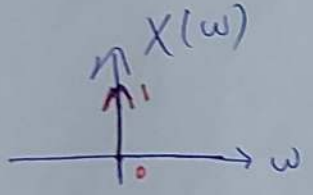
$e^{-jwt_0} = e^{+j(-wt_0 + 2\pi)}$

$= e^{-jwt_0} \cdot e^{j2\pi}$

$\Rightarrow e^{-jwt_0} = e^{-jwt_0}$ (since $e^{j2\pi} \sim 1$)

Periodic.

Ex: $X(w) = \delta(w)$. Find $x(t)$.



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dw$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(w)\, e^{jwt}\, dw$$

($w = 0$, $e^{0} = 1$)

$$\boxed{x(t) = \frac{1}{2\pi}} \#$$

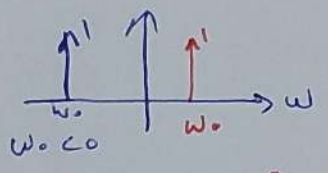
$$\frac{1}{2\pi} \overset{\mathcal{F}}{\longleftrightarrow} \delta(w)$$

$$1 \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(w)$$

$\Rightarrow$ duality properity.

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Ex: $X(\omega) = \delta(\omega - \omega_0)$. find $x(t)$.



$$\Rightarrow x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

Sol:

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega-\omega_0)\, e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi} e^{j\omega_0 t}\int_{-\infty}^{\infty} \delta(\omega-\omega_0)\, d\omega$$

$$\left[x(t) = \frac{1}{2\pi} e^{j\omega_0 t} \right] \#.$$

$$\Rightarrow \frac{1}{2\pi} e^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} \delta(\omega-\omega_0)$$

$$\left[e^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi\,\delta(\omega-\omega_0) \right] \#.$$

$$e^{-j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi\,\delta(\omega+\omega_0)$$

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Summary :-

① $\delta(t) \xleftrightarrow{F} 1$

② $1 \xleftrightarrow{F} 2\pi\delta(w)$

③ $\delta(t-t_0) \xleftrightarrow{F} e^{-jwt_0}$

④ $\delta(t+t_0) \xleftrightarrow{F} e^{jwt_0}$

⑤ $e^{jw_0t} \xleftrightarrow{F} 2\pi\delta(w-w_0)$

⑥ $e^{-jw_0t} \xleftrightarrow{F} 2\pi\delta(w+w_0)$

⑦ $\text{rect}\left(\frac{t}{T}\right) \xleftrightarrow{F} T\,\text{sinc}\left(\frac{T}{2}w\right)$

Fourier Transform properties

① Linearity :-

$x_1(t) \xleftrightarrow{F} X_1(w)$

$x_2(t) \xleftrightarrow{F} X_2(w)$

$a_1x_1(t) + a_2x_2(t) \xleftrightarrow{F} a_1X_1(w) + a_2X_2(w)$.

Ex: $x(t) = \cos(w_0t)$. Find $X(w)$.

Sol: $x(t) = \frac{1}{2}\left(e^{jw_0t} + e^{-jw_0t}\right)$

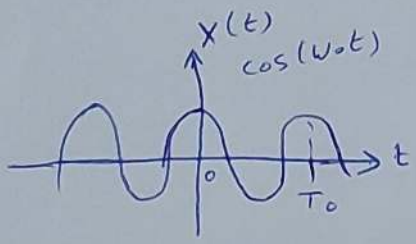
$= \frac{1}{2}\left[e^{jw_0t} + e^{-jw_0t}\right]$

$X(w) = \frac{1}{2}\left[2\pi\delta(w-w_0) + 2\pi\delta(w+w_0)\right]$

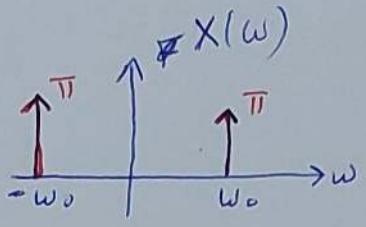
$X(w) = \pi\left[\delta(w-w_0) + \delta(w+w_0)\right]$

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$$\Rightarrow \cos(\omega_0 t) \xleftrightarrow{F} \pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$$



$$T_0 = \frac{2\pi}{\omega_0}$$



Ex: $x(t) = \sin(\omega_0 t)$ Find $X(\omega)$.

Sol:

$$x(t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$X(\omega) = \frac{1}{2j}\left[2\pi\delta(\omega-\omega_0) - 2\pi\delta(\omega+\omega_0)\right]$$

$$X(\omega) = \frac{\pi}{j}\left[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)\right]$$

(2) Time scaling

$$x(t) \xleftrightarrow{F} X(\omega)$$

$$x(at) \xleftrightarrow{F} \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

Time domain		Frequency domain
$\lvert a\rvert > 1$ ⇒ compress	⇒	Expand
$\lvert a\rvert < 1$ ⇒ expand	⇒	compress

Proof:-

$$F[x(at)] = \int_{-\infty}^{\infty} x(at)\, e^{-j\omega t}\, dt$$

$$\Rightarrow \text{let } \tau = at \rightarrow \begin{cases} t=\infty \rightarrow \tau = \infty\,(a) \\ t=-\infty \rightarrow \tau = -\infty\,(a) \end{cases} \Rightarrow \tau = \begin{cases} \pm\infty, & a>0 \\ \mp\infty, & a<0 \end{cases}$$

$$d\tau = a\, dt$$

① $a > 0$.

$$F[x(at)] = \int_{-\infty}^{\infty} x(\tau)\, e^{-j\omega\frac{\tau}{a}}\, \frac{d\tau}{a}$$

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$$\Rightarrow \mathcal{F}\{x(at)\} = \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

(2) $a < 0$

$$\mathcal{F}\{x(at)\} = \int_{+\infty}^{-\infty} x(\tau)\, e^{-j\omega\frac{\tau}{2}} \frac{d\tau}{a}$$

$$= \frac{-1}{a} \int_{-\infty}^{\infty} x(\tau)\, e^{-j\omega\frac{\tau}{2}} d\tau$$

$$= \frac{-1}{a} X\left(\frac{\omega}{2}\right)$$

$\leftarrow -|a|$

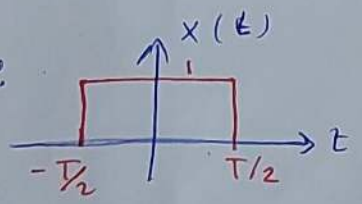
$a < 0 \Rightarrow -|a|$

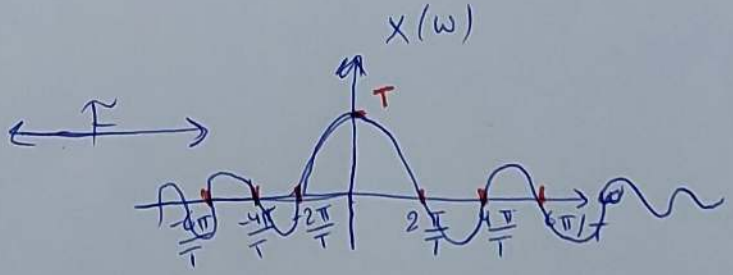
$$= \frac{1}{|a|} X\left(\frac{\omega}{2}\right) \quad \#$$

$\Rightarrow$ ~~$\mathcal{F}\{x(at)\} \longleftrightarrow$~~

$$\left[\begin{array}{l} x(2t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} X\left(\frac{\omega}{2}\right) \\ x(-2t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} X\left(\frac{-\omega}{2}\right) \\ x(-t) \xleftrightarrow{\mathcal{F}} X(-\omega) \end{array}\right] \quad \#$$

Ex:

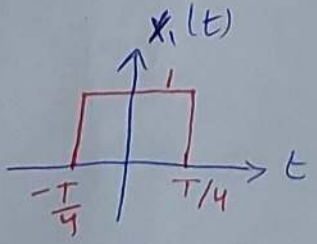




$$x(t) = \text{rect}\left(\frac{t}{T}\right) \qquad X(\omega) = T\,\text{sinc}\left(\frac{T\omega}{2}\right)$$

$$T\omega/2 = \pm n\pi$$

$$\omega = \pm \frac{2n\pi}{T}$$



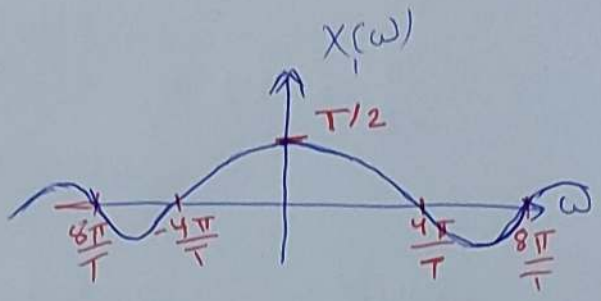
$$x_1(t) = x(2t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} X\left(\frac{\omega}{2}\right)$$

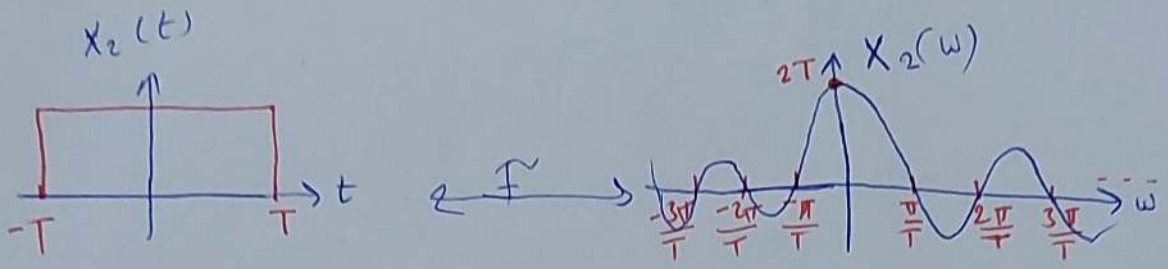
$$= \frac{1}{2} T\,\text{sinc}\left(\frac{T\omega}{4}\right)$$

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$\Rightarrow \frac{Tw}{4} = \pm n\pi$

$w = \pm 4n\frac{\pi}{T}$





$x_2(t) = x\left(\frac{t}{2}\right) \xleftrightarrow{F} \frac{1}{1/2} X(2w)$

$2T\,\mathrm{sinc}(Tw)$

$Tw = \pm n\pi$

$w = \pm \frac{n\pi}{T}$

3 Time Shifting

$x(t) \xleftrightarrow{F} X(w)$

$x(t-t_0) \xleftrightarrow{F} X(w)\, e^{-jwt_0}$

$x(t+t_0) \xleftrightarrow{F} X(w)\, e^{jwt_0}$

Ex: $x(t) = 10\cos\left(200\pi\left(t - 1.25*10^{-3}\right)\right)$

Find $X(w)$.

Sol:

$X(w) = 10\pi\left[\delta(w-200\pi) + \delta(w+200\pi)\right] e^{-jw\,1.25*10^{-3}}$

$w = 200\pi$

$= 10\pi\left[\delta(w-200\pi)\, e^{-j\pi/4} + \delta(w+200\pi)\, e^{j\pi/4}\right]$

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4 Time Transformation.

$$x(t) \xleftrightarrow{F} X(w).$$

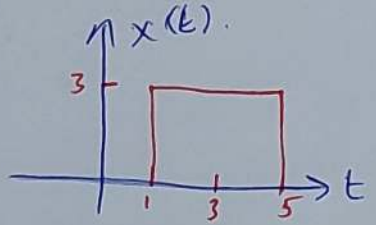
time transformation.

$$x(at+b) \xleftrightarrow{F} \frac{1}{|a|} X\left(\frac{w}{a}\right) e^{+jw\frac{b}{a}}$$

$at+b=0$

$t=-\frac{b}{a}$

Ex:



Find $X(w)$ use $\underbrace{rect(t)}_{x(t)} \xleftrightarrow{F} \underbrace{sinc\left(\frac{w}{2}\right)}_{X(w)}$.

$x_1(t) = 3\, rect\left(\frac{t-3}{4}\right)$ → shift.

$$x_1(t) = 3\, x\left(\underbrace{\frac{t}{4}}_{a} - \underbrace{\frac{3}{4}}_{b/a}\right) \rightarrow \frac{t}{4} - \frac{3}{4} = 0$$

$\frac{t}{4} = \frac{3}{4}$

$t=3 \rightarrow b=3$

$$X_1(w) = \frac{3}{1/4}\, x\left(\frac{w}{1/4}\right) e^{-jw\frac{b}{a}}$$

$$= 12\, sinc(2w)\, e^{-jw3}$$

5 Duality:

⇒ This property is known as symmetry property.

$$x(t) \xleftrightarrow{F} X(w)$$

$$X(t) \xleftrightarrow{F} 2\pi x(-w).$$

$$X(t) = X(w)\Big|_{w=t}, \qquad x(-w) = x(t)\Big|_{t=-w}$$

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Ex: $\delta(t) \xleftrightarrow{\mathcal{F}} 1$

$1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(-\omega)$; $\delta \to$ even.

$= 2\pi\delta(\omega)$

Ex: $x(t) = \text{sinc}(t)$. find $X(\omega)$.

Sol^{n} – using duality property.

$\text{rect}\left(\frac{t}{T}\right) \xleftrightarrow{\mathcal{F}} T\,\text{sinc}\left(T\frac{\omega}{2}\right)$

$T\,\text{sinc}\left(T\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} 2\pi e\,\text{rect}\left(\frac{-\omega}{T}\right) = 2\pi\,\text{rect}\left(\frac{\omega}{T}\right)$, even

$T=2$

$2\,\text{sinc}(t) \xleftrightarrow{\mathcal{F}} 2\pi\,\text{rect}\left(\frac{\omega}{2}\right)$

$\text{sinc}(t) \xleftrightarrow{\mathcal{F}} \pi\,\text{rect}\left(\frac{\omega}{2}\right)$ #

6 Convolution.

$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(\omega) \cdot X_2(\omega)$

$x_1(t) \cdot x_2(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$

Proof: –

$$\mathcal{F}\{x_1(t) * x_2(t)\} = \int_{-\infty}^{\infty} (x_1(t) * x_2(t))\, e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_1(\tau)\, x_2(t-\tau)\, d\tau\right] e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} x_1(\tau) \underbrace{\left[\int_{-\infty}^{\infty} x_2(t-\tau)\, e^{-j\omega t}\, dt\right]}_{X_2(\omega)\, e^{-j\omega\tau}} d\tau$$

$\Rightarrow$

$$= \int_{-\infty}^{\infty} x_1(\tau)\, X_2(\omega)\, e^{-j\omega\tau}\, d\tau$$

$$= X_2(\omega) \int_{-\infty}^{\infty} x_1(\tau)\, e^{-j\omega\tau}\, d\tau$$

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$\Rightarrow \boxed{X_1(\omega)\, X_2(\omega)}$ ##

⇒ Proof:

$$\mathcal{F}^{-1}\left[\frac{1}{2\pi} X_1(\omega) * X_2(\omega)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{2\pi} X_1(\omega) * X_2(\omega)\, e^{j\omega t}\, d\omega.$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{2\pi}\left[\int_{-\infty}^{\infty} X_1(\sigma)\, X_2(\omega-\sigma)\, d\sigma\right] e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\sigma)\left[\int_{-\infty}^{\infty} \frac{1}{2\pi} X_2(\omega-\sigma)\, e^{j\omega t}\, d\omega\right] d\sigma$$

$$\Rightarrow \quad X_2(t)\, e^{j\sigma t}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\sigma)\, x_2(t)\, e^{j\sigma t}\, d\sigma$$

$$= x_2(t)\, x_1(t)$$

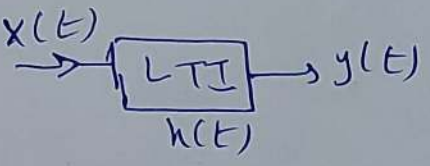
$$= x_1(t)\, x_2(t) \quad \#_\#$$

Note: For LTI system.

$$y(t) = x(t) * h(t)$$

$$Y(\omega) = X(\omega) \cdot H(\omega).$$

$$\Rightarrow y(t) = \mathcal{F}^{-1}\left[Y(\omega)\right].$$



Ex: Find $\int_{-\infty}^{\infty} \text{sinc}(\tau)\, \text{sinc}(t-\tau)\, d\tau$

⇒ Sol: ⇓ $\text{sinc}(t) * \text{sinc}(t)$.

or $\frac{\sin(t)}{t} * \frac{\sin(t)}{t}$

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⇒ Duality

$\text{rect}(t/T) \xleftrightarrow{F} T\,\text{sinc}(T\frac{w}{2})$.

$T\,\text{sinc}(T\frac{t}{2}) \xleftrightarrow{F} 2\pi\,\text{rect}(\frac{w}{T})$; $y(-w)$ even

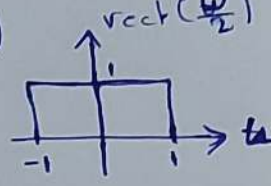
$T=2$

$(0.5)*\ 2\,\text{sinc}(t) \xleftrightarrow{F} 2\pi\,\text{rect}(\frac{w}{2})$

$\text{sinc}(t) \xleftrightarrow{F} \pi\,\text{rect}(\frac{w}{2})$.

$\Rightarrow \text{sinc}(t) * \text{sinc}(t) \xleftrightarrow{F} \pi^2\,\text{rect}^2(\frac{w}{2})$

$= \pi^2\,\text{rect}(\frac{w}{2})$.



$\Rightarrow \mathcal{F}^{-1}\{\pi^2\,\text{rect}(\frac{w}{2})\} = \pi\,\text{sinc}(t)$

$\Rightarrow$ $\text{sinc}(t) * \text{sinc}(t) = \pi\,\text{sinc}(t)$ #

7 Frequency shifting :-

$x(t) \xleftrightarrow{F} X(w)$

$x(t)\,e^{-jw_0t} \xleftrightarrow{F} X(w+w_0)$

$x(t)\,e^{jw_0t} \xleftrightarrow{F} X(w-w_0)$

Ex: Find $F_3(w)$ if $f_3(t) = f_1(t) \cdot f_2(t)$.

$f_1(t) = 2\cos(200\pi t)$

$f_2(t) = 5\cos(1000\pi t)$

Sol: $f_3(t) = 2\cos(200\pi t)\,5\cos(1000\pi t)$

$= 10\cos(200\pi t)\left[\frac{e^{j1000\pi t} + e^{-j1000\pi t}}{2}\right]$

$= 5\cos(200\pi t)\,e^{j1000\pi t} + 5\cos(200\pi t)\,e^{-j1000\pi t}$

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$$\Rightarrow F_3(w) = 5\pi\left[\delta(w-200\pi \pm 1000\pi) + \delta(w+200\pi \pm 1000\pi)\right]$$

$$= 5\pi\left[\delta(w-1200\pi) + \delta(w+800\pi) + \delta(w+1200\pi) + \delta(w-800\pi)\right] \quad \#$$

OR :-

$$f_1(t) \cdot f_2(t) \xleftrightarrow{F} \frac{1}{2\pi} F_1(w) * F_2(w)$$

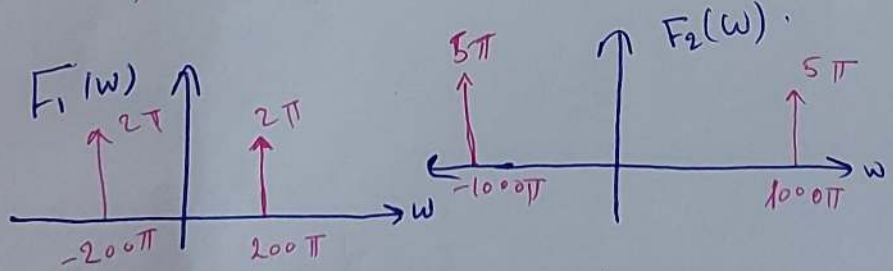
$$F_1(w) = 2\pi\left[\delta(w-200\pi) + \delta(w+200\pi)\right]$$

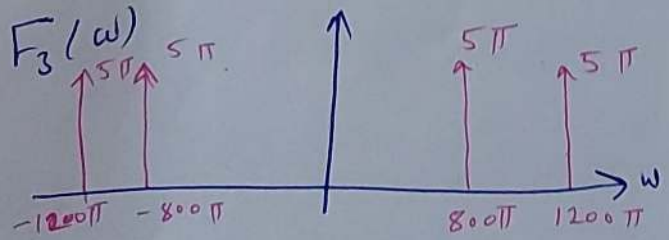
$$F_2(w) = 5\pi\left[\delta(w-1000\pi) + \delta(w+1000\pi)\right]$$

$$F_3(w) = \frac{F_1(w) * F_2(w)}{2\pi}$$

$$= 5\pi\left[\left(\delta(w-200\pi) + \delta(w+200\pi)\right) * \left(\delta(w-1000\pi) + \delta(w+1000\pi)\right)\right]$$

$$F_3(w) = 5\pi\left[\delta(w-1200\pi) + \delta(w+800\pi) + \delta(w-800\pi) + \delta(w+1200\pi)\right] \quad \#$$





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8 Time-differentiation :-

$$x(t) \xleftrightarrow{F} X(w).$$

c

$$g(t) = \frac{d\,x(t)}{dt}.$$

$$g(t) \xleftrightarrow{F} G(w).$$

$$G(w) = jwX(w)$$

$$X(w) = \frac{G(w)}{jw} + 2\pi \underbrace{c}_{\text{Constant}} \delta(w). \rightarrow c : \text{DC-value of } x(t)$$

$$\frac{d^n x(t)}{dt^n} \xleftrightarrow{F} (jw)^n X(w).$$

Proof :-

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dw = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\underbrace{X(w)}_{\text{const}} \frac{d\,e^{jwt}}{dt}\right) dw$$

$$\frac{d\,x(t)}{dt} = \frac{1}{2\pi}\int_{-\infty}^{\infty} jw\,X(w)\, e^{jwt}\, dw$$

$$\frac{d\,x(t)}{dt} = F^{-1}\{jwX(w)\}.$$

$$F\left\{\frac{d\,x(t)}{dt}\right\} = jwX(w)$$

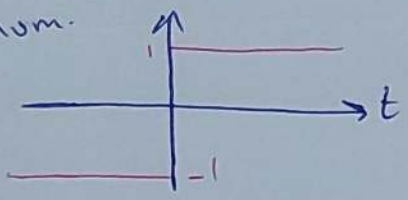
$$\Rightarrow \frac{d\,x(t)}{dt} \xleftrightarrow{F} jwX(w)$$

$$\frac{d^n x(t)}{dt^n} \xleftrightarrow{F} (jw)^n . X(w).$$

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Ex:- $x(t) = \text{sgn}(t)$ Find $X(w)$

Signum.



C: dc value = Zero.

⇒ Sol:-

$$\frac{dx(t)}{dt} = 2\delta(t).$$

$\mathcal{F}$ ↓

$$jw\,X(w) = 2$$

$$X(w) = \frac{2}{jw} + 2\pi C\,\delta(t).$$

(zero)

$$X(w) = \frac{2}{jw}$$

$$\Rightarrow \quad \text{sgn}(t) \xleftrightarrow{\mathcal{F}} \frac{2}{jw}$$

Ex: $x(t) = u(t)$, Find $X(w)$.

Sol: $\int_{-\infty}^{\infty} |u(t)|\,dt = \int_{0}^{\infty} |u(t)|\,dt = t\Big]_0^{\infty} = \infty \Rightarrow$ not absolutely integrable.

⇒ ① using $\frac{d\,u(t)}{dt}$

$$\Rightarrow \quad \frac{d(u(t))}{dt} = \delta(t).$$

$\mathcal{F}$ ↓

$$jw\,\mathcal{F}[u(t)] = 1$$

$$\mathcal{F}[u(t)] = \frac{1}{jw} + 2\pi\,\delta(w)\,(u_{dc})_{1/2}$$

$$\mathcal{F}[u(t)] = \frac{1}{jw} + \pi\delta(w).$$

u_{dc} ⇒ DC-value of the unit step. fun.

$$\Rightarrow u_{dc} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} u(t)\,dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{0}^{T/2} 1\,dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \cdot \frac{T}{2} = \frac{1}{2}$$

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⇒ (2) using sgn(t).

$$u(t) = \frac{1}{2}[1 + \text{sgn}(t)]$$

$$\mathcal{F}[u(t)] = \frac{1}{2}\left[2\pi\delta(\omega) + \frac{2}{j\omega}\right]$$

$$\mathcal{F}[u(t)] = \pi\delta(\omega) + \frac{1}{j\omega}\Big] \quad \#$$

[9] Time integration:-

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \xleftrightarrow{\mathcal{F}} \frac{X(\omega)}{j\omega} + \pi \underbrace{X(0)}_{\omega=0}\delta(\omega)$$

Proof:-

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau)\,\underbrace{u(t-\tau)}_{-\infty \to t}\,d\tau$$

$$= \int_{-\infty}^{t} x(\tau)\,d\tau$$

$$\Rightarrow \mathcal{F}\left[\int_{-\infty}^{t} x(\tau)\,d\tau\right] = \mathcal{F}[x(t) * u(t)]$$

$$= X(\omega)\cdot\left[\frac{1}{j\omega} + \pi\delta(\omega)\right]$$

$$= \frac{X(\omega)}{j\omega} + \pi X(\omega)\delta(\omega).$$

$$\boxed{\mathcal{F}\left[\int_{-\infty}^{t} x(\tau)\,d\tau\right] = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)} \quad \#.$$

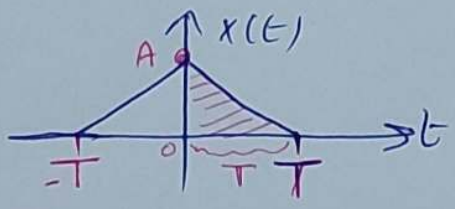
Ex: Find $\mathcal{F}[u(t)]$ using the time integration property.

Sol: $u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$.

$$\boxed{\mathcal{F}[u(t)] = \frac{1}{j\omega} + \pi\delta(\omega).} \quad \#$$

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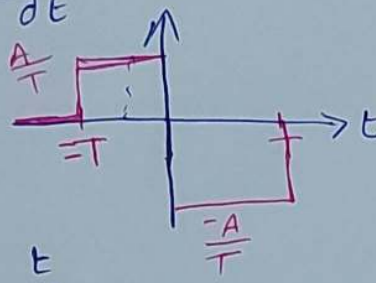
Ex: Find $X(\omega)$.



$$x(t) = A\,tri\left(\frac{t-0}{T}\right) = A\,tri\left(\frac{t}{T}\right).$$

Solve

$$\frac{dx(t)}{dt} = g(t).$$



$$\Rightarrow g(t) = \frac{A}{T}\,rect\left(\frac{t+\frac{T}{2}}{T}\right) - \frac{A}{T}\,rect\left(\frac{t-\frac{T}{2}}{T}\right)$$

$$x(t) = \int_{-\infty}^{t} g(\tau)\,d\tau.$$

$$X(\omega) = \frac{G(\omega)}{j\omega} + \pi G(0)\,\delta(\omega).$$

$$\Rightarrow g(t) = \frac{A}{T}\,rect\left(\frac{t+T/2}{T}\right) - \frac{A}{T}\,rect\left(\frac{t-T/2}{T}\right)$$

$$G(\omega) = A\,sinc\left(T\frac{\omega}{2}\right)e^{j\omega T/2} - A\,sinc\left(T\frac{\omega}{2}\right)e^{-j\omega T/2}$$

$$= A\,sinc\left(T\frac{\omega}{2}\right)\left[e^{j\omega T/2} - e^{-j\omega T/2}\right]\frac{j2}{j2}\left(\frac{\omega T/2}{\omega T/2}\right)$$

$$= Aj\frac{2\,\omega T}{2}\,sinc\left(T\frac{\omega}{2}\right)\left[\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j\,\omega T/2}\right]$$

$\left[\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j\,\omega T/2}\right] = sinc(T\omega/2)$

$$\boxed{G(\omega) = j\omega AT\,sinc^2\left(T\frac{\omega}{2}\right)}$$

$$\Rightarrow X(\omega) = AT\,sinc^2\left(T\frac{\omega}{2}\right) + \pi \underbrace{G(0)}_{zero}\,\delta(\omega)$$

$$X(\omega) = AT\,sinc^2\left(T\frac{\omega}{2}\right)$$

$$\boxed{\begin{aligned} A\,tri\left(\frac{t}{T}\right) &\xleftrightarrow{F} AT\,sinc^2\left(T\frac{\omega}{2}\right) \\ tri\left(\frac{t}{T}\right) &\xleftrightarrow{F} T\,sinc^2\left(T\frac{\omega}{2}\right) \end{aligned}}$$

#

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10] Frequency Differentiation:-

$$x(t) \xleftrightarrow{\mathcal{F}} X(w).$$

$$-jt\,x(t) \xleftrightarrow{\mathcal{F}} \frac{dX(w)}{dw}$$

$$(-jt)^n x(t) \xleftrightarrow{\mathcal{F}} \frac{d^n X(w)}{dw^n}$$

Proof:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dw.$$

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt.$$

$$\frac{dX(w)}{dw} = \int_{-\infty}^{\infty} x(t)\, \frac{de^{-jwt}}{dw}\, dt$$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\,(-jw)\, dt.$$

$$= \mathcal{F}\{(-jt)\,x(t)\}$$

$$\left[\mathcal{F}^{-1}\left\{\frac{dX(w)}{dw}\right\} = -jt\,x(t)\right] \quad \#$$

Ex: Find $\mathcal{F}\{t\,x(t)\}$

$$t\,x(t) \xleftrightarrow{\mathcal{F}} j\frac{dX(w)}{dw}$$

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$\Rightarrow$ $x(t) \xleftrightarrow{F} X(w)$.

x_{dc} ↓

$g(t) = \frac{dx(t)}{dt} \xleftrightarrow{F} G(w)$.

g_{dc} ↓

$\frac{dg(t)}{dt} = h(t) \xleftrightarrow{F} H(w)$.

Find $X(w)$:-

① $h(t) = \frac{dg(t)}{dt}$

F ↓

$H(w) = jw\,G(w)$

$G(w) = \frac{H(w)}{jw} + 2\pi g_{dc}\,\delta(w)$.

② $g(t) = \frac{dx(t)}{dt}$

F ↓

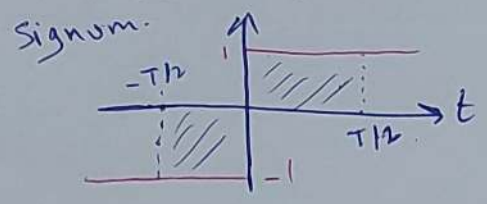
$G(w) = jw\,X(w)$

$X(w) = \frac{G(w)}{jw} + 2\pi x_{dc}\,\delta(w)$

#

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Ex:- $x(t) = \text{sgn}(t)$ Find $X(w)$



C: dc value = Zero.

$$dc = \lim_{T\to\infty} \frac{1}{T}\left[\int_{-T/2}^{0} -1\,dt + \int_{0}^{T/2} 1\,dt\right]$$

[dc = zero]

⇒ Sol:-

$$\frac{dx(t)}{dt} = 2\delta(t).$$

$\mathcal{F}\downarrow$

$$jw\,X(w) = 2$$

$$X(w) = \frac{2}{jw} + 2\pi c\,\delta(t).$$

(zero)

$$X(w) = \frac{2}{jw}$$

$$\Rightarrow \left[\left[\text{sgn}(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{2}{jw}\right]\right] \#$$

Ex: $x(t) = u(t)$, Find $X(w)$.

Sol: $\int_{-\infty}^{\infty} |u(t)|\,dt = \int_{0}^{\infty} |u(t)|\,dt = t\Big]_0^{\infty} = \infty \Rightarrow$ not absolutely integrable.

⇒ ① using $\frac{d\,u(t)}{dt}$

$$\Rightarrow \frac{d(u(t))}{dt} = \delta(t).$$

$\mathcal{F}\downarrow$

$$jw\,\mathcal{F}[u(t)] = 1$$

$$\mathcal{F}[u(t)] = \frac{1}{jw} + 2\pi\,\delta(w)\,(u_{dc})_{1/2}$$

$u_{dc} \Rightarrow$ DC-value of the unit step. fun.

$$\left[\left[\mathcal{F}[u(t)] = \frac{1}{jw} + \pi\delta(w).\right]\right] \#$$

$$\Rightarrow u_{dc} = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} u(t)\,dt$$

$$= \lim_{T\to\infty} \frac{1}{T}\int_{0}^{T/2} 1\,dt$$

$$= \lim_{T\to\infty} \frac{1}{T}\cdot\frac{T}{2} = \frac{1}{2}$$

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⇒ (2) Using sgn(t).

$$u(t) = \frac{1}{2}[1 + \text{sgn}(t)]$$

$$\mathcal{F}[u(t)] = \frac{1}{2}\left[2\pi\delta(\omega) + \frac{2}{j\omega}\right]$$

$$\boxed{\mathcal{F}[u(t)] = \pi\delta(\omega) + \frac{1}{j\omega}} \quad \#$$

[9] Time integration:-

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \xleftrightarrow{\mathcal{F}} \frac{X(\omega)}{j\omega} + \pi X(0)\,\delta(\omega)$$

($X(0)$ means $X(\omega)$ at $\omega = 0$)

Proof:-

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau)\,u(t-\tau)\,d\tau$$

($u(t-\tau)$: $-\infty \to t$)

$$= \int_{-\infty}^{t} x(\tau)\,d\tau$$

$$\Rightarrow \mathcal{F}\left[\int_{-\infty}^{t} x(\tau)\,d\tau\right] = \mathcal{F}[x(t) * u(t)]$$

$$= X(\omega)\cdot\left[\frac{1}{j\omega} + \pi\delta(\omega)\right]$$

$$= \frac{X(\omega)}{j\omega} + \pi X(\omega)\delta(\omega).$$

$$\boxed{\mathcal{F}\left[\int_{-\infty}^{t} x(\tau)\,d\tau\right] = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)} \quad \#$$

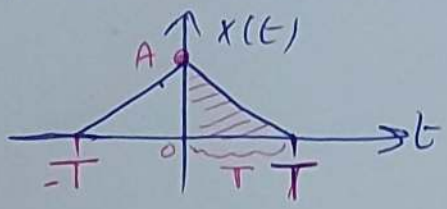
Ex: Find $\mathcal{F}[u(t)]$ using the time integration property.

Sol: $u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$.

$$\boxed{\mathcal{F}[u(t)] = \frac{1}{j\omega} + \pi\delta(\omega).} \quad \#$$

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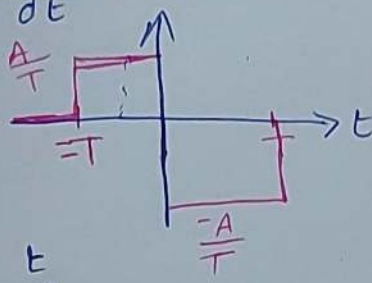
Ex: Find $X(w)$.



$$x(t) = A\ tri\left(\frac{t-0}{T}\right) = A\ tri\left(\frac{t}{T}\right).$$

⇒ Solⁿ

$$\frac{dx(t)}{dt} = g(t).$$



$$\Rightarrow g(t) = \frac{A}{T}\ rect\left(\frac{t+\frac{T}{2}}{T}\right) - \frac{A}{T}\ rect\left(\frac{t-\frac{T}{2}}{T}\right)$$

$$x(t) = \int_{-\infty}^{t} g(\tau)\, d\tau.$$

$$X(w) = \frac{G(w)}{jw} + \pi G(0)\,\delta(w).$$

$$\Rightarrow g(t) = \frac{A}{T}\ rect\left(\frac{t+T/2}{T}\right) - \frac{A}{T}\ rect\left(\frac{t-T/2}{T}\right)$$

$$G(w) = A\ sinc\left(T\frac{w}{2}\right) e^{jwT/2} - A\ sinc\left(T\frac{w}{2}\right) e^{-jwT/2}$$

$$= A\ sinc\left(T\frac{w}{2}\right)\left[e^{jwT/2} - e^{-jwT/2}\right] \frac{j2}{j2}\left(\frac{wT/2}{wT/2}\right)$$

$$= Aj\frac{wT}{2}\ sinc\left(T\frac{w}{2}\right)\left[\underbrace{\frac{e^{jwT/2} - e^{-jwT/2}}{2j\ wT/2}}_{sinc(Tw/2)}\right]$$

$$\boxed{G(w) = jwAT\ sinc^2\left(T\frac{w}{2}\right)}.$$

$$\Rightarrow X(w) = AT\ sinc^2\left(T\frac{w}{2}\right) + \pi \underbrace{G(0)}_{zero}\,\delta(w)$$

$$X(w) = AT\ sinc^2\left(T\frac{w}{2}\right)$$

$$\boxed{\begin{aligned} A\ tri\left(\frac{t}{T}\right) &\xleftrightarrow{F} AT\ sinc^2\left(T\frac{w}{2}\right) \\ tri\left(\frac{t}{T}\right) &\xleftrightarrow{F} T\ sinc^2\left(T\frac{w}{2}\right) \end{aligned}} \#$$

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Ex: Evaluate $\text{sinc}(t) * \text{sinc}^2(t)$

OR $\int_{-\infty}^{\infty} \text{sinc}^2(\tau)\ \text{sinc}(t-\tau)\, d\tau$.

Sol:

① $\text{rect}\left(\frac{t}{1}\right) \xleftrightarrow{\mathcal{F}} T\,\text{sinc}\left(T\frac{w}{2}\right)$ Duality #

$T\,\text{sinc}\left(T\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} 2\pi\,\text{rect}\left(\frac{w}{T}\right)$.

$T=2$, $*\frac{1}{2}$ $\Rightarrow$ $\text{sinc}(t) \xleftrightarrow{\mathcal{F}} \pi\,\text{rect}\left(\frac{w}{2}\right)$ #

② $\text{tri}\left(\frac{t}{T}\right) \xleftrightarrow{\mathcal{F}} T\,\text{sinc}^2\left(T\frac{w}{2}\right)$ $\Rightarrow$ Duality

$T\,\text{sinc}^2\left(T\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} 2\pi\,\text{tri}\left(\frac{w}{T}\right)$ $\Rightarrow$ even

$T=2$, $\times\frac{1}{2}$ $\Rightarrow$ $\text{sinc}^2(t) \xleftrightarrow{\mathcal{F}} \pi\,\text{tri}\left(\frac{w}{2}\right)$ #

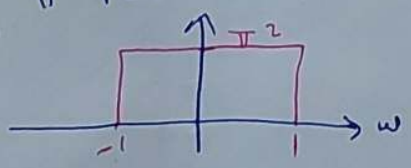
$\Rightarrow \text{sinc}(t) * \text{sinc}^2(t) \xleftrightarrow{\mathcal{F}} \pi\,\text{rect}\left(\frac{w}{2}\right) \cdot \pi\,\text{tri}\left(\frac{w}{2}\right)$

$= \pi^2\,\text{rect}\left(\frac{w}{2}\right)\text{tri}\left(\frac{w}{2}\right)$

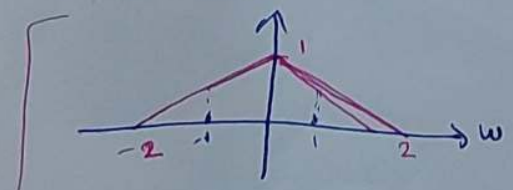
$\Downarrow$

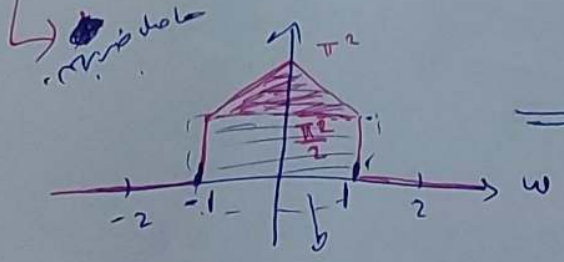
$= \frac{\pi}{2}^2\,\text{rect}\left(\frac{w}{2}\right) + \frac{\pi}{2}^2\,\text{tri}\left(\frac{w}{2}\right)$.

$\Rightarrow$ $\pi^2\,\text{rect}(w/2)$



$\text{tri}\left(\frac{w}{2}\right)$





$\Rightarrow \frac{\pi}{2}^2\,\text{rect}\left(\frac{w}{2}\right) + \frac{\pi}{2}^2\,\text{tri}(w)$

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$$\Rightarrow \quad T\,\mathrm{sinc}\left(T\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} 2\pi\,\mathrm{rect}\left(\frac{w}{T}\right)$$

خاصية الثنائية Duality.

$$\xrightarrow{T=2,\ *\pi/4} \boxed{\frac{\pi}{2}\,\mathrm{sinc}(t) \xleftrightarrow{\mathcal{F}} \frac{\pi}{2}^{2}\,\mathrm{rect}\left(\frac{w}{2}\right)} \quad \#$$

$$\Rightarrow \quad T\,\mathrm{sinc}^2\left(T\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} 2\pi\,\mathrm{tri}\left(\frac{w}{T}\right)$$

$$\xrightarrow{T=1,\ *\frac{\pi}{4}} \boxed{\frac{\pi}{4}\mathrm{sinc}^2\left(\frac{t}{2}\right) \xleftrightarrow{\mathcal{F}} \frac{\pi}{2}^{2}\,\mathrm{tri}(w).} \quad \#$$

$$\Rightarrow \quad \mathrm{sinc}(t) * \mathrm{sinc}^2(t) = \frac{\pi}{2}\mathrm{sinc}(t) + \frac{\pi}{4}\mathrm{sinc}^2\left(\frac{t}{2}\right).$$

(10) Frequency Differentiation:-

$$x(t) \xleftrightarrow{\mathcal{F}} X(w).$$

$$-jt\,x(t) \xleftrightarrow{\mathcal{F}} \frac{d\,X(w)}{dw}$$

$$(-jt)^n x(t) \xleftrightarrow{\mathcal{F}} \frac{d^n X(w)}{dw^n}$$

*** Proof:**

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w)\,e^{jwt}\,dw.$$

$$\Rightarrow \quad X(w) = \int_{-\infty}^{\infty} x(t)\,e^{-jwt}\,dt.$$

$$\frac{d\,X(w)}{dw} = \int_{-\infty}^{\infty} x(t)\,\frac{d\,e^{-jwt}}{dw}\,dt$$

$$\frac{d\,X(w)}{dw} = \int_{-\infty}^{\infty} x(t)\,e^{-jwt}(-jt)\,dt.$$

$$\frac{d\,X(w)}{dw} = \mathcal{F}\{-jt\,x(t)\} \rightarrow \boxed{\mathcal{F}^{-1}\left\{\frac{d\,X(w)}{dw}\right\} = -jt\,x(t)} \quad \#$$

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Ex: $t\,x(t) \xleftrightarrow{\mathcal{F}} j\frac{dX(w)}{dw}$.

Ex: ① DC level.

$$e^{jw_0 t} \xleftrightarrow{\mathcal{F}} \delta(w-w_0)$$

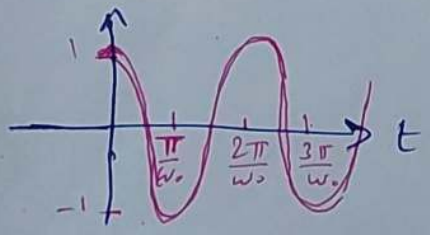
$$1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(w).$$

$$k \xleftrightarrow{\mathcal{F}} 2\pi k\,\delta(w)$$

② Switched Cosine.

⇒ Cosine wave that is turned on at $t=0$

⇒ $f(t) = \cos(w_0 t)\,u(t)$.



$w_0 t = \pm n\frac{\pi}{2} \Rightarrow t = \pm n\frac{\pi}{2w_0}$ ⇒ zeros

$$F(w) = \frac{1}{2}u(t)\,e^{jw_0 t} + \frac{1}{2}u(t)\,e^{-jw_0 t}$$

$$= \frac{1}{2}\left[u(t)\,e^{jw_0 t} + u(t)\,e^{-jw_0 t}\right]$$

$$= \frac{1}{2}\left[\frac{1}{j(w-w_0)} + \pi\delta(w-w_0) + \frac{1}{j(w+w_0)} + \pi\delta(w+w_0)\right]$$

$$= \frac{\pi}{2}\left[\delta(w-w_0)+\delta(w+w_0)\right] + \frac{1}{2}\frac{1}{j(w-w_0)}\,(w+w_0) + \frac{1}{2}\frac{1}{j(w+w_0)}\,(w-w_0)$$

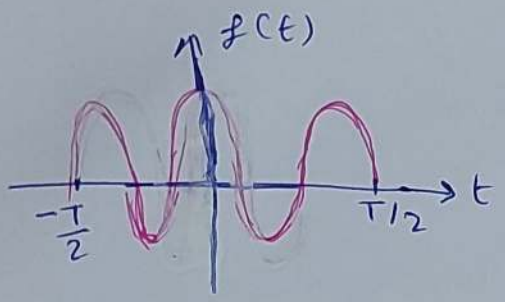
$$= \frac{\pi}{2}\left[\delta(w-w_0)+\delta(w+w_0)\right] + \frac{w}{j(w^2-w_0^2)}$$

$$\frac{2w}{2j(w^2-w_0^2)} = \frac{w}{j(w^2-w_0^2)}$$

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(3) Pulsed cosine:

$$f(t) = \cos(w_0 t)\ \text{rect}\left(\frac{t}{T}\right).$$

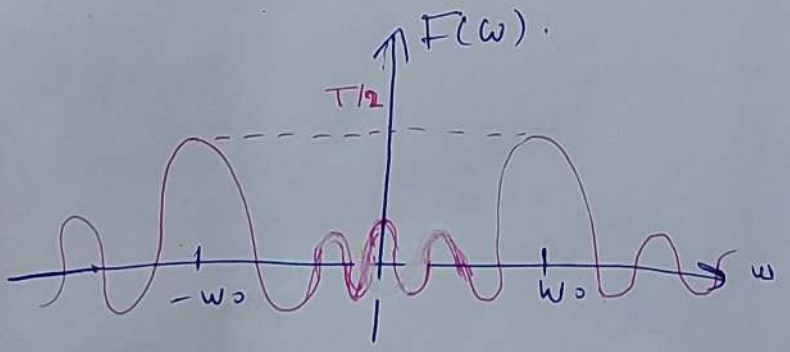


$$F(w) = \frac{1}{2\pi} F\{\cos(w_0 t)\} * F\left\{\text{rect}\left(\frac{t}{T}\right)\right\}$$

$$= \frac{1}{2\pi}\left[\pi(\delta(w-w_0)+\delta(w+w_0))\right) * T\ \text{sinc}\left(T\frac{w}{2}\right).$$

$$= \frac{1}{2}\left[\delta(w-w_0)+\delta(w+w_0)\right] * T\ \text{sinc}\left(T\frac{w}{2}\right)$$

$$= \frac{T}{2}\ \text{sinc}\left(\frac{T(w-w_0)}{2}\right) + \frac{T}{2}\ \text{sinc}\left(\frac{T(w+w_0)}{2}\right)$$



(4) Exponential pulse.

$$f(t) = e^{-at}\,u(t).$$

$$F(w) = \int_{-\infty}^{\infty} e^{-at}\,u(t)\,e^{-jwt}\,dt.$$

$$= \int_{0}^{\infty} e^{-at}\,e^{-jwt}\,dt = \int_{0}^{\infty} e^{-t(a+jw)}\,dt$$

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$$F(w) = \frac{e^{-t(a+jw)}}{-(a+jw)} \Big]_0^{\infty}$$

$$= \frac{1}{a+jw}$$

$$\Rightarrow \left[\begin{array}{l} e^{-at} u(t) \xleftrightarrow{F} \frac{1}{a+jw} \\ \quad x(t) \\ e^{at} u(-t) \xleftrightarrow{F} \frac{1}{a-jw} \\ \quad x(-t) \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{l} x(t) \xleftrightarrow{F} X(w) \\ x(-t) \xleftrightarrow{F} \frac{1}{|a|} X\left(\frac{w}{a}\right) = X(-w) \end{array} \right] \#$$

(5) $x(t) = e^{-a|t|}$, $a > 0$

$$X(w) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-jwt} \, dt.$$

$$= \int_{-\infty}^{0} e^{at} e^{-jwt} \, dt + \int_{0}^{\infty} e^{-at} e^{-jwt} \, dt.$$

$$= \int_{-\infty}^{0} e^{(a-jw)t} \, dt + \int_{0}^{\infty} e^{-(a+jw)t} \, dt$$

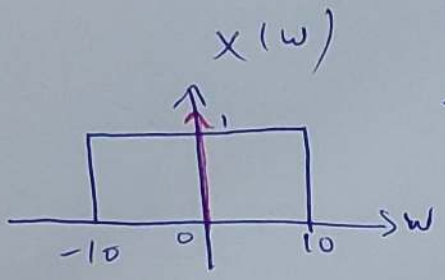
$$= \frac{e^{(a-jw)t}}{a-jw} \Big]_{-\infty}^{0} + \frac{e^{-(a+jw)t}}{-(a+jw)} \Big]_{0}^{\infty}$$

$$= \frac{1}{a-jw} + \frac{1}{a+jw}$$

$$= \frac{a+jw+a-jw}{a^2+w^2} = \frac{2a}{a^2+w^2}, \quad a > 0$$

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Ex:



Find x(t).

$$X(w) = \text{rect}\left(\frac{w}{20}\right) + \delta(w)$$

$$\text{rect}\left(\frac{t}{T}\right) \xleftrightarrow{F} T\,\text{sinc}\left(T\frac{w}{2}\right)$$

$$T\,\text{sinc}\left(t\frac{T}{2}\right) \xleftrightarrow{F} 2\pi\,\text{rect}\left(\frac{w}{T}\right) \quad \Rightarrow \; T=20 \;\; \times \frac{1}{2\pi}$$

$$\frac{20}{2\pi}\,\text{sinc}\left(t(10)\right) \xleftrightarrow{F} \text{rect}\left(\frac{w}{20}\right)$$

$$\left[\frac{10}{\pi}\,\text{sinc}(10t) \xleftrightarrow{F} \text{rect}\left(\frac{w}{20}\right)\right] \#$$

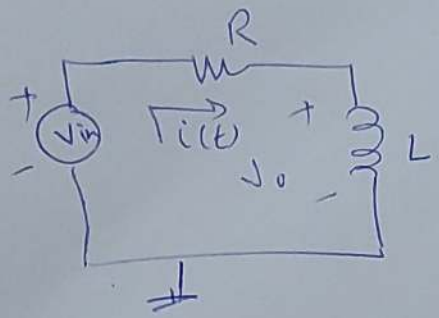
$$1 \xleftrightarrow{F} 2\pi\delta(w)$$

$$\left[\delta(w) \Leftrightarrow \frac{1}{2\pi}\right]$$

$$\Rightarrow \boxed{x(t) = \frac{1}{2\pi} + \frac{10}{\pi}\,\text{sinc}(10t)} \#$$

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Circuit Application of the Fourier Transform.



$$V_{in}(t) = R\,i(t) + L\frac{d\,i(t)}{dt} \Rightarrow (1)$$

$$V_o(t) = L\frac{d\,i(t)}{dt} \Rightarrow (2)$$

⇒ Fourier Transform used to simplify the calculation of the response of linear system to input signals.

⇒ using Fourier Transform:-

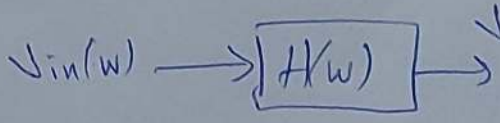
$$V_{in}(w) = R\,I(w) + jw\,L\,I(w).$$

$$V_o(w) = jw\,L\,I(w).$$

$$I(w) = \frac{V_{in}(w)}{R + jwL}$$

$$V_o(w) = \frac{jwL}{R + jwL} V_{in}(w).$$

$$V_o(w) = H(w)\,V_{in}(w)$$



$$V_o(w) = V_{in}(w)\,H(w).$$

$$H(w) = \frac{V_o(w)}{V_{in}(w)}$$

$$H(w) = \frac{jwL}{R + jwL}$$

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⇒ $H(w)$: is the transfer function.
or frequency response function.

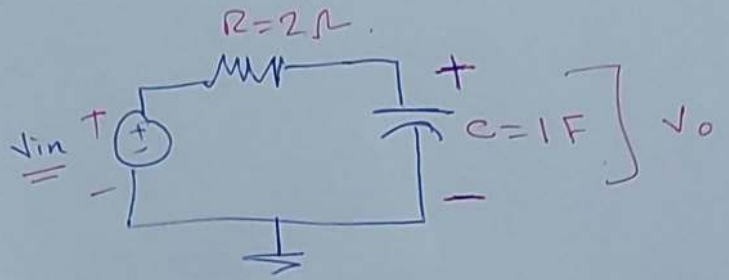
⇒ if the input is impulse function.

$v_{in}(t) = \delta(t) \rightarrow V_{in}(w) = 1$

⇒ $V_o(w) = H(w)$

⇒ $h(t) = \mathcal{F}^{-1}\{H(w)\}$.

Ex: Find $v_o(t)$ for the following circuit if $v_{in}(t) = 2e^{-3t}u(t)$.



⇒ Using F.T

$V_o(w) = V_{in}(w)\, H(w)$.

$H(w) = \dfrac{V_o(w)}{V_{in}(w)}$.

$$\left[V_{in}(w) = \frac{2}{3+jw} \right] \#.$$

$$V_o(w) = \frac{V_{in}(w)\ \frac{1}{jwc}}{R + 1/jwc} \quad \Big] \Rightarrow \text{voltage divider.}$$

$R=2\,\Omega$ & $C=1$ $\Rightarrow$

$$\left[V_o(w) = V_{in}(w)\ \frac{1/jw}{2 + 1/jw} \right]$$

$$\left[H(w) = \frac{V_o(w)}{V_{in}(w)} = \frac{1}{1+2jw} \right]$$

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$$\Rightarrow V_o(w) = \frac{2}{(3+jw)(1+2jw)} = \frac{2}{(3+jw)\,2(\frac{1}{2}+jw)}$$

$$V_o(w) = \frac{1}{(3+jw)(\frac{1}{2}+jw)}$$

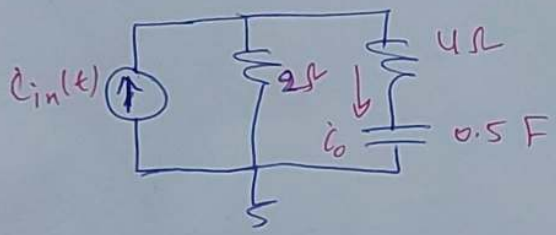
⇒ Using partial fraction.

$$V_o(w) = \frac{-0.4}{3+jw} + \frac{0.4}{\frac{1}{2}+jw} = 0.4\left(\frac{1}{\frac{1}{2}+jw} - \frac{1}{3+jw}\right).$$

$$\mathcal{F}^{-1}$$

$$\boxed{v_o(t) = 0.4\left(e^{-1/2\,t} - e^{-3t}\right)u(t)} \quad \#.$$

EX° Find $i_o(t)$ for the following circuit, if $i_{in}(t) = 10\sin(2t)$



$$\Rightarrow H(w) = \frac{I_o(w)}{I_{in}(w)} \Rightarrow I_o(w) = \frac{I_{in}(w)\,(2)}{2+4+\frac{2}{jw}}$$ ⇒ current divider.

$$I_{in}(w) = \mathcal{F}\left[10\sin(2t)\right] \qquad \frac{I_o(w)}{I_{in}(w)} = \frac{2}{6+\frac{2}{jw}}$$

$$= 10\,j\pi\left[\delta(w+2) - \delta(w-2)\right]. \qquad \boxed{H(w) = \frac{jw}{1+3jw}}$$

$$\Rightarrow I_o(w) = H(w)\,I_{in}(w)$$

$$I_o(w) = \frac{10\pi jw\left[\delta(w-2) - \delta(w+2)\right]}{1+3jw}$$

$$i_o(t) = \mathcal{F}^{-1}\left\{I_o(w)\right\}$$

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$$\Rightarrow i_o(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{10\pi w\,\delta(w-2)}{1+3jw} e^{jwt}\,dw - \int_{-\infty}^{\infty} \frac{10\pi w\,\delta(w+2)}{1+3jw} e^{jwt}\,dw$$

$$= \frac{1}{2\pi}\left[\int_{-\infty}^{\infty} \frac{10\pi(2)\,\delta(w-2)}{1+6j} e^{j2t}\,dw - \int_{-\infty}^{\infty} \frac{10\pi(-2)\,\delta(w+2)}{1-6j} e^{-j2t}\,dw\right]$$

$$= \frac{1}{2\pi}\left[\frac{20\pi}{1+6j} e^{j2t}\underbrace{\int_{-\infty}^{\infty}\delta(w-2)\,dw}_{1} + \frac{20\pi}{1-6j} e^{-j2t}\underbrace{\int_{-\infty}^{\infty}\delta(w+2)\,dw}_{1}\right]$$

$$= 10\left[\frac{e^{j2t}}{1+6j} + \frac{e^{-j2t}}{1-6j}\right]$$

$$\Rightarrow 1+j6 = \sqrt{37}\angle 80.5^\circ \approx 6.1\, e^{j80.5^\circ}$$

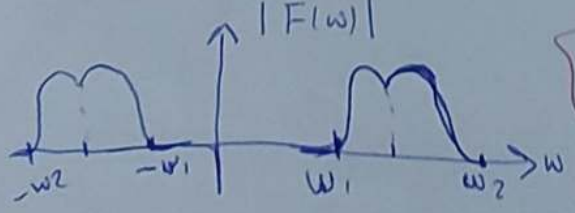
$$= 10\left[\frac{1}{6.1} e^{j(2t-80.5)} + \frac{1}{6.1} e^{-j(2t-80.5^\circ)}\right]$$

$$= 1.64\left[e^{j(2t-80.5)} + e^{-j(2t-80.5)}\right]\frac{2}{2}$$

$$\boxed{i_o(t) = 3.28\cos(2t-80.5^\circ)}\ \#$$

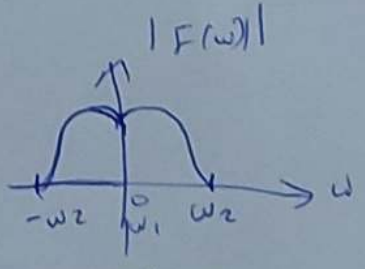
* Bandwidth Relationships.

BW is defined by the range of positive frequencies for which the frequency spectrum of the signal is non zero.



Band pass signal.

$$BW = w_2 - w_1$$



$w_1 = 0$

(Baseband signal)

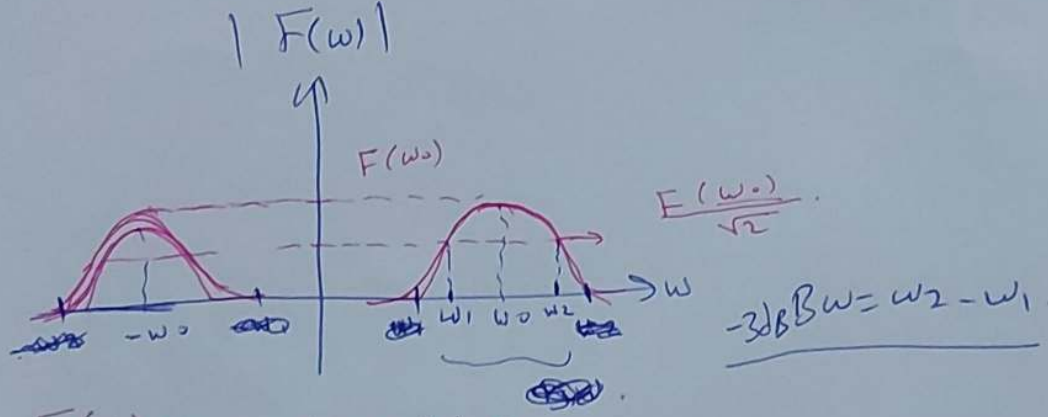
$BW = w_2$.

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⇒ 3-dB Bandwidth. OR Half-power bandwidth.

⇒ is defined as the range of frequencies for which the magnitude of the frequency spectrum is no less than $\frac{1}{\sqrt{2}}$ times of maximum value within the range.

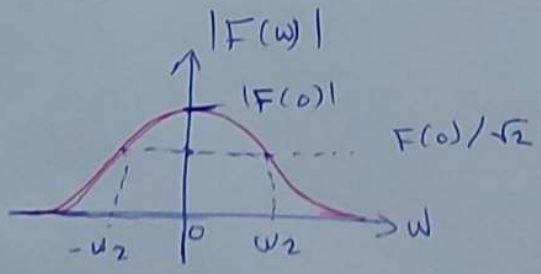
$$-3\,dB = 20 \log_{10}\left(\frac{1}{\sqrt{2}}\right).$$



$-3dB\ Bw = w_2 - w_1$.

$F(w_0)$ ⇒ maximum value.

$\frac{F(w_0)}{\sqrt{2}}$ → $-3\,dB$.



$-3dB\ Bw = w_2$.

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* Chapter 5 ⇒ Table 5.2 : Fourier Transform pairs. Page 223 | Table 5.1 page 207. properties.

* Chapter 7: The Laplace Transform: (LT)

⇒ Laplace transform is the general case of Fourier transform.

$$f(t) \Longleftrightarrow L\{f(t)\} = F(s).$$

$$F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt.$$

$s = \sigma + jw$ → complex variable / frequency.

$$F(w) = \int_{-\infty}^{\infty} f(t)\, e^{-jwt}\, dt. \rightarrow F.T. \rightarrow w \rightarrow \text{real value / frequency}$$

⇒ $F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt$ ⇒ Bilateral Laplace Transform. (Two-sided LT).

$F(s) = L[f(t)]$.

$f(t) = L^{-1}[F(s)]$. ⇒ Inverse Laplace Transform.

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - jw}^{\sigma + jw} F(s)\, e^{st}\, ds.$$ ⇒ Usually we use the partial fractions.

⇒ if $f(t)$, $t > 0$.

$$F(s) = \int_{0}^{\infty} f(t)\, e^{-st}\, dt$$ ⇒ Unilateral LT. (single-sided LT).

⇒ For $f(t) \rightarrow F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-(\sigma + jw)t}\, dt$

$$= \int_{-\infty}^{\infty} f(t)\, e^{-\sigma t}\, e^{-jwt}\, dt$$

$$= \int_{-\infty}^{\infty} f_1(t)\, e^{-jwt}\, dt \Rightarrow \int_{-\infty}^{\infty} |f_1(t)|\, dt < \infty$$

$F(s) = F\{f(t)\, e^{-\sigma t}\}$

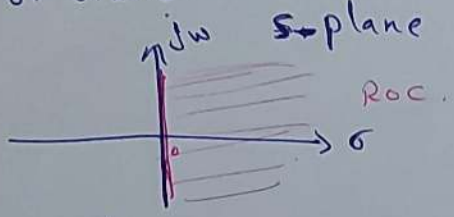
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$\Rightarrow \int_{-\infty}^{\infty} |f(t) e^{-\sigma t}| dt < \infty \rightarrow$ range of $\sigma \Rightarrow$ RoC.

if $\sigma > 0 \Rightarrow \int_{-\infty}^{\infty} |f(t) e^{-\sigma t}| dt < \infty$ (#)

RoC: Region of Convergence $\Rightarrow$ where the LT of $x(t)$ is exists or finite.

$s = \sigma + j\omega$



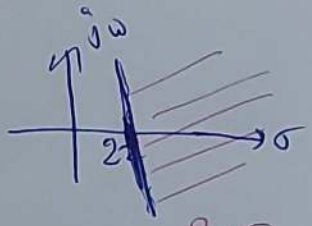
Ex: RoC of $f(t) = e^{2t} u(t)$.

$\Rightarrow \int_{-\infty}^{\infty} |f(t) e^{-\sigma t}| dt = \int_{-\infty}^{\infty} |e^{2t} e^{-\sigma t} u(t)| dt$

$= \int_{0}^{\infty} e^{(2-\sigma)t} dt.$

$\Rightarrow \int_{0}^{\infty} |e^{(2-\sigma)t}| dt < \infty \Rightarrow$ if $2 - \sigma < 0$

$\boxed{\sigma > 2}$



Ex:-

$\Rightarrow x(t) = e^{-at} u(t).$

Sol: $X(s) = \int_{-\infty}^{\infty} e^{-at} \cdot e^{-st} u(t) dt.$

$= \int_{0}^{\infty} e^{-t(a+s)} dt$

$= \frac{e^{-(a+s)t}}{-(a+s)} \Big]_0^{\infty}$

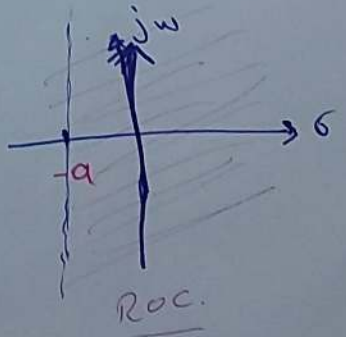
$\boxed{X(s) = \frac{1}{a+s}}$ #

$\mathcal{L}\{e^{-at} u t\} \longleftrightarrow \frac{1}{a+s}$

RoC: $Re\{s\} > -a$

$s + a > 0$

$\boxed{s > -a}$



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Ex: Find $F(s)$ for $f(t) = u(t)$.

$$\Rightarrow F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt.$$

$$= \int_{0}^{\infty} u(t)\, e^{-st}\, dt.$$

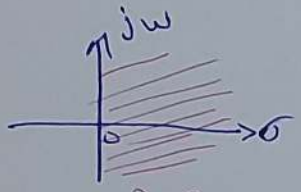
$$= \frac{e^{-st}}{-s}\Big]_{0}^{\infty} = \frac{1}{s} - \underbrace{e^{-s\infty}}_{\text{zero.}} \rightarrow e^{-s\infty} = e^{-\infty(\sigma + j\omega)}$$

$$= e^{-\infty\sigma} \cdot e^{-j\omega\infty}$$

$$\sim \text{zero}$$

$$F(s) = \frac{1}{s}$$

ROC: $Re\{s\} > 0$



ROC.

$$\Rightarrow \left\{ \begin{array}{l} u(t) \overset{L}{\longleftrightarrow} \frac{1}{s} \\ e^{-at} \overset{L}{\longleftrightarrow} \frac{1}{s+a} \\ e^{at} \overset{L}{\longleftrightarrow} \frac{1}{s-a} \end{array} \right\} \#$$

Ex: Find $F(s)$ for $f(t) = \delta(t)$.

$$\Rightarrow F(s) = \int_{-\infty}^{\infty} \delta(t)\, e^{-st}\, dt \quad (\delta(t) \neq 0 \text{ only at } t = 0)$$

$$= \int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

$$F(s) = 1$$

$$\delta(t) \overset{L}{\longleftrightarrow} 1$$

* if $\delta(t - t_0) \overset{L}{\longleftrightarrow} e^{-at_0}$

* Ex: Find RoC for $x(t) = 3e^{-2t}u(t) + 4e^{t}u(-t)$

$$X(s) = 3\left(\frac{1}{s+2}\right) + 4\left(\frac{1}{1-s}\right)$$

$$= \frac{3}{s+2} + \frac{4}{1-s}$$

$$= \frac{3}{s+2} - \frac{4}{s-1}$$

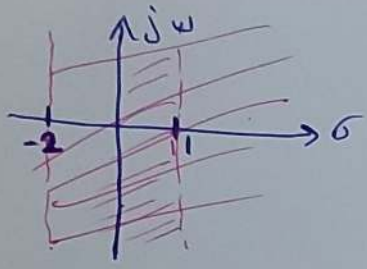
$ROC_1 \Rightarrow (s+2) > 0$

$Re\{s\} > -2$

$ROC_2 \Rightarrow (s-1) < 0$

$Re\{s\} < 1$

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⇒ 

∴ RoC ⇒ $-2 < Re\{s\} < 1$

*** Properties of LT** ⇒ ① Linearity.

⇒ If $f_1(t) \xleftrightarrow{L} F_1(s)$.

$a_1 f_1(t) \xleftrightarrow{L} a_1 F_1(s)$.

$f_2(t) \xleftrightarrow{L} F_2(s)$.

$a_2 f_2(t) \xleftrightarrow{L} a_2 F_2(s)$.

$$[a_1 f_1(t) + a_2 f_2(t)] \xleftrightarrow{L} a_1 F_1(s) + a_2 F_2(s)$$

* Linearity property. ##

Ex: Find F(s) for $f(t) = \cos(at)$.

⇒ $$\cos(at) = \frac{e^{jat} + e^{-jat}}{2}$$

$$= \frac{1}{2} e^{jat} + \frac{1}{2} e^{-jat}$$

$$L[\cos(at)] = L\left\{\frac{1}{2} e^{jat}\right\} + L\left[\frac{1}{2} e^{-jat}\right]$$

$$= \frac{1}{2}\left(\frac{1}{s - ja}\right) + \frac{1}{2}\left(\frac{1}{s + ja}\right)$$

$$= \frac{1}{2}\left[\frac{s + ja + s - ja}{(s+ja)(s-ja)}\right]$$

$$= \frac{1}{2}\left[\frac{2s}{s^2 + a^2}\right]$$

$$\boxed{L\{\cos(at)\} = \frac{s}{s^2 + a^2}} \; \#$$

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Ex: Find Laplace Transform for $x(t) = \sin(bt)$.

$$\Rightarrow \quad \sin(bt) = \frac{1}{2j}\left[e^{jbt} - e^{-jbt} \right]$$

$$\mathcal{L}\left[\sin(bt)\right] = \mathcal{L}\left[\frac{1}{2j} e^{jbt}\right] - \mathcal{L}\left[\frac{1}{2j} e^{-jbt}\right]$$

$$= \frac{1}{2j}\left(\frac{1}{s-jb}\right) - \frac{1}{2j}\left(\frac{1}{s+bj}\right)$$

$$= \frac{s+jb-(s-jb)}{2j\,(s-jb)(s+jb)}$$

$$= \frac{2jb}{2j\,(s^2+b^2)}$$

$$\boxed{\mathcal{L}\left[\sin(bt)\right] = \frac{b}{s^2+b^2}} \quad \#$$

(2) S-Domain Shifting $\Rightarrow$ $x(t) \xleftrightarrow{\mathcal{L}} X(s)$

$e^{at}\,x(t) \xleftrightarrow{\mathcal{L}} X(s-a)$

$e^{-at}\,x(t) \xleftrightarrow{\mathcal{L}} X(s+a)$

Ex: $e^{-at}\cos(bt)$ $\Rightarrow$ Find the LT

$$\cos(bt)\,e^{-at} = \frac{1}{2} e^{jbt} e^{-at} + \frac{1}{2} e^{-jbt} e^{-at}$$

$$= \frac{1}{2} e^{t(jb-a)} + \frac{1}{2} e^{-t(a+jb)}$$

$$\mathcal{L}\left\{\cos(bt)\,e^{-at}\right\} = \frac{1}{2}\left(\frac{1}{s-(jb-a)}\right) + \frac{1}{2}\left(\frac{1}{s+(a+jb)}\right)$$

$(s+a-jb)$ $\qquad$ $(s+a+jb)$

$$= \frac{1}{2}\left[\frac{s+a+jb+s+a-jb}{(s+a)^2+b^2}\right]$$

$$= \frac{1}{2}\left[\frac{2(s+a)}{(s+a)^2+b^2}\right]$$

$$\boxed{\mathcal{L}\left\{\cos(bt)\,e^{-at}\right\} = \frac{s+a}{(s+a)^2+b^2}} \quad \#$$

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$\Rightarrow f(t)\, e^{-at} \xleftrightarrow{L} F(s+a)$.

(3) Time-shifting OR real-shifting.

$$u(t-t_0) \xleftrightarrow{L} \frac{1}{s} e^{-t_0 s}$$

$$f(t-t_0) \xleftrightarrow{L} F(s)\, e^{-t_0 s}$$

Ex: Find LT for $u(t) - u(t-2a)$

$\Rightarrow L\{u(t)\} - L\{u(t-2a)\}$

$= \frac{1}{s} - \frac{e^{-2as}}{s}$

Ex: Find LT for $f(t) = 5\, e^{-0.3(t-2)}\, u(t-2)$

$\Rightarrow F(s) = 5\left(\frac{1}{s+0.3}\right) e^{-2s}$

$= \frac{5e^{-2s}}{s+0.3}$

(4) Time-Scalling:-

$x(t) \xleftrightarrow{L} X(s)$.

$x(at) \xleftrightarrow{L} \frac{1}{(a)} X\left(\frac{s}{a}\right)$.

Ex: If $x(t) = \sin(3t)$ Find LT for $x_1(t) = \sin\left(3\left(4t - \frac{\pi}{6}\right)\right)$

a: Time Scalling; Time shifting

$\Rightarrow X(s) = L[\sin(3t)] = \frac{3}{s^2+9}$

$x_1(t) \xleftrightarrow{L} \frac{1}{a}\left[X\left(\frac{s}{a}\right)\right]$

$L\left\{\sin\left(3\left(4t - \frac{\pi}{6}\right)\right)\right\} = \frac{1}{4}\left(\frac{3}{(s/4)^2+9}\right) e^{-s\pi/24}$

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$$\Rightarrow \frac{12\, e^{-s\pi/24}}{s^2+144}$$

(5) Time Differentiation:

$$f(t) \xleftrightarrow{\mathcal{L}} F(s)$$

$$\frac{df(t)}{dt} \xleftrightarrow{\mathcal{L}} sF(s) - f(0)$$

$$\frac{d^n f(t)}{d^n t} \xleftrightarrow{\mathcal{L}} s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - s f^{(n-2)}(0) - f^{(n-1)}(0)$$

Ex: Using the time differentiation property to find the LT for $\sin(bt)$

$$\Rightarrow \quad \sin(bt) = -\frac{1}{b}\frac{d}{dt}(\cos(bt))$$

$$\mathcal{L}\{\sin(bt)\} = \mathcal{L}\left\{-\frac{1}{b}\frac{d}{dt}(\cos(bt))\right\}$$

$$= -\frac{1}{b}\left[s\mathcal{L}\{\cos(bt)\} - \cos(bt)\Big|_{t=0}\right]$$

$$= -\frac{1}{b}\left[s\left(\frac{s}{s^2+b^2}\right) - 1\right]$$

$$\left[\mathcal{L}\{\sin bt\} = \frac{b}{s^2+b^2}\right] \#.$$

(6) Time ~~differentiation~~ integration

~~$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} \longleftrightarrow sF(s) - f(0)$~~

$$f(t) = \int_0^t g(\tau)\, d\tau$$

$$\mathcal{L}\left\{\int_0^t g(\tau)\, d\tau\right\} = \frac{1}{s}F(s).$$

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Ex: Find LT for ramp function.

- $\int_0^t u(\tau)\,d\tau = t$

$$\Rightarrow \mathcal{L}\{t\} = \mathcal{L}\left[\int_0^t u(\tau)\,d\tau\right].$$

$$= \frac{1}{s}\,\frac{1}{s}$$

$$= \frac{1}{s^2}$$

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* Chapter 5 ⇒ Table 5.2 : Fourier Transform pairs. Page 223 | Table 5.1 page 207. properties.

* Chapter 7: The Laplace Transform: (LT)

⇒ Laplace transform is the general case of Fourier transform.

$$f(t) \Leftrightarrow \mathcal{L}\{f(t)\} = F(s).$$

$$F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt.$$

$s = \sigma + jw$. → complex variable / frequency.

$$F(w) = \int_{-\infty}^{\infty} f(t)\, e^{-jwt}\, dt. \rightarrow \text{F.T.} \rightarrow w \rightarrow \text{real value / frequency}$$

⇒ $F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt$ ⇒ Bilateral Laplace Transform. (Two-sided LT).

$F(s) = \mathcal{L}[f(t)]$.

$f(t) = \mathcal{L}^{-1}[F(s)]$. ⇒ Inverse Laplace Transform.

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - jw}^{\sigma + jw} F(s)\, e^{st}\, ds.$$ ⇒ Usually we use the partial fractions.

⇒ if $f(t)$, $t > 0$.

$$F(s) = \int_{0}^{\infty} f(t)\, e^{-st}\, dt$$ ⇒ Unilateral LT. (single-sided LT).

⇒ For $f(t) \rightarrow F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-(\sigma + jw)t}\, dt$

$$= \int_{-\infty}^{\infty} f(t)\, e^{-\sigma t}\, e^{-jwt}\, dt$$

$$= \int_{-\infty}^{\infty} f_1(t)\, e^{-jwt}\, dt \Rightarrow \int_{-\infty}^{\infty} |f_1(t)|\, dt < \infty$$

$F(s) = F\{f(t)\, e^{-\sigma t}\}$

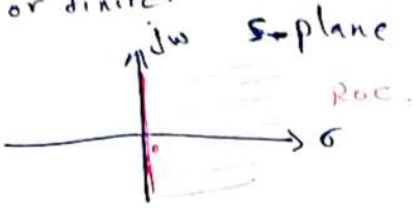
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$\Rightarrow \int_{-\infty}^{\infty} |f(t)\, e^{-\sigma t}|\, dt < \infty \rightarrow$ range of $\sigma \Rightarrow$ RoC.

if $\sigma > 0 \Rightarrow \int |f(t)\, e^{\sigma t}|\, dt < \infty$ (*)

RoC: Region of Convergence $\Rightarrow$ where the LT of $x(t)$ is exists or finite.

$s = \sigma + j\omega$



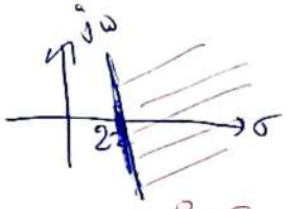
Ex: RoC of $f(t) = e^{2t} u(t)$.

$\Rightarrow \int_{-\infty}^{\infty} |f(t)\, e^{-\sigma t}|\, dt = \int_{-\infty}^{\infty} |e^{2t}\, e^{-\sigma t}\, u(t)|\, dt$

$= \int_{0}^{\infty} e^{(2-\sigma)t}\, dt.$

$\Rightarrow \int_{0}^{\infty} |e^{(2-\sigma)t}|\, dt < \infty \Rightarrow$ if $2 - \sigma < 0$

$\boxed{\sigma > 2}$



Ex:-

$\Rightarrow x(t) = e^{-at} u(t).$

Sol: $X(s) = \int_{-\infty}^{\infty} e^{-at}\, e^{-st}\, u(t)\, dt.$

$= \int_{0}^{\infty} e^{-t(a+s)}\, dt$

$= \frac{e^{-(a+s)t}}{-(a+s)} \Big]_{0}^{\infty}$

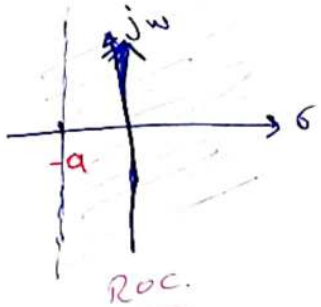
$\boxed{X(s) = \frac{1}{a+s}}$ #

$\mathcal{L}\{e^{-at} u(t)\} \longleftrightarrow \frac{1}{a+s}$

RoC: $\mathrm{Re}\{s\} > -a$

$s + a > 0$

$\boxed{s > -a}$



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Ex: Find $F(s)$ for $f(t) = u(t)$.

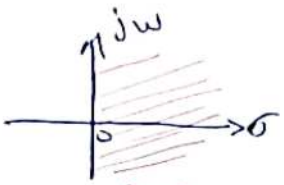
$$\Rightarrow F(s) = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt$$

$$= \int_{0}^{\infty} u(t)\, e^{-st}\, dt$$

$$= \frac{e^{-st}}{-s}\Big|_0^{\infty} = \frac{1}{s} - \underbrace{e^{-s\infty}}_{\text{zero.}} \rightarrow e^{-s\infty} = e^{-\omega(\sigma + j\omega)} = e^{-\omega\sigma}\cdot e^{-j\omega\infty}$$

$$F(s) = \frac{1}{s}$$

ROC: $Re\{s\} > 0$



ROC.

$$\Rightarrow \left\{ \begin{array}{l} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \\ e^{-at} \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a} \\ e^{at} \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-a} \end{array} \right\} \#$$

Ex: Find $F(s)$ for $f(t) = \delta(t)$.

$$\Rightarrow F(s) = \int_{-\infty}^{\infty} \delta(t)\, e^{-st}\, dt \quad (e^{-s0} = 1)$$

$$= \int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

$$F(s) = 1$$

$$\delta(t) \overset{\mathcal{L}}{\longleftrightarrow} 1$$

* if $\delta(t - t_0) \overset{\mathcal{L}}{\longleftrightarrow} e^{-at_0}$.

* Ex: Find ROC for $x(t) = 3e^{-2t}u(t) + 4e^{t}u(-t)$

$$X(s) = 3\left(\frac{1}{s+2}\right) + 4\left(\frac{1}{1-s}\right)$$

$$= \frac{3}{s+2} + \frac{4}{1-s}$$

$$= \frac{3}{s+2} - \frac{4}{s-1}$$

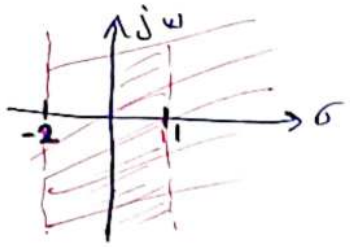
$ROC_1 \Rightarrow \{s+2\} > 0$

$Re\{s\} > -2$

$ROC_2 \Rightarrow (s-1) < 0$

$Re\{s\} < 1$

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⇒ 

∴ ROC ⇒ $-2 < Re\{s\} < 1$

*** Properties of LT** ⇒ ① Linearity.

⇒ If $f_1(t) \xleftrightarrow{L} F_1(s)$.

$a_1 f_1(t) \xleftrightarrow{L} a_1 F_1(s)$.

$f_2(t) \xleftrightarrow{L} F_2(s)$.

$a_2 f_2(t) \xleftrightarrow{L} a_2 F_2(s)$.

$$[a_1 f_1(t) + a_2 f_2(t)] \xleftrightarrow{L} a_1 F_1(s) + a_2 F_2(s)$$

* Linearity property. ##

Ex: Find F(s) for $f(t) = \cos(at)$.

⇒ $$\cos(at) = \frac{e^{jat} + e^{-jat}}{2}$$

$$= \frac{1}{2} e^{jat} + \frac{1}{2} e^{-jat}$$

$$L[\cos(at)] = L\left\{\frac{1}{2} e^{jat}\right\} + L\left[\frac{1}{2} e^{-jat}\right]$$

$$= \frac{1}{2}\left(\frac{1}{s - ja}\right) + \frac{1}{2}\left(\frac{1}{s + ja}\right)$$

$$= \frac{1}{2}\left[\frac{s + ja + s - ja}{(s+ja)(s-ja)}\right]$$

$$= \frac{1}{2}\left[\frac{2s}{s^2 + a^2}\right]$$

$$\boxed{L\{\cos(at)\} = \frac{s}{s^2 + a^2}} \#$$

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Ex: Find Laplace Transform for $x(t) = \sin(bt)$.

$$\Rightarrow \quad \sin(bt) = \frac{1}{2j}\left[e^{jbt} - e^{-jbt}\right]$$

$$\mathcal{L}\left[\sin(bt)\right] = \mathcal{L}\left[\frac{1}{2j} e^{jbt}\right] - \mathcal{L}\left[\frac{1}{2j} e^{-jbt}\right]$$

$$= \frac{1}{2j}\left(\frac{1}{s-jb}\right) - \frac{1}{2j}\left(\frac{1}{s+bj}\right)$$

$$= \frac{s+jb-(s-jb)}{2j\,(s-jb)(s+jb)}$$

$$= \frac{2jb}{2j(s^2+b^2)}$$

$$\boxed{\mathcal{L}\left[\sin(bt)\right] = \frac{b}{s^2+b^2}} \quad \#$$

(2) S-Domain Shifting $\Rightarrow$ $x(t) \xleftrightarrow{\mathcal{L}} X(s)$

$e^{at}x(t) \xleftrightarrow{\mathcal{L}} X(s-a)$

$e^{-at}x(t) \xleftrightarrow{\mathcal{L}} X(s+a)$

Ex: $e^{-at}\cos(bt)$ $\Rightarrow$ Find the LT

$$\cos(bt)\,e^{-at} = \frac{1}{2} e^{jbt} e^{-at} + \frac{1}{2} e^{-jbt} \cdot e^{-at}$$

$$= \frac{1}{2} e^{t(jb-a)} + \frac{1}{2} e^{-t(a+jb)}$$

$$\mathcal{L}\left\{\cos(bt)\,e^{-at}\right\} = \frac{1}{2}\left(\frac{1}{s-(jb-a)}\right) + \frac{1}{2}\left(\frac{1}{s+(a+jb)}\right)$$

$(s+a-jb)$ $\qquad$ $(s+a+jb)$

$$= \frac{1}{2}\left[\frac{s+a+jb+s+a-jb}{(s+a)^2+b^2}\right]$$

$$= \frac{1}{2}\left[\frac{2(s+a)}{(s+a)^2+b^2}\right]$$

$$\boxed{\mathcal{L}\left\{\cos(bt)e^{-at}\right\} = \frac{s+a}{(s+a)^2+b^2}} \quad \#$$

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$\Rightarrow \quad f(t)\, e^{-at} \xleftrightarrow{L} F(s+a)$.

(3) Time-shifting OR real-shifting.

$$u(t-t_0) \xleftrightarrow{L} \frac{1}{s} e^{-t_0 s}$$

$$f(t-t_0) \xleftrightarrow{L} F(s)\, e^{-t_0 s}$$

$$f(t+t_0) \xleftrightarrow{L} F(s)\, e^{st}$$

Ex: Find LT for $u(t) - u(t-2a)$

$$\Rightarrow L\{u(t)\} - L\{u(t-2a)\}$$

$$= \frac{1}{s} - \frac{e^{-2as}}{s}$$

Ex: Find LT for $f(t) = 5\, e^{-0.3(t-2)}\, u(t-2)$

$$\Rightarrow F(s) = 5\left(\frac{1}{s+0.3}\right) e^{-2s}$$

$$= \frac{5e^{-2s}}{s+0.3}$$

(4) Time-Scalling:-

$$x(t) \xleftrightarrow{L} X(s).$$

$$x(at) \xleftrightarrow{L} \frac{1}{(a)} X\left(\frac{s}{a}\right). \rightarrow \text{Unilateral LT.} \quad a>0$$

Ex: If $x(t) = \sin(\beta t)$ Find LT for $x_1(t) = \sin\left(3\left(4t - \frac{\pi}{6}\right)\right)$

a: Time scalling; Time shifting

$$\Rightarrow X(s) = L[\sin(\beta t)] = \frac{3}{s^2+9}$$

$$x_1(t) \xleftrightarrow{L} \frac{1}{a}\left[X\left(\frac{s}{a}\right)\right]$$

$$L\left\{\sin\left(3\left(4t - \frac{\pi}{6}\right)\right)\right\} = \frac{1}{4}\left(\frac{3}{(s/4)^2+9}\right) e^{-s\pi/24}$$

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$$\Rightarrow \frac{12\, e^{-s\pi/24}}{s^2+144} \quad \#$$

(5) Time Differentiation:

$$f(t) \xleftrightarrow{L} F(s)$$

$$\frac{df(t)}{dt} \xleftrightarrow{L} sF(s) - f(0^+)$$

$$\frac{d^n f(t)}{d^n t} \xleftrightarrow{L} s^n F(s) - s^{n-1} f(0^+) - s^{n-2} f'(0^+) - \dots - s f^{(n-2)}(0) - f^{(n-1)}(0^+)$$

Ex: Using the time differentiation property to find the LT for $\sin(bt)$

$$\Rightarrow \quad \sin(bt) = -\frac{1}{b}\frac{d}{dt}(\cos(bt))$$

$$\mathcal{L}\{\sin(bt)\} = \mathcal{L}\left\{-\frac{1}{b}\frac{d}{dt}(\cos(bt))\right\}$$

$$= -\frac{1}{b}\left[sL\{\cos(bt)\} - \cos(bt)\Big|_{t=0}\right]$$

$$= -\frac{1}{b}\left[s\left(\frac{s}{s^2+b^2}\right) - 1\right]$$

$$\left[\mathcal{L}\{\sin bt\} = \frac{b}{s^2+b^2}\right] \#.$$

(6) Time ~~differentiation~~ integration:-

~~$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} \longleftrightarrow sF(s) - f(0)$~~

$$f(t) = \int_0^t g(\tau)\, d\tau$$

$$\mathcal{L}\left\{\int_0^t g(\tau)\, d\tau\right\} = \frac{1}{s}F(s).$$

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Ex: Find LT for ramp function: $f(t)=t$, $t>0$

Sol: $\int_0^t u(\tau)\,d\tau = t$

$$\Rightarrow \mathcal{L}[t] = \mathcal{L}\left[\int_0^t u(\tau)\,d\tau\right].$$

$$= \frac{1}{s}\cdot\frac{1}{s}$$

$$\# \boxed{\mathcal{L}[t] = \frac{1}{s^2}} \Rightarrow \boxed{\text{In general } \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}} \#$$

Ex: If $f(t)=t^2$, find the LT for $\ddot{f}(t)$.

Sol:-

$$\mathcal{L}\{\ddot{f}(t)\} = s^2F(s) - \underbrace{sf(0)}_{0} - \underbrace{\dot{f}(0)}_{0}$$

$$\mathcal{L}\{f^2(t)\} = s^2\frac{2}{s^3} \Rightarrow \boxed{\mathcal{L}\{\ddot{f}(t)\} = \frac{2}{s}} \#$$

$f(t) = t^2$
$\dot{f}(t) = 2t$
$\ddot{f}(t) = 2$

$$F(s) = \mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$$

$f(0) = 0$

$\dot{f}(0) = 0$

7] Multiplication by t or differentiation in S-Domain

$$f(t) \xleftrightarrow{\mathcal{L}} F(s).$$

$$-t\,f(t) \xleftrightarrow{\mathcal{L}} \frac{dF(s)}{ds} \quad \text{OR} \quad t\,f(t) \xleftrightarrow{\mathcal{L}} -\frac{dF(s)}{ds}.$$

$$(-t)^n f(t) \xleftrightarrow{\mathcal{L}} \frac{d^nF(s)}{ds^n} \quad \text{OR} \quad t^n f(t) \xleftrightarrow{\mathcal{L}} (-1)^n \frac{d^nF(s)}{ds^n}$$

Ex: Find LT for $t^2 \Rightarrow$ let $f(t) = t.t$

$$\mathcal{L}\{t^2\} = -\frac{d}{ds}\{\mathcal{L}\{t\}\}$$

$$= -\frac{d}{ds}\left\{\frac{1}{s^2}\right\} = \frac{2}{s^3} \#$$

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Ex: Find LT for $f(t) = t\cos(bt)$

Sol: $\mathcal{L}\{\cos(bt)\} = \frac{s}{s^2+b^2}$

$$\mathcal{L}[t\cos(bt)] = -\frac{dF(s)}{ds}$$

$$= -\frac{d}{ds}\left[\frac{s}{s^2+b^2}\right].$$

$$= \frac{(s^2+b^2) - s(2s)}{(s^2+b^2)^2}$$

$$\boxed{\mathcal{L}[t\cos(bt)] = \frac{s^2-b^2}{(s^2+b^2)^2}} \quad \#$$

[8] Initial value:-

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s).$$

OR $f(0^+) = \lim_{s\to\infty} sF(s).$

$f^n(0^+) = \lim_{s\to\infty} s^{n+1}F(s).$

$\Rightarrow$ ① $f(t) = 0$, $t<0$

② $f(t)$ must not contain any impulse or higher order singularities at $t=0$ discontinuity.

[9] Final value:-

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s).$$

OR $f(\infty)$

$\Rightarrow$ ① $f(t) = 0$, $t<0$

② $sF(s)$ must have poles in the left half of the s-plane.

Ex: The transfer function of the control system is given by.

$$H(s) = \frac{c}{s(s+b)+c}$$ & the input $x(t) = Au(t)$.

Find the final value of the output $y(t)$.

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⇒ Sol: $X(s) = \frac{A}{s}$.

$$H(s) = \frac{C}{s(s+b)+c}$$

⇒ $Y(s) = X(s)\,H(s)$.

$$Y(s) = \frac{AC}{s^2(s+b)+cs}$$

⇒ final value. $\lim_{t\to\infty} y(t)$.

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$$

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} s\,\frac{AC}{s^2(s+b)+cs} = \frac{AC}{s(s+b)+c}$$

$$s(s)(s+b+c)$$

$$= \frac{AC}{c}$$

$$\boxed{\lim_{t\to\infty} y(t) = A}$$ ##

Ex: Find the initial values for the unit step function.

final

⇒ Sol: $f(t) = u(t)$

(1) ⇒ $f(0^+) = \lim_{s\to\infty} sF(s)$

$$\left[f(0^+) = \lim_{s\to\infty} s\,\frac{1}{s} = 1 \right]$$ ## initial value.

(2) $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

$$\left[\lim_{t\to\infty} f(t) = \lim_{s\to 0} s\,\frac{1}{s} = 1 \right]$$ ## final value.

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⇒ **Remark:** The final value for ($\sin(bt)$ & $\cos(at)$) are undefined.

⇒ So, the final value property is applicable only if $f(t)$ has a final value.

10] Convolution in time:-

For LTI system.

$x(t) \rightarrow [h(t)] \rightarrow y(t)$.

$y(t) = x(t) * h(t)$.

$Y(s) = X(s) H(s)$.

$\Rightarrow x(t) * h(t) \xleftrightarrow{\mathcal{L}} H(s) \cdot X(s)$.

11] Multiplication in time or (Convolution in s-Domain):-

$f_1(t) \xleftrightarrow{\mathcal{L}} F_1(s)$.

$f_2(t) \xleftrightarrow{\mathcal{L}} F_2(s)$.

$f_1(t) \cdot f_2(t) \xleftrightarrow{\mathcal{L}} \frac{1}{2\pi j} [F_1(s) * F_2(s)]$.

12] Divided by (t) or (integration in s-Domain):-

$$\frac{f(t)}{t} \xleftrightarrow{\mathcal{L}} \int_s^\infty F(s)\, ds.$$

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* [Table 7.2 : Laplace Transform.] page 348 #

* [Table 7.3: Laplace Transform properties] page 361 #

* Examples:-

(1) If $X(s) = \frac{cs+d}{(s-a)(s-b)}$, $a \neq b$.

Find the initial value of $x(t)$.

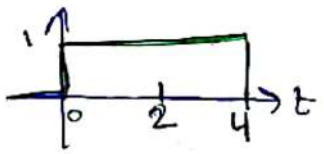
⇒ Sol: $x(0^+)$ or $\lim_{t\to 0} x(t) = \lim_{s\to\infty} sX(s)$.

$$x(0^+) = \lim_{s\to\infty} \frac{s(cs+d)}{(s-a)(s-b)} = \lim_{s\to\infty} \frac{cs^2}{s^2} = c$$

$$\Rightarrow [x(0^+) = c] \#$$

(2) Find $F(s)$ for $f(t) = rect\left(\frac{t-2}{4}\right)$.

⇒ Sol:



$$f(t) = u(t) - u(t-4).$$

$$F(s) = \frac{1}{s} - \frac{e^{-4s}}{s}$$

(3) Find $F(s)$ for $f(t) = \cos(\omega_0 t)\,u(t)$ & Find ROC.

Sol: $f(t) = \frac{1}{2} e^{j\omega_0 t} u(t) + \frac{1}{2} e^{-j\omega_0 t} u(t)$.

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$$\Rightarrow F(s) = \frac{1}{2(s-j\omega_0)} + \frac{1}{2(s+j\omega_0)}$$

$$\left[F(s) = \frac{s}{s^2+\omega_0^2} \right] \#$$

$$\left[\text{Roc: } Re[s] > 0 \right] \#.$$

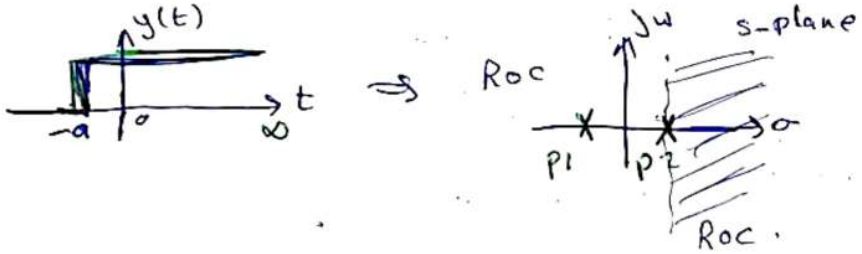
⇒ ROC properties:

① Roc don't include any poles.

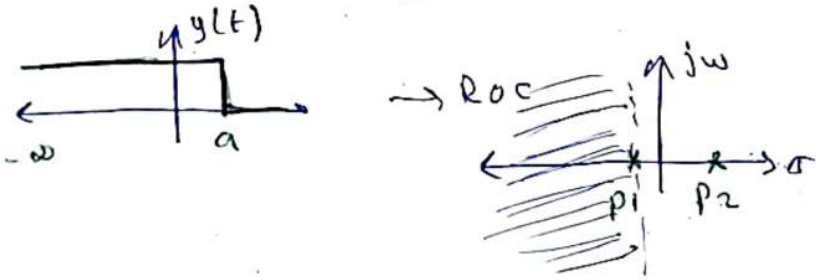
$$\Rightarrow F(s) = \frac{1}{s+2} \rightarrow \text{Poles} \rightarrow s+2=0 \rightarrow \boxed{s=-2}$$

$$\left[\begin{matrix} \text{at } s=-2 \\ F(s)=\infty \end{matrix} \right] \Rightarrow \text{Roc must not include } \underline{\underline{-2}}$$

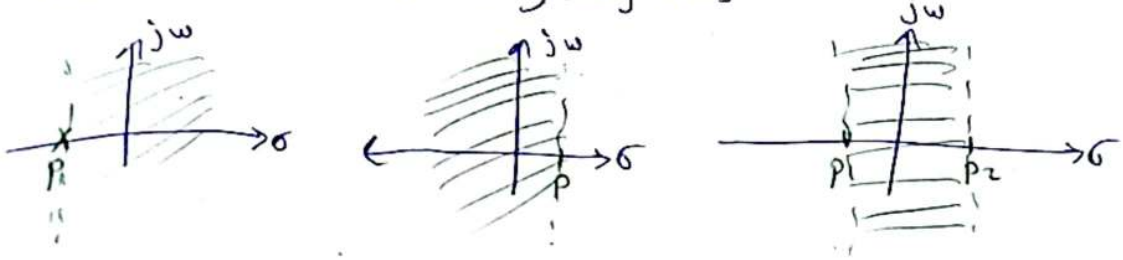
② For Right sided signals, Roc is right side to the rightmost pole



③ For left sided signals, Roc is left side to the leftmost pole.

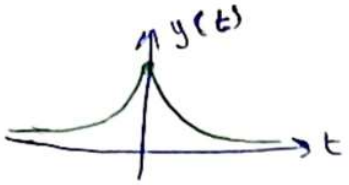


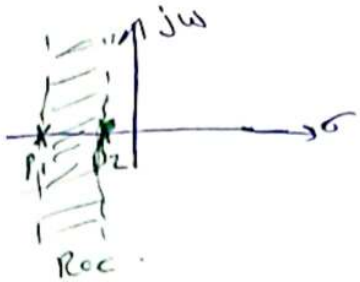
④ For absolute integrability of a signal or the stability of a system Roc should include imaginary axis.



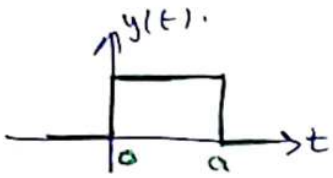
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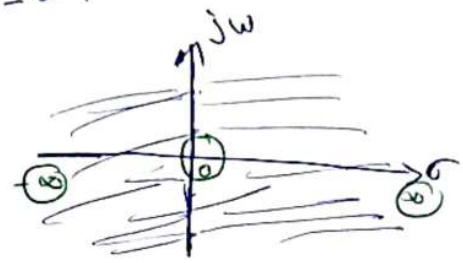
(5) For both sided signal, Roc is a strip in s-plane.





(6) For finite duration signals, Roc is the entire s-plane excluding $s=0$ & OR $s=\pm\infty$.





Roc : s-plane $-\{0, \pm\infty\}$.

Ex: Find $f(t)$ if $F(s) = \dfrac{s-5}{s(s+2)^2}$

⇒ Sol: using partial fraction.

$$F(s) = \frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2}$$

$$A(s+2)^2 + BS(s+2) + CS = s-5.$$

⇒ At $s=0$

$$A(2)^2 = -5$$

$$4A = -5 \rightarrow \boxed{A = \frac{-5}{4} = -1.25.}$$

⇒ At $s=-2$

$$-2C = -7 \rightarrow \boxed{C = \frac{7}{2} = 3.5.}$$

⇒ At $s=1$

$$9\left(\frac{-5}{4}\right) + 3B + \frac{7}{2} = -4$$

$$3B = -4 - \frac{7}{2} + \frac{45}{4} \rightarrow \boxed{B = 5/4 = 1.25}$$

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$$\Rightarrow F(s) = \frac{-5/4}{s} + \frac{5/4}{(s+2)} + \frac{7/2}{(s+2)^2}$$

$\mathcal{L}^{-1}$ ⇓

$$f(t) = \frac{-5}{4}u(t) + \left[5/4\left(e^{-2t}\right) + \frac{7}{2}te^{-2t}\right]u(t)$$

$$f(t) = \left[\frac{-5}{4} + \frac{5}{4}e^{-2t} + \frac{7}{2}te^{-2t}\right], \quad \underline{t>0}$$

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* The Z-Transform :-

⇒ For Fourier Transform → continuous-time FT.
→ Discrete-time FT.

⇒ Laplace Transform ⇒ is a general case of continuous-time FT. used for continuous-time signals & system.

⇒ The Z-transform ⇒ is general case of DTFT.
Discrete-time Fourier transform.

⇒ Z-transform : Mathmatical tool used to convert discrete-time signal to complex frequency domain.

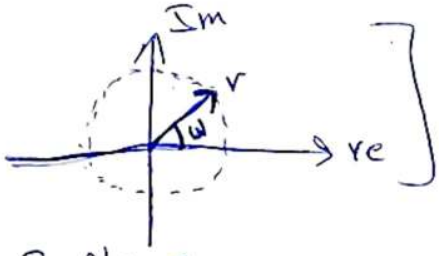
⇒ $x[n] \underset{\text{inverse}}{\overset{Z}{\rightleftarrows}} X[z]$

- Where $\left[X[z] = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}\right]$ #

- where $z = re^{j\omega}$ ⇒ complex function of ω. & n: integer.

$x[n]$ & $X[z]$ ⇒ are a Z-transform pair.

⇒ $r = |z|$



Z-plane.

⇒ The circle in Z-plane because $|z|$ or r is remain the same when ω change

⇒ $X[z] = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$ → Bidirectional ZT

$X[z] = \sum_{n=0}^{\infty} x[n]\, z^{-n}$ → Unidirectional ZT

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⇒ Ex: Find $X[z]$ for $x[n] = a^n u[n]$. ⇒ Right-sided sequence.

Sol:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (a z^{-1})^n = \frac{(az^{-1})^0 - (az^{-1})^{\infty+1}}{1 - az^{-1}}$$

$(az^{-1})^0 = 1$

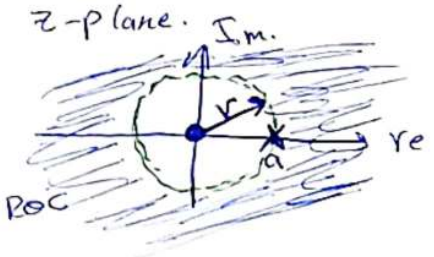
$(az^{-1})^{\infty} = \begin{cases} 0, & \text{when } |az^{-1}| < 1 \rightarrow |z| > |a|, \text{ convergence.} \\ \infty, & \text{otherwise. , divergence} \end{cases}$

$$\Rightarrow X[z] = \frac{1}{1 - az^{-1}}, \quad |z| > |a|.$$

$$= \frac{z}{z - a}, \quad |z| > a, \quad a \text{ is real and positive}$$

⇒ $r = |z| \Rightarrow r > a \rightarrow$ ROC] #

[$X[z]$ has one zero at $z = 0$ & one pole at $z = a$] #



zero → represented by ○
pole → represented by X

(a) ROC ⇒ outside the circle → in this region the z-transform is exists and converges.
is finite function.

(b) Inside the circle the z-transform is not exists & diverges ⇒ $X[z]$ is equal to ∞

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Ex: For $x[n] = -a^n u[-n-1]$ Find $X[z]$.

Sol: $x[n] \Rightarrow$ left sided sequence.

$$X[z] = \sum_{n=-\infty}^{\infty} -a^n u[-n-1] z^{-n}$$

($-\infty \to -1$)

$$= \sum_{n=-\infty}^{-1} -a^n z^{-n}.$$

$$= -\sum_{n=-\infty}^{-1} (az^{-1})^n$$

$$= -\left[\frac{(az^{-1})^{-\infty} - (az^{-1})^{-1+1}}{1-az^{-1}}\right]$$

($(az^{-1})^{-\infty} \to 0$)

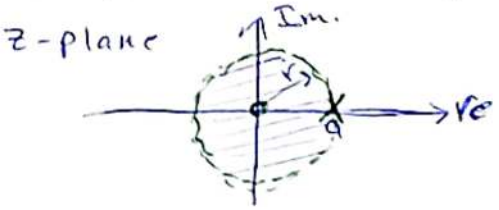
$$= -\left[\frac{0-1}{1-az^{-1}}\right]$$

$$= \frac{1}{1-az^{-1}}, \quad |z| < |a|.$$

$\Rightarrow (az^{-1})^{-\infty} = (a^{-1}z)^{\infty} = \begin{cases} 0, & \text{when } |a^{-1}z| < 1 \Rightarrow |z| < |a| \\ \infty, & \text{otherwise.} \end{cases}$

$$\Rightarrow X[z] = \frac{z}{z-a}, \quad |z| < |a|. \quad \to a: \text{real and positive.}$$

$r = |z| \to r < |a|$, $X[z] \to$ one zero at $z=0$, one pole at $z=a$.



Roc $\Rightarrow$ inside the circle $\Rightarrow$ The Z.T is exists & converges.

Ex:

$$a^n u[n] \xleftrightarrow{Z} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |z| > |a|$$

$$-a^n u[-n-1] \xleftrightarrow{Z} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |z| < |a|$$

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⇒ Remarks: ✳

① For right-sided exponential signal $\left(\overset{Ex}{X[n] = a^n u[n]}\right)$.

The ROC of $X[z]$ is outside the circle.

⇒ The circle is not included in ROC.

② For left-sided exponential signal $\left(\overset{Ex}{X[n] = -a^n u[-n-1]}\right)$

The ROC of $X[z]$ is inside the circle.

→ The circle is not included.

③ The ROC doesn't include any pole because the pole makes $X[z] = \infty$.

④ The ROC may include zero or not because the zero makes $X(z) = 0$, which is finite.

Ex: Sum of two exponential sequences.

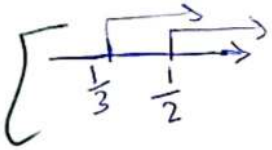
$x[n] = \left[\frac{1}{2}\right]^n u[n] + \left[\frac{-1}{3}\right]^n u[n]$. Find $X[z]$.

Sol:

$$X[z] = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - \frac{1}{3}z^{-1}}$$

$$X[z] = \frac{z}{z - \frac{1}{2}} + \frac{z}{z + \frac{1}{3}}, \quad |z| > \left|\frac{1}{2}\right| \text{ & } |z| > \left|\frac{-1}{3}\right|$$

⇒ The ROC is the common region between $|z| > \frac{1}{2}$ & $|z| > \frac{1}{3}$.



→ common region is $|z| > \frac{1}{2}$

⇒ ROC is $|z| > \frac{1}{2}$.

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$$\Rightarrow X[z] = \frac{z}{z-\frac{1}{2}} + \frac{z}{z+\frac{1}{3}}$$

$$= \frac{z(z+\frac{1}{3}) + z(z-\frac{1}{2})}{(z-\frac{1}{2})(z+\frac{1}{3})}$$

$$X[z] = \frac{2z^2 - \frac{z}{6}}{z^2 - \frac{z}{6} - \frac{1}{6}}, \quad \text{ROC } |z| > \frac{1}{2}$$

$\Rightarrow$ 2-Zeros $\rightarrow$ $2z^2 - \frac{z}{6} = 0$

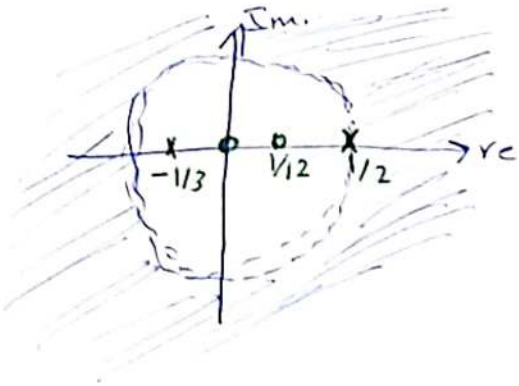
$$z(2z - \frac{1}{6}) = 0$$

$$\left[z = 0 \;,\; z = \frac{1}{12} \right] \text{ \#}$$

2-Poles $\rightarrow$ $z^2 - \frac{z}{6} - \frac{1}{6} = 0$

$$(z-\frac{1}{2})(z+\frac{1}{3}) = 0$$

$$\left[z = \frac{1}{2} \;,\; z = -\frac{1}{3} \right] \text{ \#}$$



ROC $|z| > \frac{1}{2}$

Ex: Two-Sided exponential signal $x[n] = \left[\frac{1}{3}\right]^n u[n] - \left[\frac{1}{2}\right]^n u[-n-1]$.

Find $X[z]$.

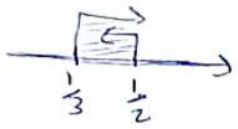
Sol:

$$X[z] = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{2}z^{-1}}$$

$|z| > |-\frac{1}{3}|$ $\qquad$ $|z| < |\frac{1}{2}|$.

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$\Rightarrow |z| > \frac{1}{3}$ & $|z| < \frac{1}{2}$



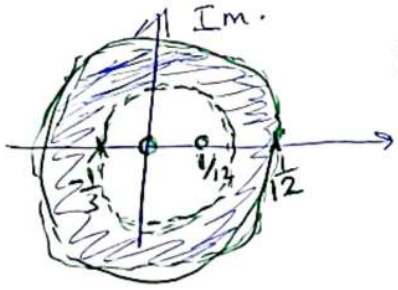
$\Rightarrow \frac{1}{3} < |z| < \frac{1}{2}$

$$\Rightarrow X[z] = \frac{z}{z+\frac{1}{3}} + \frac{z}{z-\frac{1}{2}}$$

$$X[z] = \frac{2z^2 + \frac{1}{6}z}{(z^2 - \frac{z}{6} - \frac{1}{6})}$$

$\Rightarrow$ 2 Zeros at $z=0$ & $z=\frac{1}{12}$.

2 poles at $z=\frac{1}{2}$ & $z=-\frac{1}{3}$.



Roc $\frac{-1}{3} < |z| < \frac{1}{2}$.

Roc: Between 2 circles. (ring).

Ex: Finite length sequence.

$$x[n] = \begin{cases} a^n, & 0 \le n \le N-1 \\ 0, & \text{otherwise.} \end{cases}$$

Find $X[z]$. if $N = 16$.

Sol:-

$$X[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=0}^{N-1} a^n z^{-1}$$

$$= \frac{1 - (az^{-1})^N}{1 - az^{-1}} * \frac{z^N}{z^N}$$

$$= \frac{z^N - a^N}{z^N - az^{N-1}} = \frac{z^N - a^N}{z^{N-1}(z-a)}$$

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⇒ At $z=a$ → zeros cancels the pole

⇒ So, there is no zero nor pole.

→ The roots of $z^N - a^N$ → can be obtained by.

$$z_k = a e^{j2\pi k/N}, \quad k = 0, 1, 2, \ldots N-1$$

So, we have N roots

$z_0 = a e^{0} = a$

$z_1 = a e^{j2\pi/N}$ ⇒

$z_2 = a e^{j4\pi/N}$ ⇒ . } $N=16$.

$\vdots$

$z_{N-1} = a e^{j2\pi\frac{N-1}{N}}$

$$\Rightarrow X[z] = \frac{z^N - a^N}{z^{N-1}(z-a)} = \frac{(z-z_0)(z-z_1)\cdots(z-z_{15})}{z^{N-1}(z-a)} \quad \rightarrow N-1$$

$N=16$, $z_0 = a$.

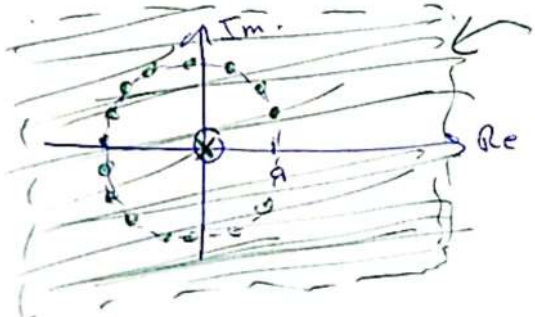
$$X[z] = \frac{(z-a)(z-z_1)\cdots(z-z_{15})}{z^{15}(z-a)}$$

$$X[z] = \frac{(z-z_1)\cdots(z-z_{15})}{z^{15}}$$

⇒ $X[z]$ has 15 zeros at $z = z_1, z_2 \ldots z_{15}$.

& 15 poles at $z=0$

⇒ For a real & positive.



← Roc.

⇒ The Roc is the z-plane except at $z=0$ because it is a pole

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* Remarks:

① For two-sided exponential sequences $x[n]$, the ROC of $X[z]$ is a ring and the two circles are not included.

② For finite-length sequence $x[n]$, the ROC of $X[z]$ is all z values (all z-plane) except at $z=0$ or at $z=\infty$

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